

Thin-wall Quillwort
(Isoetes microvela)
Species Status Assessment
Version 1.0



Thin-wall quillwort. Photo credit: Jess Schaner (North Carolina Natural Heritage Program).

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EXECUTIVE SUMMARY

This report summarizes the results of a species status assessment (SSA) conducted for thin-wall quillwort, an at-risk species for which the U.S. Fish and Wildlife Service (Service) received a petition to list in 2010. A 90-day finding published in the Federal Register on September 27, 2011, determined this species may be warranted for listing. This SSA report contains the best scientific and commercial information available regarding the species' biology and factors that influence the species' viability.

Thin-wall quillwort (*Isoetes microvela*) is a primitive spore-bearing plant endemic to blackwater river habitats in the Coastal Plain of southeastern North Carolina that are periodically inundated or scoured by water flow. The greatest threats to the species appear to be associated with actions that influence the natural hydrological cycles upon which thin-wall quillwort depends, as well as those actions that modify or destroy habitat or water quality, such as concentrated animal feeding operations (CAFOs) and compound storm events. Other stressors that may negatively affect the species include local recreational activity, non-native invasive species encroachment, and climate change.

For the purposes of this SSA, we defined resiliency as the ability of thin-wall quillwort to sustain populations in the habitats where it occurs over time. To have high resiliency, populations must have abundant suitable habitat for successful growth and development of mature plants and establishment of dispersing propagules, adequate water quality, enough individuals to support effective reproduction and gene flow, sufficient connectivity between populations for successful dispersal, gene flow, and rescue (when necessary), and can recover from stochastic events or disturbances (e.g., drought, flooding, etc.) over time. Accordingly, to assess current resiliency, we considered a combination of factors including suitability of habitat, water quality, population size (number of individual plants), and population connectivity.

At present, two of seven known thin-wall quillwort populations (28.5%) have high resiliency, three populations (43%) have moderate resiliency, and two (28.5%) have low resiliency. These medium and highly resilient populations represent 90-97% of the known plants and 95% of the total area currently occupied by the species. Over two-thirds of all known plants (3,228 of 3,882) occur at the two largest thin-wall quillwort populations. The site occupied by the largest population is owned by The Nature Conservancy, where lands are protected from future development and are thus expected to remain in good condition. With regards to current representation, thin-wall quillwort has multiple moderately to highly resilient populations across its known range, thus enabling the species to withstand catastrophic events. The species also possesses various traits, which support its overall adaptive capacity. Additionally, no populations are known to be historical or extirpated.

For our future conditions analysis, we looked at how the threats of habitat loss and modification, water quality, and severe storms may affect the resiliency of thin-wall quillwort populations out to 2100 under two scenarios. The number and distribution of highly and moderately resilient populations is expected to remain stable through 2100. One highly resilient population is situated wholly on protected lands and will be managed and likely maintain high resiliency into the future. One low resiliency population on private land may become extirpated by continued recreational impacts.

While thin-wall quillwort occupies a narrow geographic range, it occurs across a modest expanse of the North Carolina Coastal Plain in two ecoregions and experts indicate that with increased sampling more populations may be discovered and the true extent of the species more clearly understood.

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CHAPTER 1: INTRODUCTION

1.1 Background

This report summarizes the results of a species status assessment (SSA) conducted for thin-wall quillwort (*Isoetes microvela*) using the best scientific and commercial information available. Thin-wall quillwort was included in a listing petition from the Center for Biological Diversity et al. (CBD 2010, pp. 628–629) in April 2010. The petition requested that the Service list 404 aquatic, riparian, and wetland species as endangered or threatened under the Act and in 2011 the Service found that this petition presented substantial scientific or commercial information indicating that listing may be warranted for 374 species, including thin-wall quillwort.

1.2 Analytical Framework

The SSA Framework is an analytical process developed by the U.S. Fish and Wildlife Service (Service) to deliver foundational science to inform Endangered Species Act (16 U.S.C. 1531 *et seq.*) (Act) decisions (Service 2016, entire; Smith et al. 2018, entire). Intended to support more transparent and defensible decision-making, an SSA is a focused, repeatable, and rigorous scientific review of the species' biology and threats, an evaluation of its biological status, and an assessment of the resources and conditions relevant to maintaining viable populations through time. While not a decisional document, the resulting SSA report is the document upon which other decision documents—listing rules, recovery plans, and 5-year reviews—are based if the species is determined to warrant listing under the Act. Using the conservation biology principles of resiliency, redundancy, and representation (i.e., the “3Rs”) as a lens to evaluate the current and future condition of the species, the SSA report provides a thorough account of known biology and natural history and assesses demographic risks, stressors, and limiting factors in the context of determining the viability and risks of extinction for the species (Wolf et al. 2015, pp. 204-206).

Resiliency is the ability of a species to withstand environmental stochasticity (normal, year-to-year variations in environmental conditions such as temperature, rainfall), periodic disturbances within the normal range of variation (fire, floods, storms), and demographic stochasticity (normal variation in demographic rates such as mortality and fecundity) (Smith et al. 2018, p. 304, 306). At the population scale, resiliency reflects an organism's capacity to persist across the natural range of favorable and unfavorable conditions.

Redundancy describes the ability of a species to withstand catastrophes—stochastic events expected to lead to population collapse regardless of population health and for which adaptation is unlikely (Evans et al. 2016, pp. 6, 21; Smith et al. 2018, pp. 304, 306). Herein evaluated at the species-level, redundancy analysis entails assessing the cumulative risk of catastrophes transpiring over time.

Representation is the ability of a species to adapt to both near-term and long-term changes in its physical (climate and habitat conditions, habitat structure, etc.) and biological (pathogens, competitors, predators, etc.) environments (Smith et al. 2018, p. 306). Referred to as adaptive capacity, representation occurs via either (1) migration to new, suitable environments or (2) altering physical or behavioral traits (phenotypes) to match the new environmental conditions (Thurman et al. 2020, entire). Such acclimatization is essential for viability, as species need to continually adapt to their continuously changing environments (Nicotra et al. 2015, pp. 1269-1270; Beever et al. 2016, p. 132). The latter (evolution) occurs via the evolutionary processes of natural selection, gene flow, mutations, and genetic drift (Crandall et al. 2000, pp. 290-291; Sgro et al. 2011, p. 327).

This document is a compilation of the best available scientific and commercial information (and associated assumptions and uncertainties regarding that information) used to assess the viability of the

species. We herein define viability as the ability of the species to sustain populations in the wild within a meaningful timeframe. For thin-wall quillwort, we determined that 75 years is an appropriate period of time, which enables reasonable forecasting of the effects of potential stressors in conjunction with species' response to said stressors.

To most accurately portray species and habitat conditions in this evaluation, we made a concerted effort to restrict citations wherever possible—geographically to the southeastern United States, and southeastern North Carolina, specifically; biologically to known *Isoetes*; to ecologically similar or proxy habitats; and ecologically to amphibious/semi-aquatic quillwort taxa (i.e., those adapted to environments where water is present for only part of the year and may live submerged for long periods, but can also produce spores once emerged), with the consideration that characters may subtly differ from wholly aquatic taxa (i.e., aquatic species inhabiting perennial water bodies that only produce spores if permanently submerged) (Azzella et al. 2024, p. 1421).

CHAPTER 2: SPECIES INFORMATION

2.1 Taxonomy

Extensive field and cytological (cellular) investigations of *Isoetes* in the southeastern United States began in 1990 (Brunton and Britton 1998, pp. 262-263). This work revealed a new hexaploid (having six complete sets of chromosomes) species in North Carolina with unusually large spores (microscopic reproductive units), which was subsequently described as *Isoetes microvela* {thin-wall quillwort} (Brunton and Britton 1998, pp. 262-263). The type specimen was collected in 1995 from the banks of the White Oak River near Maysville, North Carolina (Jones/Onslow counties), where thousands of clumps were observed. To date, few specimens of thin-wall quillwort are known, with the earliest dating to 1950 and collected by A.E. Radford from Bolivia Branch of Middle Swamp outside of Bolivia, North Carolina (Brunswick County) (Brunton and Britton 1998, p. 271). Subsequent collections date to 1995, 1999, 2004-2006, 2009, 2010, and 2013 (Schaner et al. 2021, p. 6; NCNHP 2026a, pp. 1-4, 6, 9-11, 15, 17, 19-21, 25; SERNEC 2026, unpaginated).

Best available information indicates that *Isoetes microvela* is a valid taxon and, like many *Isoetes*, demonstrates unique allelic patterns (Schafran 2019, p. 15; WFO 2026, unpaginated; IT IS 2026, unpaginated). More precisely, Schafran (2019 pp. 160, 193) proposed that thin-wall quillwort is a complex taxon of mixed (*I. melanopoda* (2n) × *I. melanopoda* ssp. *silvatica* (2n)), and somewhat unknown, genetic parentage. He found that the genotype of individuals collected from the type-locality on the White Oak River contains an extra allele from a parent that was not found to be present in all other populations of thin-wall quillwort sequenced (Schafran 2019, p. 193). This comes as no surprise, as hybridization between quillwort species is accepted as the rule, rather than the exception (Hickey et al. 1989, p. 88; Hoot et al. 2004, pp. 899, 903; Schafran 2019, p. 11; Larsén et al. 2022, p. 2; Suissa et al. 2022, entire).

Isoetes is the only genus in the quillwort family (Isoëtaceae). Members of this plant family are found on all continents except Antarctica (Freund 2016, p. 87; Larsén and Rydin 2016, p. 157; Schafran 2019 p. 3). Like thin-wall quillwort, most *Isoetes* taxa occurring in the southeastern United States are narrow endemics with limited geographic distributions (Brunton et al. 2025, pp. 3, 5). The species name, *microvela*, is derived from the Latin words “micro” for small and “velum” that covers the sporangia (a sac-like enclosure where spores are produced and stored), an indication that the velum (thin layer of tissue on the inner leaf face) in this species is highly reduced (Brunton and Britton 1998, pp. 270-271; Brunton et al. 2025, p. 7). Thin-wall quillwort is not known by any other common names.

2.2 Historical and Current Range

Thin-wall quillwort is endemic to the Coastal Plain region of the White Oak, Lumber, and Cape Fear River basins of North Carolina. As of the 2021 status survey, thin-wall quillwort is considered extant in Bladen, Brunswick, Jones, Onslow, Pender, and Sampson counties (Schaner et al. 2021, entire; Figure 2.2). At some sites, the river represents the county border, and thus populations are considered as occurring in both counties. While it is not considered historical or extirpated from any county, individuals of one sub-element occurrence in Brunswick County have not been observed since 2015 (Schaner et al. 2021, p. 14; NCNHP 2026a, p. 3).

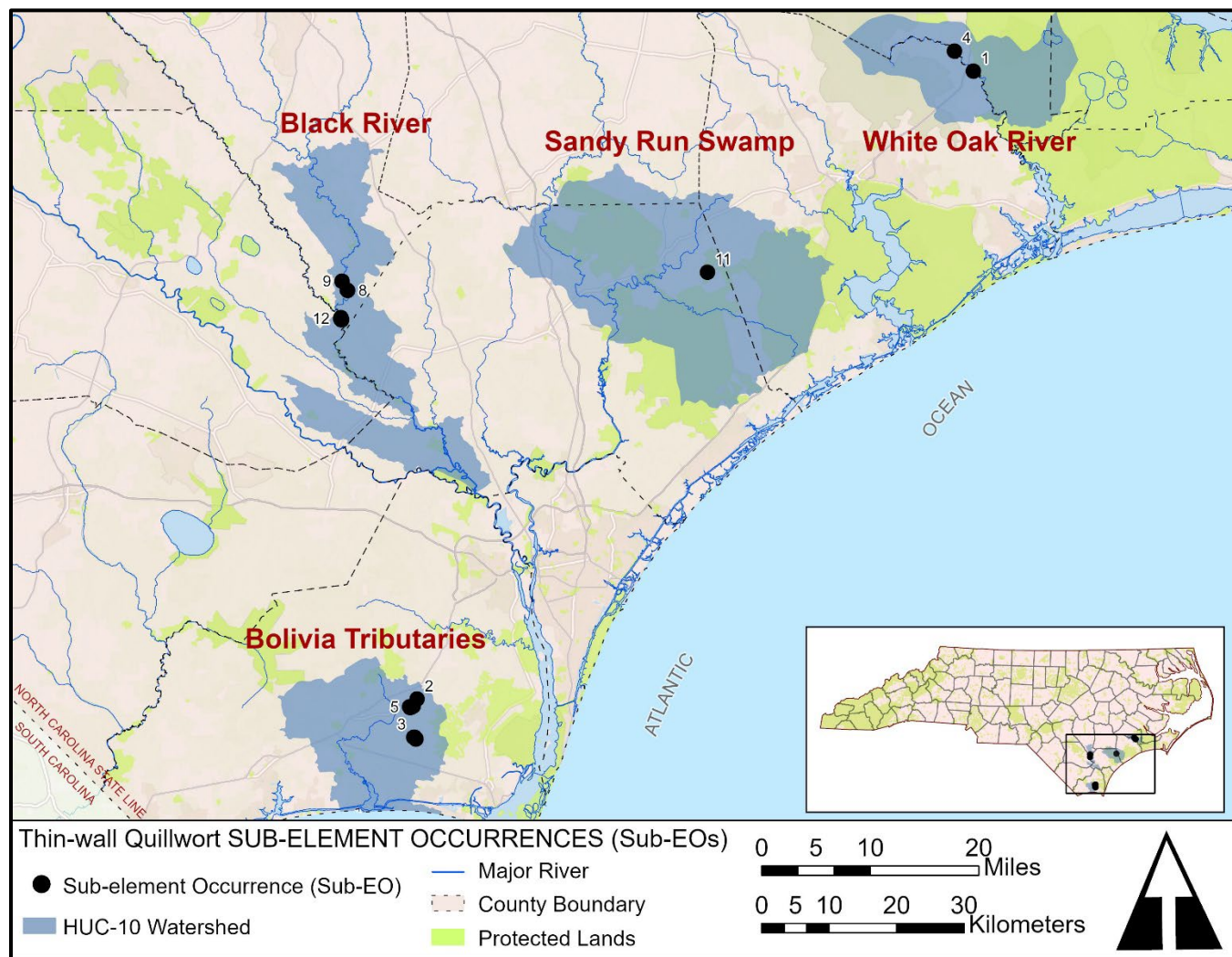


Figure 2.2 Thin-wall quillwort sub-element occurrence (sub-EO) and HUC-10 watershed map. Thin-wall quillwort is considered extant in occupied watersheds (NCNHP site names in red) (Schaner et al. 2021, entire).

2.3 Species Morphological Description

Thin-wall quillwort is a perennial, semi-aquatic, seedless vascular herb belonging to an ancient plant family known as Isoëtaceae, or commonly, the quillworts (Brunton and Britton 1998, entire). Like many quillworts, it is an obligate wetland plant (USACE 2022, unpaginated; NRCS 2026b, unpaginated). It is one of ~22 described *Isoëtes* taxa in the southeastern United States, which is a fraction of the approximately 200-300 species known worldwide (Brunton 2015, entire; Schafran et al. 2016, p. 97;

Troia et al. 2016, p. 101). Its life cycle follows that of a typical quillwort, alternating between a macroscopic sporophyte form and a minute gametophyte form.

2.3.1 Sporophyte Form

When compared to other members of the genus *Isoetes*, the sporophyte form of thin-wall quillwort is rather large, its leaves 35 to 40 cm (13.75 to 15.75 in) in length and 1.1 to 1.7 cm (0.4 to 0.7 in) wide. It grows in clumps of typically 20 to 25 simple linear leaves arising from a lobed corm (thickened, vertical, underground stem that stores food for a plant) (Figure 2.3.1). The leaves are lax (loose or open) and dark green and can wither on exposed plants during periods of prolonged drought. They have a single vein and are largely hollow, filled with four large air chambers that run the length of the leaf. The sporangia are streaked with brown and have a 3.0 to 4.5 mm (0.12 to 0.18 in) long ligule (membranous scale) at their base that is triangular-lanceolate (tapered to a sharp point) in shape and of indefinite function (Sharma and Singh 1984, entire; Brunton and Britton 1998, p. 265; Freund 2016, p. 88). The velum in thin-wall quillwort is opaque and remarkably narrow, covering only about 10% of the sporangium (Brunton and Britton 1998, pp. 261, 265, 271-272; Brunton et al. 2025, pp. 7, 66). It is the smallest known velum covering of any of the southern United States polyploid (having more than two complete sets of chromosomes) species, which is reflected in the specific epitaph of the taxon, *microvela*, and is useful in identification (Pfeiffer 1922, pp. 84, 87; Brunton and Britton 1998, pp. 270-271; Schafran et al. 2016, p. 104).

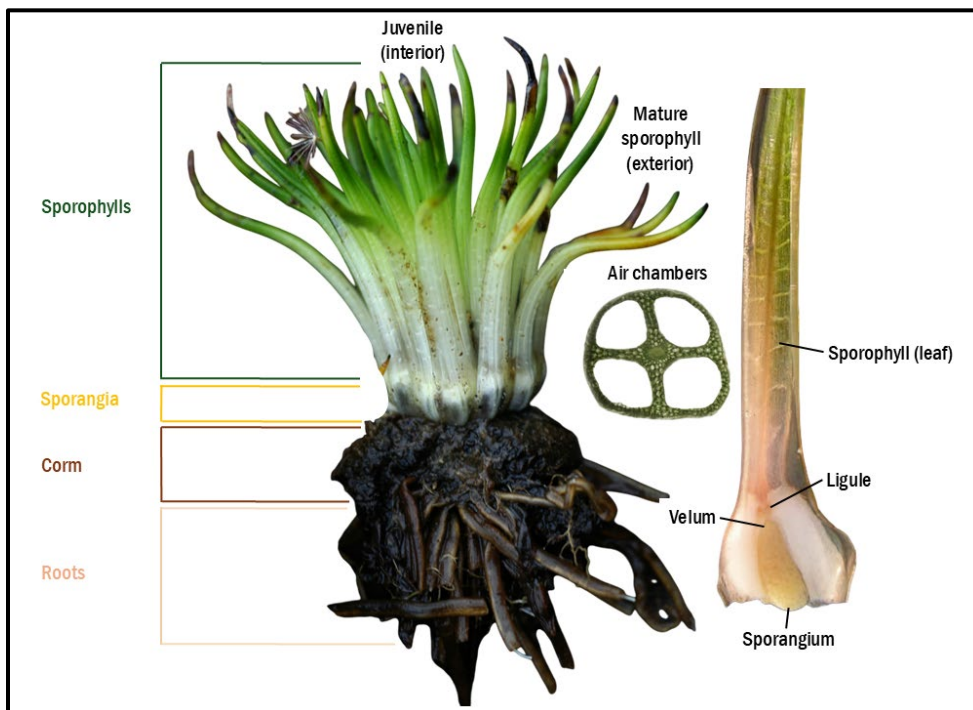


Figure 2.3.1 General *Isoetes* (sporophyte) plant morphology. Images taken from: (left-right) A. Munro (sporophyte with corm and roots), R. Golding (sporophyll cross-section), and P. Dziuk (sporophyll with ligule, velum, and sporangium).

2.3.2 Gametophyte Form

Differentiating taxa in the genus *Isoetes* is challenging due to simplicity in form, the lack of easily observable identifying structural features, and plasticity found in the characters that can be studied (Hickey 1986, p. 1; Brunton 2015, p. 86). For accurate species-level identification, examination of the spores under magnification is common and typically required (Musselman 2002, p. 475; Bray et al.

2018, p. 264; Larsén et al. 2025, p. 2). Plants in the genus *Isoetes* are heterosporous, thereby producing two types of spores (diaspores): female spores that are relatively large (megaspores) and minute male spores (microspores) (Figure 2.3.2; Freund 2022, p. 21). Megaspore ornamentation (“sculpturing”, or surface morphology) and size are the most important characters for identifying *Isoetes* (Pfeiffer 1922, p. 89; Musselman 2002, p. 475; Schafran 2019, p. 4). The size of the megaspores reflects the ploidy level (the number of sets of chromosomes present in a plant cell) of the species, with higher ploidy levels producing larger megaspores. Nearly half of the nine known *Isoetes* taxa in the southeastern United States are diploids, which have the smallest megaspores, barely visible with the naked eye (Schafran 2019, pp. 30-31). However, polyploids such as thin-wall quillwort (a hexaploid), have the largest megaspores, readily visible without magnification (Brunton and Britton 1998, entire; Brunton 2015, p. 90).

The megaspores of thin-wall quillwort average 525 μm (0.021 in) in diameter, which is considerably larger than Engelmann’s quillwort (*I. engelmannii*) (460 μm / 0.018 in), with which it was often confused, but smaller than most polyploids (Brunton and Britton 1998, pp. 265-266). Megaspores mature from May–July and possibly through September but can often be found in the soil near the base of the plant after they have dehisced (split open along a pre-existing line of weakness to release contents) (Brunton and Britton 1998, p. 270). Thin-wall quillwort has a densely reticulate (net-like pattern of veins, fibers, or lines) ornamentation pattern, more congested than other hexaploids (Hickey 1986, p. 3; Brunton and Britton 1998, p. 266). All surfaces feature irregularly radiating crests and thin tubercles (small, rounded wart-like projections), with some having ridges with short spines (Figure 2.3.2; Brunton and Britton 1998, p. 270).

Thin-wall quillwort microspores are very small (averaging 29 μm /0.001 in in diameter) and only visible with scanning electron microscope (Brunton and Britton 1998, p. 270). This is significantly smaller than microspores of other southeastern United States hexaploids and more comparable to other tetraploid *Isoetes* (Brunton and Britton 1998, pp. 266-268). The microspores of thin-wall quillwort lack surface ornamentation or are obscurely ornamented (Figure 2.3.2; Brunton and Britton 1998, pp. 261, 272).

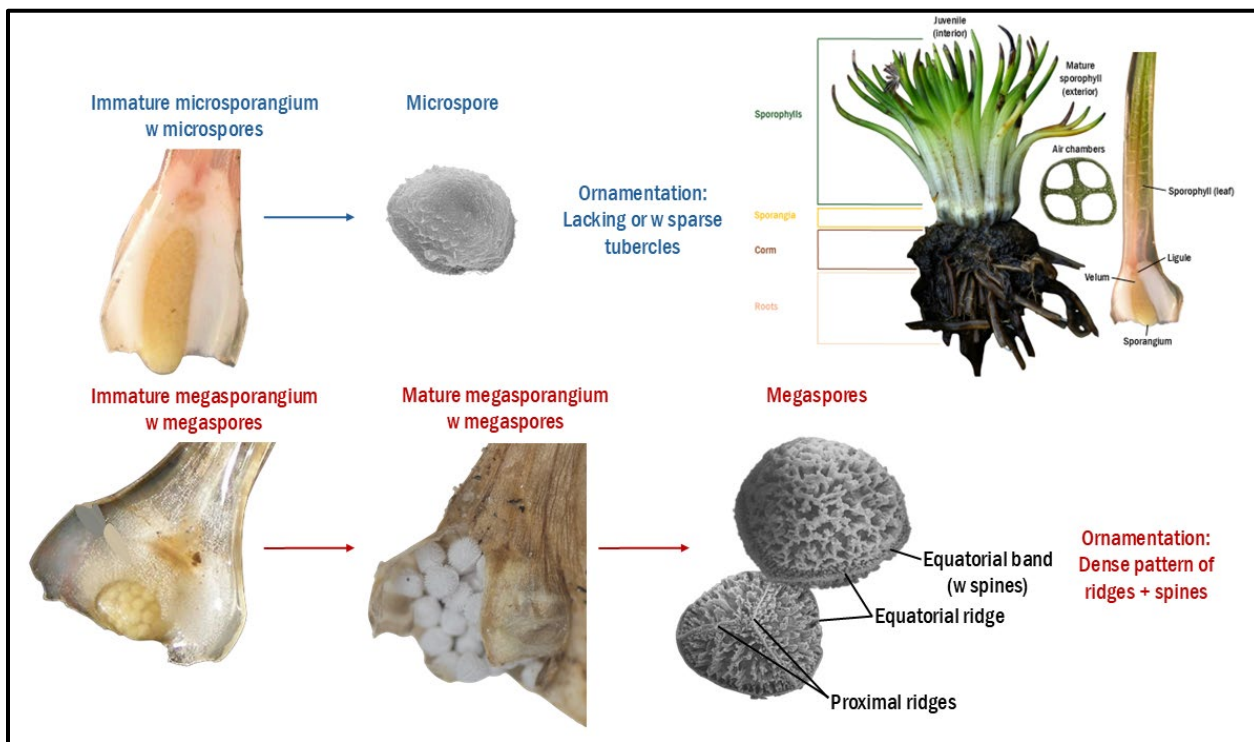


Figure 2.3.2 General *Isoëtes* microspore and megaspore (gametophytes) morphology. Images taken from: (clockwise from top-left) P. Dziuk (megaspore/microsporangium with spores), K.L. McIntosh (microspore), D.F. Brunton and K.L. McIntosh (megaspores), and S. Brinker (mature megasporangium with megaspores). Inset images taken from: (left-right) A. Munro (sporophyte with corm and roots), R. Golding (sporophyll cross-section), and P. Dziuk (sporophyll with ligule, velum, and sporangium).

2.4 Life History

Apart from its original description (Brunton and Britton 1998, entire), very little is known about the specific biology and life history of thin-wall quillwort and no targeted life history studies have been performed. Therefore, the following description is compiled using information derived from recent status surveys and encounters, expert opinion, museum collections, and observations and records of other quillwort taxa.

In brief, thin-wall quillwort follows the standard *Isoëtes* life cycle to accomplish sexual reproduction, alternating between mature plants (sporophyte stage) and haploid spores (gametophyte stage) (Figure 2.4).

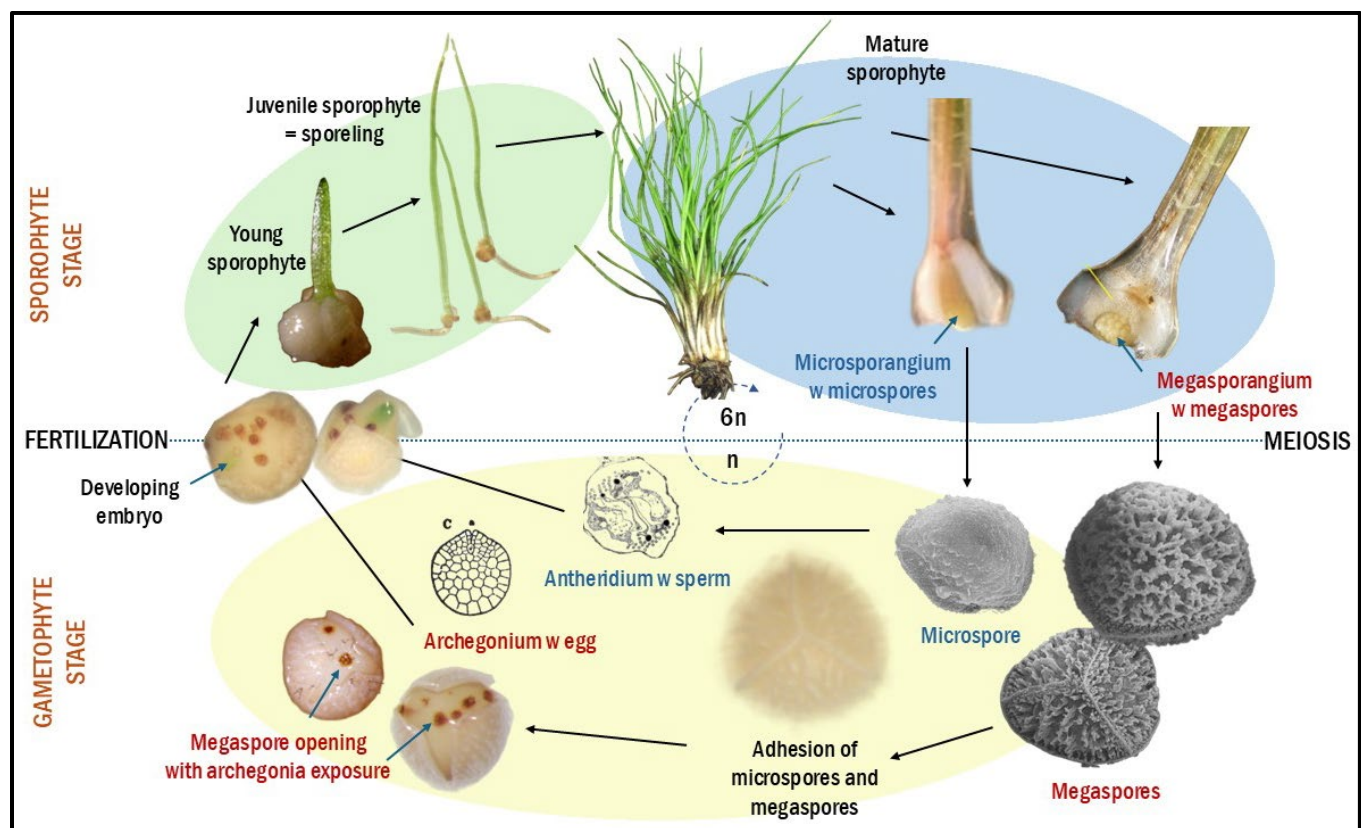


Figure 2.4 Generalized life cycle of *Isoetes macrospora*. Images taken from: (clockwise from top-left) Wikipedia (young sporophyte), Magrini et al. (juvenile sporophytes), P. Dziuk (megaspore/microsporangium with spores), K.L. McIntosh (microspore), D.F. Brunton and K.L. McIntosh (megaspores), Renzaglia et al. (antheridium with sperm), and Magrini et al. (micro/megaspore adhesion, megaspores with archegonia, and developing embryo).

More specifically, spores are produced in structures called sporangia, found at the base of the sporophyte leaf blade near the corm, with megaspores commencing development first, followed by microspores (Schafran 2019, p. 4; Larsén et al. 2025, p. 1). There are separate sporangia for male and

female spores, which via meiosis produce haploid microspores (inner leaf blades) and megaspores (outer leaves), respectively (Freund 2022, p. 21). Spores formed within microsporangia have been observed to number several hundred thousand and often more than 100 in megasporangia (Smith 1900, pp. 251, 258; Blackmore et al. 2012, p. 85). Timing of spore maturation is associated with plant growth form—while submerged aquatic spores mature gradually in summer, those of terrestrial/amphibious taxa, such as thin-wall quillwort, develop synchronously with drying of ephemeral habitats and are released when leaves senesce (deteriorate/wither) at the end of the growing season (Taylor and Hickey 1992, p. 621; Vöge 2006, p. 232). Thin-wall quillwort megaspores reportedly mature between May and July (possibly to September) (Brunton and Britton 1998, p. 270; Schaner et al. 2021, p. 7). Megaspores and microspores germinate (within the spore) and develop into short-lived gametophytes that produce eggs and sperm, which fuse to form an embryo that develops into a new sporophyte (plant) (Freund 2022, p. 21). Following the development of sterile leaves for upwards of several seasons, juvenile plants (sporelings) then begin to develop sporangia (Pfeiffer 1922, pp. 91).

Quillwort dispersal occurs via transport of various dispersal units—i.e., spores, detached leaf material, and entire plants (Small and Hickey 1997, p. 110; Brunton 2001, entire). Once released from sporangia through the drying and disintegration of plant tissue, mature spores commonly remain free on the soil surface and dispersed to nearby localities along water courses (hydrochory) and possibly may be ingested or adhere to foraging wildlife (e.g., migratory waterfowl) (zoochory) (LaMotte 1937, p. 455; Brunton 2001, p. 20; Hoot et al. 2006, p. 458; Larsén and Rydin 2016, pp. 169-170; Troia 2016, entire; Brock 2025, pp. 2121-2122). Spores, as well as entire mature *Isoëtes* plants, may also conceivably “hitch-hike” on the gales of severe spring storms (anemochory), but the likelihood of such occurrence is deemed uncertain and rare (Brunton 2001, p. 20). Thin-wall quillwort megaspores have been observed drifting downstream from established populations and/or adhered to riparian vegetation (Brunton 2001, p. 20). For those species occurring along water courses, there may be high levels of gene flow among neighboring sub-populations due to successful spore dispersal (Yang et al. 2011, entire; Ma et al. 2019, p. 343). Precise dispersal patterns are likely the result of overlapping physiological, physiographic, and habitat factors. Accurate records of quillwort dispersal distance remain scarce and further supporting research on *Isoëtes* dispersal rates and mechanisms are required (Cox and Hickey 1984, p. 438; Wood et al. 2018, p. 477; Wood et al. 2020, p. 19).

Due to the proximity of the mega- and microsporangia on an individual, it has been suggested that there is a high degree of self-fertilization among *Isoëtes* (Caplen and Werth 2000, p. 255; Abeli et al. 2026, p. 4). In fact, it is fundamental that the two spore types germinate in close proximity for successful sexual reproduction; indeed, this often transpires via synaptospory (the adhesion of microspores to megaspores) (Troia 2016, pp. 279-280). Fortunately, as observed by various authors, coordinated dispersal of quillwort megaspores and microspores appears common (Musselman 2002, pp. 475, 484; Troia 2016, p. 279). Indeed, the differing physical characteristics of the two spore types (i.e., size and ornamentation), as well as unique features adaptive for the aquatic, amphibious, or terrestrial habitats in which the taxa are found, reflect their different dispersal potentials (Link-Pérez and Laffan 2018, p. 516).

Recent research has confirmed that some members of the genus *Isoëtes* can also reproduce asexually (Caldeira et al. 2021, p. 2, 8-11). Bray et al. (2018, pp. 264-265) documented that the corms of some *Isoëtes* may branch and produce plantlets that detach and live independently. It is possible that thin-wall quillwort also vegetatively reproduces, as Brunton and Britton observed in one instance that two clumps of *I. microvela* were joined at the corm (Brunton and Britton 1998, p. 264).

Along with hybridization, polyploidy (the multiplication of complete chromosome sets per nucleus) is considered to be a major mode of speciation in quillworts. The evolutionary consequences of polyploidy are highly context dependent. In some circumstances, polyploidy may provide a selective advantage for

quillwort dispersal/colonization events, thereby bolstering wide geographic distribution, increasing inter-species range commonality, and concentration at lower elevations relative to diploid taxa (Liu et al. 2004, pp. 1240-1241, 1243-1244; Husband et al. 2013, entire; Suissa et al. 2022, entire; Troia et al. 2016, entire; Larsén and Rydin 2016, pp. 168-169). However, it has also been correlated with increased rates of extinction and lower rates of diversification in some plants (Mayrose et al. 2011, entire; Arrigo and Barker 2012, entire).

No life history study has directly assessed the longevity of mature thin-wall quillwort plants; however, some quillworts can survive for up to 8-10 years and *Isoetes lacustris* (a pond species) has been observed up to 50 years (Husák et al. 2000, p. 247; Čtvrtlíková et al. 2020, p. 9; Čtvrtlíková et al. 2023, pp. 8, 10). Similarly, the duration that free-floating quillwort spores remain viable (retain the capacity to germinate) is unknown but is expected to be considerable, as megaspores apparently containing intact and fluid contents have been observed on *Isoetes* herbarium specimens dating back to the late 19th century (Rhazi et al. 2004, p. 682; Gabriel y Galán and Prada 2010, p. 193). Quillwort spore viability has been correlated with ploidy level, age, and habitat factors (Kott and Britton 1982, pp. 1680-1682; Gabriel y Galán and Prada 2010, pp. 195-199).

2.5 Habitat

Thin-wall quillwort is a narrow endemic plant of blackwater river systems in the North Carolina Coastal Plain. In the southeastern United States and North Carolina in particular, “blackwater” describes low-gradient, highly tannic linear water bodies carrying very low suspended sediment (typically 20 mg/L or 20 ppm), but which contain large amounts of dissolved organic matter from wetland seepage (Benedetti et al. 2006, p. 264; Flotemersch 2023, p. 673; Flotemersch et al. 2024, p. 1; Weakley et al. 2025, pp. 86, 88). Within this system, which includes an array of seasonally flooded wetlands including riverine bottomland hardwood forests, the species occurs in areas that are periodically and regularly inundated or scoured by water flow (with flow velocities historically seldom exceeding 6 ft/s (1.8 m/s) even during storm events), where plants can remain submerged for extended periods of time following rain events (Simmon 1993, p. 7; Benedetti et al. 2006, p. 264; Schaner et al. 2021, p. 8). Brunton and Britton (1998, p. 268), reported that thin-wall quillwort is found in “permanent water courses under the deep shade of deciduous swamp forest. The plants grow in sandy alluvium [unconsolidated deposit left by flowing water], usually with little to no associated vascular vegetation in shallow water or (more commonly) on seasonally exposed streams or riverbanks.” Others have described its habitat as “hummocks within swamp,” “wet floodplain forest,” “*Taxodium* floodplain forest,” “blackwater swamp,” “edge of swamp stream channel,” “sluggish stream,” “steeply-sloping spring-flooded sand and silt of riverbank,” “mossy low shelf on shore of river and floodplain pool,” and “mossy, sandy point along riverbank” (Schaner et al. 2021, p. 8; SERNEC 2026, unpaginated). Good habitat condition for the species has been characterized as intact swamp forest having little (flow) gradient, gradual (not steep) stream banks, and little flood scouring (J. Bolin 2026, pers. comm.)

Most recently, during a 2021 status survey, thin-wall quillwort was found in habitats classified as Coastal Plain Small Stream Swamp, Blackwater Bottomland Hardwood Forest, Blackwater Levee/Bar Forest, and Sand and Mud Bar (Schafale 2024, pp. 666-668, 648-649, 628-630, and 676-677, respectively). Within these defined community types, plants have been observed in distinctive microhabitats in both mucky and sandy soil and in full sun as well as deep shade in 1) floodplain pool habitats, 2) riverbank habitats, 3) overflow inlets to swamp habitats, and 4) other riverine habitats (Schaner et al. 2021, p. 8).



Figure 2.5 Observed thin-wall quillwort habitat (clockwise from upper left): Coastal Plain Small Stream Swamp (Brunswick County, J. Schaner); Sand and Mud Bar habitat (Bladen/Sampson County, J. Schaner); Blackwater Bottomland Hardwood Forest habitat (Brunswick County, J. Schaner); and Other Riverine habitat, such as interwoven among tree roots or *Taxodium distichum* knees (B. Wichmann).

1. Floodplain Pool Habitat (Figure 2.5, upper left): For example, at Bolivia and Sandy Run Swamp, thin-wall quillwort inhabits small, seasonally inundated depression pools in the floodplain of small streams under a canopy of hydrophytic (“water loving”) hardwoods, with associated herbs varying depending on light levels and degree of disturbance (Schaner et al. 2021, pp. 8, 16, 18, 27, 30).
2. Riverbank Habitat (Figure 2.5, upper right): Thin-wall quillwort has been found on shady (closed canopy), seasonally exposed, very gently sloping riverbanks, as well as sunny and more densely vegetated sand and mud bars adjacent to swamps, along the Black River (Schaner et al. 2021, pp. 8, 9, 17, 29). It is not typically present on steep banks or recently developed point bar deposits, which tend to be unstable due to frequent alteration of the substrate (Schaner et al. 2021, pp. 8, 10, 13). Thin-wall quillwort similarly occurs on both the inside and outside of gentle river bends in areas where the sediment deposition is lighter, and the substrate is more stable. On the White Oak River, for example, thin-wall quillwort occurs on the riverbank as well as on low, seasonally exposed shelves in the river channel (Schaner et al. 2021, pp. 8, 18).
3. Swamp Overflow Inlet Habitat (Figure 2.5, lower right): On the Black River, thin-wall quillwort inhabits shallow overflow channels leading to submerged bottomland hardwood forest (aka backswamp) (Schaner et al. 2021, pp. 8, 17, 26). Within these overflow inlets, thin-wall quillwort occurs in bare sand under gently flowing water under a canopy of hydrophytic hardwood species, and herbs are sparse (Schaner et al. 2021, pp. 9, 17).
4. Other Riverine Habitat (Figure 2.5, lower left): Thin-wall quillwort is often associated with exposed roots and knees of bald cypress trees (*Taxodium distichum*) that may serve to anchor the plant during periods of river scour (Schaner et al. 2021, pp. 8, 17, 31).

2.6 Geology and Soils

Thin-wall quillwort inhabits both mucky and sandy alluvial soils overlaying assorted geology of the Coastal Plain (Brunton and Britton 1998, pp. 268, 275; LeBlond and Brunton 2010, p. 501; Brunton 2015, p. 94; Schaner et al. 2021, pp. 8, 9, 16-18; NRCS 2026a, unpaginated). Known subpopulations are most commonly documented on nonhydric fine sandy loams and hydric mucky loam that overlays various sedimentary rock formations: six occur on sand or loosely consolidated sand, silt, and clay; two on limestone bedrock; and one—the type-locality near Maysville, North Carolina—is associated with an uncommon matrix of sand and sedimentary (oyster) shellstone (Brunton and Britton 1998, p. 268; NRCS 2026a, unpaginated). Soils in blackwater river floodplain habitats tend to be older and shallower than soils along larger rivers due to reduced sediment loads, and because these watersheds are small, extremes in soil moisture are more likely (Hupp et al. 2000, entire; Burke et al. 2003, p. 999).

2.7 Climate

The climate of southeastern North Carolina where thin-wall quillwort populations occur is classified as humid subtropical by the Koppen Trewartha Climate Classification Model, indicating that a mean temperature at or above 10°C (50°F) is sustained for at least eight months of the year (Belda et al. 2014, entire). It is characterized by hot, humid summers and mild winters (Aucott 1996, p. E2). The mean total annual precipitation for Wilmington, North Carolina, from 2000 to 2026 is 59.10 inches (maximum 102.43 inches in 2018, minimum 33.37 inches in 2007) (NCEI 2026, unpaginated).

2.8 Needs of Thin-wall Quillwort

Using the best available information, we identified specific ecological needs (conditions necessary to complete each stage of its life cycle and reproductive phase (“generation”) for thin-wall quillwort at the individual, population, and species levels (Table 2.8; *see Figure 2.4. Generalized life cycle*). The resource needs described are considered essential to the survival and recruitment success of thin-wall quillwort, thus impacts to the condition of these resources are expected to impact the viability of the species. And, while herein described discretely, these needs typically operate concomitantly to support species’ physiological growth and its perpetuation to subsequent generations.

Table 2.8 Thin-wall quillwort resource needs by life stage (M/J/E/S) and reproductive phase (6n/n). M=mature sporophyte (adult), J=juvenile sporophyte (sporeling), E=embryo, S=spore (megaspore/microspore). H=habitat, A=resource acquisition, R=reproduction.

RESOURCE	Relevant Life Stage	Resource Function	Resource Need
HABITAT			
Sporophyte (6n)	MJ	A	Adequate sunlight to drive photosynthesis for metabolic processes
Sporophyte (6n) + Gametophyte (n)	MJES	HAR	Sufficient and consistent water/moisture at appropriate periods to prevent desiccation that leads to extirpation
	MJES	H	Substrate with structural integrity sufficient to facilitate anchoring of individual plants and nutrient uptake
	MJES	AR	Appropriate temperature for sporulation, plant germination, growth, and development
	MJES	HAR	Adequate water chemistry that promotes survival at each life stage

	MJES	HAR	Sufficient connectivity of physical habitats to support dispersal for successful reproduction and survival
DEMOGRAPHY			
Gametophyte (n)	S	R	Proximate dispersal of megaspores/microspores for successful reproduction and gene flow
Sporophyte (6n) + Gametophyte (n)	MJES	HR	Sufficient connectivity of physical habitats to prevent genetic isolation by natural or anthropogenic factors
	MJS	R	Adequate dispersal to support gene flow

2.8.1 Individual Needs

Water

While the degree and seasonality of their aquatic nature vary among species, water availability—inclusive of source, quantity, and timing—is fundamental to quillworts (Brunton et al. 2025, p. 3). The many stages of sexual reproduction, including the formation of micro- and megaspores, as well as dispersal, cross fertilization, and development, are themselves water-mediated, thus requiring ample moisture (Caplen and Werth 2000, p. 256; Oh et al. 2013, p. 18; Vander Stelt et al. 2017, p. 7; Azzella et al. 2024, p. 1421). Quillwort life stages are synchronous with seasonal water flux, which is generally exhibited as dry season dormancy (corm intact and negligible aboveground tissue) and wet season emergence (Leonard 1998, p. 2; Chappuis et al. 2016, pp. 374, 377; Santos et al. 2020, p. 10). In addition, both water flow and depth play an important role in viability and dispersal, with some researchers reporting that slower streamflow velocity (e.g., 0.01 to 0.04 m/s or 0.03 to 0.13 ft/s) and reduced depth is more tolerable for some endangered quillwort taxa (Wang et al. 2022, pp. 5, 7-8).

Sunlight

Similarly, sunlight is necessary for the growth and reproduction of thin-wall quillwort (Engelmann 1882, pp. 371-372; LaMotte 1937, p. 455; Chappuis et al. 2016, p. 380; Brunton et al. 2025, p. 3). Like many amphibious *Isoetes* that grow in both terrestrial and aquatic environments (e.g., *I. howellii*), thin-wall quillwort may utilize two differing albeit compatible photosynthetic pathways to adapt to changing environmental conditions, switching between crassulacean acid metabolism (CAM) when submerged (i.e., process involving CO₂ uptake at night, but restricted during the day) and C₃ metabolism in terrestrial settings (daytime CO₂ uptake) (Keeley 1998, p. 158; Yang and Liu 2015, p. 6; Koga et al. 2024, p. 593). While this stress adaptation response is widespread in *Isoetes*—wherein carbon accumulates as malic acid during the night and enters the Calvin cycle during the day to improve CO₂ utilization—the presence, strength, or nature of CAM in thin-wall quillwort has not yet been verified (Suissa and Green 2020, entire; Ciu et al. 2023, p. 7).

Isoetes as a whole are adapted to varying degrees of low and moderate light conditions, and some quantity of solar irradiation is therefore likely important to thin-wall quillwort, even when occurring in shaded habitats (Rorslett and Johansen 1995, p. 240; Rhazi et al. 2004, p. 682; Brizova 2011, p. 25; Čtvrtlíková et al. 2012, p. 142; Chappuis et al. 2015, entire). For semi-aquatic quillwort taxa that are ecologically similar to thin-wall quillwort (e.g., *I. louisianensis*), dappled/filtered light through a mostly closed canopy is optimal but brief periods of full sunlight are tolerable (McInnis 1991, p. 3; Leonard 1998, p. 7; Hoot et al. 2000, p. 9). Some other amphibious species (e.g., Georgia quillwort, *I. georgiana*)

appear to be limited less by vegetation cover than by the extent of sediment-free, water-scoured, emergent sand slopes existing under a particular deciduous swamp forest (Brunton and Britton 1996, p. 111).

The ability of quillwort spores to germinate in the dark has been investigated, yielding mixed results; compared to wholly aquatic species that show high percentage of spore germination and sporophyte emergence in the dark (albeit reduced relative to light conditions), amphibious species demonstrate no sign of sporulation in darkness (La Motte 1937, pp. 455-456).

Water and Soil Chemistry

The physicochemical environment (e.g., pH and mineral content) is a function of interacting processes in the water column, in soil, and at the soil-water interface, which are jointly facilitated by cyclical patterns of flooding (inundation) and dry-down (Wharton et al. 1982, p. 21). For the precipitation-dominated blackwater floodplains of southeastern North Carolina, water chemistry is characterized by relatively high concentrations of total organic solids (such as leaf litter) and reduced dissolved oxygen (DO), pH, nitrogen, phosphorus, and sediments, which fluctuate depending on numerous watershed factors (Wharton and Brinson 1978, p. 33; Wharton et al. 1982, p. 21; Hupp 2000, entire; Todd et al. 2009, entire). While no chemical parameters are described as standards for amphibious quillworts, particularly in blackwater systems, aquatic *Isoëtes* reportedly prefer generally circumneutral pH (6.5-7.5), low conductivity (average <100 $\mu\text{S}/\text{cm}$ or <64 ppm), low phosphorus (average 15.8 $\mu\text{g}/\text{L}$ or 0.0158 ppm), and low nitrogen (<0.5 mg/L or <0.5 ppm), supporting the proposition that quillworts tend to exist in oligotrophic environments (Azzella et al. 2024, entire).

Regarding sediment chemistry, the floodplain soil surfaces where quillworts become established may switch from aerobic to anaerobic conditions depending on the balance of atmospheric oxygen supply and demand by the sediments (when evapotranspiration becomes dominant) (Wharton and Brinson 1978, p. 34; Todd et al. 2010, entire). And while nutrients and organic matter content of sediments are generally not well documented for most quillwort species, aquatic taxa appear to track the same overall low-nutrient trends as for water chemistry (Azzella et al. 2024, entire, including SI_Substrate).

Temperature

For some quillwort taxa, temperature is considered the most important environmental factor, and some research suggests that a narrow temperature tolerance defines *Isoëtes* reproduction success (Čtvrtlíková et al. 2012, p. 151). While temperature and light both influence seasonal growth, temperature is regarded as far more important in controlling phenology and growth success, specifically spore germination and sporophyte emergence and development (Čtvrtlíková et al. 2023, p. 10; Abeli et al. 2026, entire).

While physiological thresholds of temperature tolerance are likely species and habitat-specific, *Isoëtes* spore production has been generally observed to occur above a minimum temperature of approximately 10° C (50° F), and germination increases with rising temperature until reaching a maximum high temperature (Vöge 2006, pp. 228, 230; Chappuis et al. 2016, p. 382; Čtvrtlíková et al. 2012, entire; Abeli et al. 2026, entire). Water temperature may regulate spore germination (Azzella et al. 2024, p. 1422). No information pertaining to the specific influence of temperature on amphibious species phenology has been found. Temperature greatly albeit inconsistently influences vegetative (leaf, corm) growth of individual semi-aquatic plants, with damage or mortality reported when subjected to either markedly high or low temperatures (i.e., *I. butleri*, *I. melanospora*) (Baskin and Baskin 1979, entire; Service 2019, p. 19).

Substrate

Although *Isoëtes* taxa are adapted to different edaphic (i.e., soil) conditions and are documented in many different soil types, North American quillwort species commonly occur in acidic to circumneutral substrates that are, at minimum, seasonally waterlogged (Azzella et al. 2024, p. 1421; Brunton et al. 2025, p. 4). For example, amphibious species, like *I. butleri* and *I. louisianensis*, are present in shallow, wet substrate throughout winter and early spring, but also survive (albeit underground) in extremely dry soils throughout the fall/summer growing season and remain capable of completing their life cycle (Baskin and Baskin 1978, p. 7; Taylor and Schwegman 1992, p. 83; Service 1996, p. 7). Robust clumps of thin-wall quillwort have been observed in a dry roadside depression/ephemeral pool in full sun, suggesting that thin-wall quillwort may tolerate periods of soil drying; however, the precise period of saturation needed for optimal species growth is not known (Schaner et al. 2021, pp. 18, 30).

As quillworts are slow-growing plants, early sporeling establishment is necessary for successful development and dormancy (Caldeira et al. 2021, entire). Sporelings must develop a stable root system (which includes the underground corm structure, see *Figure 2.3.1. General plant morphology*) before exhausting nascent food supplies (quillwort megaspores contain sufficient food reserves to support developing life stages for several months) (Campbell 1891, pp. 242, 251; Taylor and Luebke 1986, pp. 23-24). An extensive root system not only provides nutrients and carbon dioxide from the sediment, but also mechanically anchors the otherwise buoyant plant to its environs, which is of particular importance in habitats frequently disturbed by changing water levels (Richardson et al. 1984, p. 119; Boston et al. 1987, entire; Sandquist and Keeley 1990, entire; Sorrell 2004, p. 1109; Chappuis et al. 2016, p. 380). Soil stability (“firmness”), which is linked to soil texture, has been observed to influence quillwort plant retention and is frequently cited as a factor supporting thin-wall quillwort presence along water courses and floodplains (Sand-Jensen and Møller 2014, p. 435; Schaner et al. 2021, pp. 8, 31).

2.8.2 Population and Species Needs

At the population level, thin-wall quillwort requires the same key habitat-based resources as individuals, although at a scale sufficient to support population health and reproduction, as well as a sustainable population size and connectivity between populations. Thin-wall quillwort populations need to reproduce (asexually via tillers or sexually via gametophytes) and survive to replace senescing individuals, thereby maintaining a sustainable population size (Caldeira et al. 2021, entire). Like other *Isoëtes* taxa, its reproductive success largely depends on the successful germination of spores (Čtvrtlíková et al. 2014, p. 279). While sporophyte regeneration via sexual reproduction has been more frequently observed, asexual propagation is also a viable means of reproduction (Caldeira et al. 2021, p. 2). In general, larger and more closely connected populations are expected to have increased opportunities for sexual reproduction to maintain genetic diversity. However, as unique heterosporous entities, *Isoëtes* are readily able to propagate by both cross- and self-fertilization (analogous to self-pollination in seed plants) (Santos et al. 2020, p. 10). However, such intergametophytic selfing requires physical adjacency of micro- and megaspores, as well as seasonal synchrony (i.e., concurrent germination), to successfully complete sexual reproduction (Larsén and Rydin 2016, p. 169; Troia 2016, p. 280; Link-Pérez and Laffan 2018, p. 516; Freund 2022, p. 21).

Population size represents the minimum estimated total number of plants within a population based on the most recently available data (visual counts/estimates, qualitative assessments) and is an important indicator of resiliency. While estimating population size is challenging due to influences of location (spatial distribution), hydrology, and seasonality, documented local population sizes of amphibious quillworts may vary from isolated clusters of fewer than 10 plants to larger swaths of approximately 2,500, with 919 to 2,195 individuals reported as the minimum viable population range for one aquatic quillwort (no reports found for semi-aquatics) (Kang et al. 2005, p. 1268; Wang et al. 2005, pp. 4072-

4073, 4075; Gentili et al. 2010, pp. 149-151; Heydon 2011, p. 10; Leonard 2011, pp. 16-18, 23; Ma et al. 2019, p. 340; Dalapicola et al. 2021, pp. 10126-10127).

Like all quillworts, populations of thin-wall quillwort can vary substantially in abundance from year to year as well as within a given year, depending on the hydrologic condition in previous years or seasons (Larke 1997, p. 3; D. Brunton 2021, pers. comm.). Additionally, population estimates have commonly been approximated in areas where colonies of quillwort occurred in dense patches, as well as when observed from a single location (i.e., bridges) without survey of the stream itself (Schaner et al. 2021, pp. 14-18).

While it is recognized that connectivity of populations is directly related to gene flow and genetic structure, the true extent of gene migration among most quillwort populations within a given watershed remains unknown but has been assumed to be low and may decrease with distance (Hoot et al. 2000, p. 11; Chen et al. 2005, p. 96; Chen et al. 2012, p. 55; Wickell et al. 2023, p. 262). Genetic relationships among these organisms are complex due to their frequent hybridization and polyploidy, which is described by some as both a novel method of diversification as well as an evolutionary dead end (Suissa et al. 2022, entire; Wickell et al. 2023, pp. 261-262, 271). However, it is generally accepted that gene flow among diploid (progenitor) quillwort taxa is higher than for descendant polyploids (such as thin-wall quillwort), with population structure being inversely related to gene flow (Chen et al. 2010, p. 136; Wickell et al. 2023, p. 269).

No targeted phylogenetics have been performed for thin-wall quillwort, although some work has demonstrated that it is a descendent of multiple lineages, most likely including blackfoot quillwort (*I. melanopoda*) and Eastern blackfoot quillwort (*I. melanopoda* ssp. *silvatica*) (Brunton and Britton 2006, entire; Brunton and Troia 2018, p. 314; Schafran 2019, p. 193). Nonetheless, model taxa can be used to elucidate possible demographic patterns. For instance, polyploid species of the Appalachian quillwort (*I. appalachiana*) complex occurring in riverine swamp habitats along the eastern United States seaboard demonstrate distinct clustering by locality with little evidence of genetic mixing, although their diploid ancestors show some evidence of long-distance dispersal (within and between watersheds) (Brunton and Britton 1997, entire; Hoot et al. 2004, p. 902; Wickell et al. 2023, entire). Abundant research indicates that polyploid species have reduced genetic diversity relative to diploid relations (Kim et al. 2009, pp. 496, 499-503; Suissa et al. 2022, p. 11). By these examples, we interpret that polyploid quillworts are fundamentally genetically restricted, and it is chiefly through supplementary measures (e.g., artificial in-stream barriers) that dispersal and other demographic variables are disrupted and thereby enhance population isolation (Jones et al. 2020, entire).

For the species' viability as a whole, there must be adequate redundancy (suitable number of populations, distribution of populations, and connectivity between populations to allow the species to withstand catastrophic events, such as hurricanes and severe, prolonged drought) and representation (suitable genetic and environmental diversity to allow the species to adapt to changing environmental conditions). Redundancy improves with increasing numbers of populations at moderate and high resiliency conditions and connectivity (either natural or anthropogenically influenced) among those populations to allow populations to "rescue" each other after catastrophic events. Representation improves with increased genetic diversity and environmental conditions within and among populations. While we do not have specific genetic, morphological, or ecological niche information to inform where there may be differences in the species within its known range, we know enough about quillwort taxa as a whole to describe representation using known traits that support adaptive capacity. Representation is inherently low for narrow endemic taxa.

CHAPTER 3: FACTORS INFLUENCING SPECIES VIABILITY

3.1 Background

The following discussion provides a summary of plausible past, current, and future factors (both adverse and beneficial) that may affect the current and future condition of thin-wall quillwort throughout some or all its range. These factors are summarized in a conceptualized model (Figure 3.1) and are discussed in more detail below, with exploratory tables included in Appendix D. While these factors are examined discretely, it is important to recognize that they typically work cooperatively, whether paired, overlapping, or as a gross composite, to produce ultimate effects.

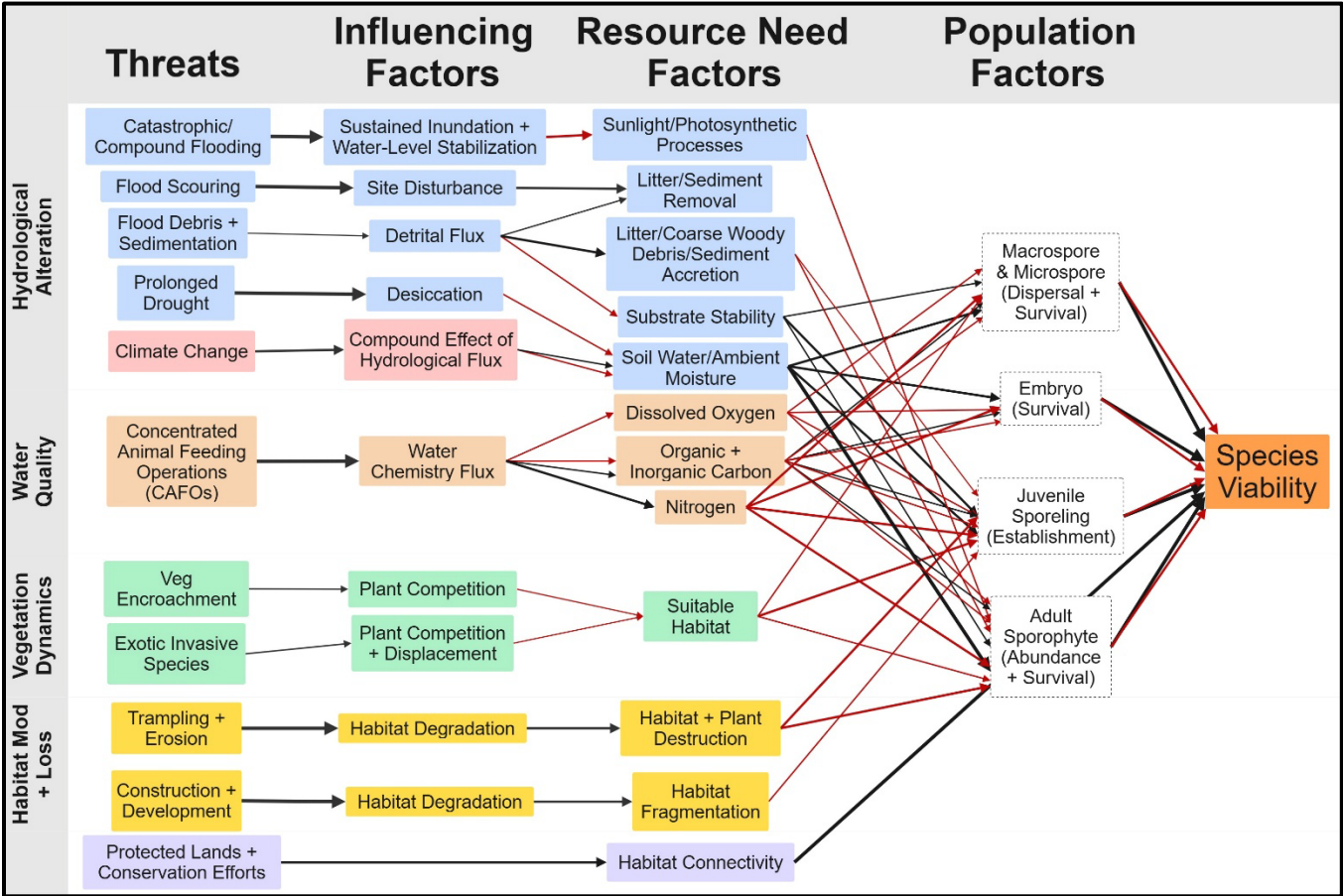


Figure 3.1 Conceptualized influence diagram for thin-wall quillwort displaying plausible relationships between factors and species’ viability. Black arrows indicate an influence that has a heightening effect. Red arrows indicate an influence having a diminishing effect.

3.2 Hydrological Alteration

Thin-wall quillwort has evolved in a dynamic “flood-pulse” habitat; however, it is not known how well it responds to more extreme changes in hydrology, such as (storm-induced) flooding (and submersion), flood scour and erosion, and desiccation and drought (Junk et al. 1989, entire; Brinson 1993, entire; Benke et al. 2000, entire). The structure of these dynamic blackwater environments is largely regulated by precipitation, with flood events being an important factor influencing the spatial and temporal heterogeneity of floodplain areas and their plant communities (Wharton et al. 1982, p. 21; Burandt et al. 2023, p. 2; Bren et al. 2025, pp. 128, 130). These systems become increasingly tidally influenced as rivers near their coastal outlet, enhancing their dynamic nature via storm surge during storm events (Gori et al. 2020, entire; Phillips 2022b, entire).

3.2.1 Flooding and Inundation

Due to the prevailing ambiguity in common parlance of the terms flooding and inundation, we herein define these terms following Majszak et al. (2026, entire) and Flick et al. (2012, entire). A flood is thus defined as a transient physical phenomenon in which pools and fluxes of water move across the landscape and are temporarily elevated beyond a specified threshold, but which return to some pre-exceedance baseline. Conversely, inundation is denoted as a prolonged or permanent state of submersion, often with a high increase in water turbidity (Bornette and Puijalon 2011, p. 6).

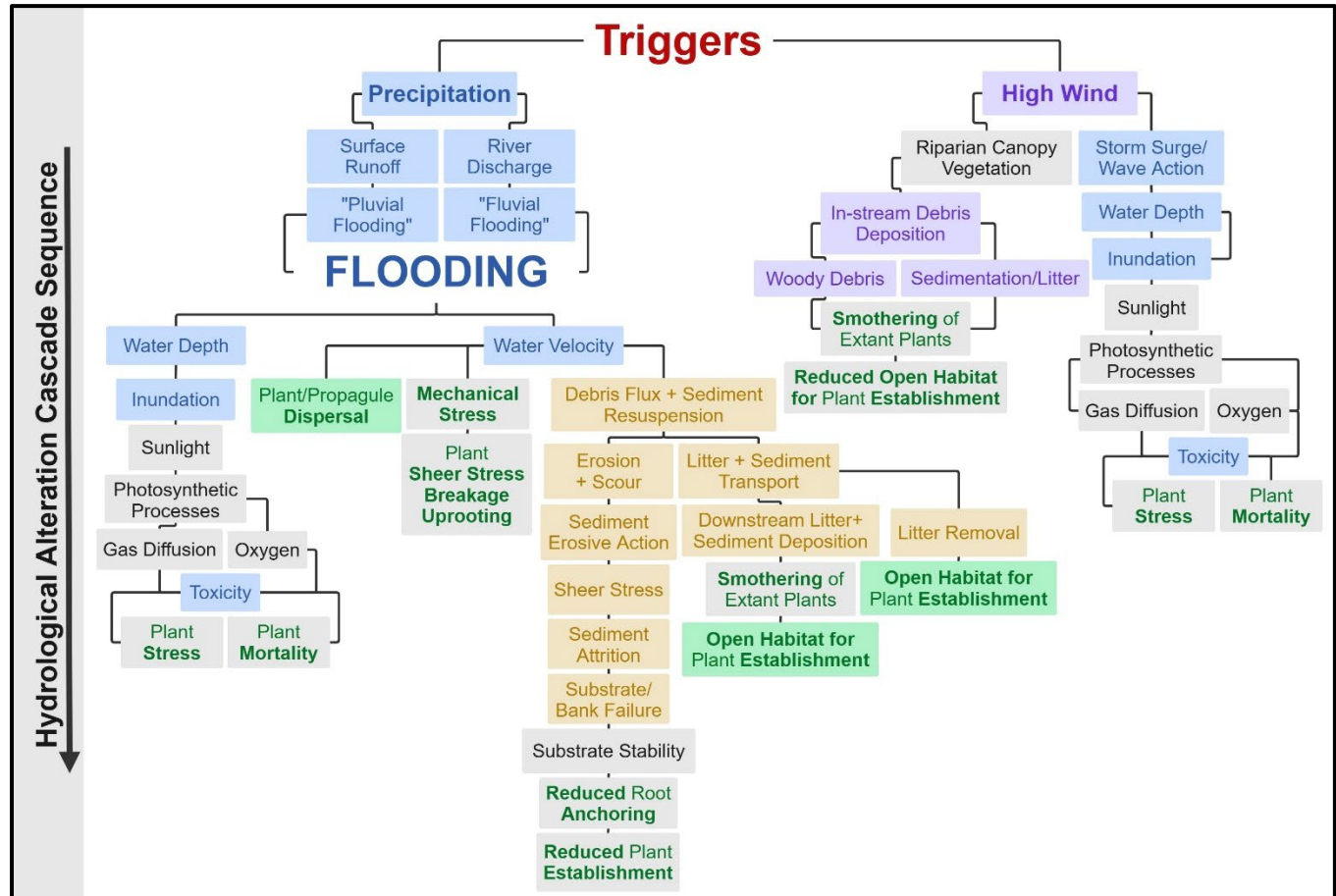


Figure 3.2.1 Generalized hydrological alteration cascade sequence triggered by storm events. Colored boxes (blue, purple, tan) are those influences that are amplified by preceding factor(s) in the cascade sequence. Gray boxes with black font indicate an influence diminished by the antecedent factor(s). Green and gray boxes with bold green font represent the end result for plants under these conditions.

A. Flooding

A variety of atmospheric and meteorological mechanisms are responsible for producing riverine flood events by transporting precipitation and overland surface water in excess of that which a watershed can absorb or store (Lecce 2000, p. 168). During a flood event, high water velocities cause physical and mechanical stress on macrophyte tissues due to increased drag forces (hydrodynamic resistance force opposing an object's motion through water), which promote shear stress, breakage, and uprooting (Bornette and Puijalon 2011, pp. 5-7). Furthermore, rapid water flow mobilizes in-stream sediments, with resulting erosion causing root detachment and thus plants get washed downstream (Bornette and Puijalon 2011, p. 6). When floods occur frequently, decreased riparian macrophyte presence has been documented (Riis and Biggs 2003, p. 19; Bornette and Puijalon 2011, p. 6).

A-1. Compound Flooding

While floods themselves are characteristic of blackwater systems of North Carolina, they are generally periodic and of short duration, with deep and widespread floodwaters, and followed by extensive periods of low discharge (“flood-pulse”) (Hupp 2000, p. 12; Burke et al. 2003, pp. 989, 991; Schafale 2024, pp. 628, 648, 666, 676). However, compound flood events, resulting from multiple co-occurring flood drivers (e.g., extreme precipitation, increased river discharge (“fluvial flooding”), and direct runoff (“pluvial flooding”)) during hurricanes and tropical storm systems (“storms”), are becoming more common, with interactions among these variables leading to impacts that are often of longer duration and/or greater spatial extent than events induced by single drivers (Wahl et al. 2015, entire; Deb et al. 2023, entire; Ali et al. 2025, p. 1). In addition, much research indicates that coastal North Carolina, including Brunswick, Onslow, and Pender counties, sustain the 5th highest percentage (83.7%) of compound flood events along the East Coast, resulting from significant contributions of precipitation, riverine discharge, and soil moisture during storms spanning 1980 to 2018 (Gori et al. 2020, entire; Yin et al. 2021, entire; Ali et al. 2025, pp. 2-5; Grimley et al. 2025, entire).

Ultimately, quillwort survival to flooding depends on the timing, frequency, depth, and duration of flood events (Suir and Berkowitz 2021, entire; Burandt et al. 2023, p. 2). Improved spore and sporeling development are improved in intermittent streams due to episodic flow dynamics, which allow them to become established; with strong, continuous flow, young plants may be dislodged and swept downriver (Hoot et al. 2000, p. 9). Further, as quillwort spores are buoyant and incredibly small, and not likely to remain stationary in flowing water, high-velocity flow associated with compound storm events may reduce the probability of a species’ necessary megaspore-microspore contact for fertilization (Wang et al. 2022, p. 8). In this manner, it is conceivable that such altered flood patterns, which are forecasted to increase in intensity in the future, may have a significant deleterious effect on plants and populations, particularly if they are already reduced in fitness and vigor from previous events (Bornette and Puijalon 2011, p. 6; de Ruiter et al. 2020, entire; Grimley et al. 2026, entire).

B. Inundation/Submergence

Inundation dynamics are defined as the spatial and temporal pattern of floodplain submersion that occur regularly within a year, or over many years, which are dependent on various flow processes and the extent of adjacent floodplain (Benke et al. 2000, p. 2730). Variable inundation of ecotonal blackwater floodplain systems in the Southeast has been shown to be a common event of high frequency that greatly expands the effective river width nearly every year, with depths varying from zero to greater than 2 m during a flood cycle (Cuffney 1988, pp. 306, 316; Benke et al. 2000, p. 2737; Stegen et al. 2025, entire). Naturally, therefore, the timing, frequency, and duration of floodwater movement and inundation is a principal component driving resident plant community composition and distribution (Schneider and Sharitz 1989, entire; Burke et al. 2003, entire).

B-1. Seasonal and Variable Inundation (Floods)

Isoetes taxa span the ecological gamut from fully submerged (“aquatic”) to temporarily emerged (“terrestrial”), with many straddling the intermediate realm of semi-aquatic (“amphibious”) (Azzella et al. 2024, p. 1421). These amphibious quillwort forms tend to be highly resilient to fluctuating water levels and possess specialized adaptations that allow them to thrive under both conditions (Casanova and Brock 2000, entire; Chappuis et al. 2016, pp. 374, 380; Koga et al. 2024, entire). For example, individual plants of federally endangered Louisiana quillwort (*I. louisianensis*), a species of similar riparian habitats, are regularly inundated following rains and may remain submerged for extended periods during flooding (McInnis 1991, p. 3; Service 1996, p. 7; Larke 1997, p. 2). Similarly, the in-stream swamp habitat within which thin-wall quillwort is known to occur may sustain prolonged floodplain inundation from winter to early spring, followed by extensive drying (sometimes completely) during the summer and autumn months (Burke et al. 2003, pp. 988-989; Todd et al. 2009, pp. 1493;

Flotemersch 2023, p. 672). In many cases, such seasonal inundation creates opportunities for dispersal, sustained germination, and increased abundance (de Oliveira et al. 2015, entire). It is likely that thin-wall quillwort can remain fully submerged through its life cycle (including sporulation) as long as water is shallow enough for plants to receive adequate sunlight through the tannin-rich medium (J. Bolin 2026, pers. comm.).

B-2. Prolonged Inundation (Impoundments)

Being regularly subject to stress from inundation and hydraulic disturbance, riparian vegetation has evolved a multiplicity of traits to facilitate existing in both terrestrial and aquatic conditions (Bornette and Puijalon 2011, pp. 6-7; Catford and Jansson 2014, entire). Nonetheless, prolonged and complete submergence will eventually greatly slow the rate of oxygen diffusion to root tissues, prevent the aerial influx of CO₂ and light, thereby largely decreasing photosynthesis (Striker 2012, entire; Catford and Jansson 2014, p. 19). Eventually, the environment becomes hypoxic (insufficient oxygen), then anoxic (no oxygen), with the accumulation of toxic compounds in the root environment constraining plant survival (Striker 2012, pp. 1, 5, 16; Tamang and Fukao 2015, pp. 30164-30165).

While most quillwort taxa are capable of some degree of underwater metabolism using CAM (see Section 2.8.1), and there are clearly wholly aquatic species that persist completely submerged for their lifespan (e.g., *I. lacustris*, *I. sabatina*, *I. malinverniana*), it remains unclear how amphibious or “flood-pulse”-adapted quillworts, such as *I. microvela*, may respond to long-term submersion (Bolpagni et al. 2021, entire; Azella et al. 2024, entire). What information does exist suggests a species-specific response; for example, Mueller’s quillwort (*I. muelleri*), which spends most of the year in water depths of 0.3-0.5 m, but in summer emerges above the water surface, is documented to favor constructed impoundments (Garrett and Kantvilas 1992, pp. 121-122). Similarly, Orcutt’s quillwort (*I. orcuttii*) demonstrates increased frequency of occurrence under prolonged submergence (Bauder 2000, pp. 47, 49, 51-52). However, some amphibious species exhibit limited growth and survival, possibly due to lack of functional stomata in submerged conditions (e.g., *I. echinospora*) (Yang and Liu 2015, p. 9; Mørk et al. 2018, entire). These studies provide no clear indication as to how thin-wall quillwort may respond under complete submergence.

The 2010 CBD petition listed anthropogenically derived dams and diversions as being harmful to thin-wall quillwort, likely due to their stabilization of water levels (CBD 2010, p. 628). While none of the water bodies upon which the taxon occurs are known to be dammed by a hydroelectric and/or flood control dam, a sizeable beaver impoundment was documented at one site in 2021 and is surmised to have flooded the resident population (Schaner et al. 2021, p. 14). While beaver dams and ponds are common in North Carolina Coastal Plain river systems and are often described as providing useful services (e.g., sediment trapping, organic matter and trace element sequestration), such activity has possibly led to thin-wall quillwort’s decline or disappearance at one locale (Kroes and Bason 2015, entire; Bason et al. 2017, p. 595; Fletcher et al. 2019, pp. 8-9).

3.2.2 Sediment and Debris Flux

A. Erosion and Scour (Degradation)

Rivers and streams are dynamic in terms of both water flow and sediment transport. In these systems, initiation of localized sediment movement and subsequent transport may occur gradually along a meandering riverbed or around obstacles such as coarse woody debris or man-made hydraulic structures (Inamdeen et al. 2021, entire). However, hydrological alteration associated with catastrophic storm events are capable of elevating these transport dynamics by several orders of magnitude in these small basins depending on the storm intensity, rate of storm decay, land cover type, and associated factors

(Lecce et al. 2005, p. 1; Bell and Lackmann 2025, entire; Sheil 2026, entire). Such swings have the potential to flush out sediments, thereby inducing much erosion and scour (Phillips 2022a, p. 8).

Erosion refers to a multiplicity of processes by which particles are removed via wind, water, or wave action (May et al. 2002, p. 14). Scour, then, is an erosive process resulting from shear forces associated with flowing water and wave action, which alters the vertical position of a stream/riverbed and/or floodplain as material is eroded and degrades (May et al. 2002, p. 17). Both suggest removal of soil and plants and although blackwater systems of the North Carolina Coastal Plain typically carry low suspended sediment loads (relative to other riverine systems), during seasonal floods any scour material tends to be deposited locally (Lecce et al. 2004, p. 341). The result is a transient form of localized scour in which removed sediments are not necessarily lost permanently but often transported downstream and deposited in lower-energy areas, resulting in a constant, natural cycle of scour and deposition (Colby 1964, entire; Culbertson et al. 1967, p. 9; Lecce et al. 2004, entire; Schafale 2024, pp. 648, 666, 676).

Riparian plants cope with the associated physical disturbance using resistance strategies like stem, leaf, and root flexibility, increased density, and/or brittle tissues that enable self-thinning (Hupp 2000, p. 10; Bornette and Puijalon 2011, pp. 6-7; Janauer et al. 2013, pp. 247-249; Catford and Jansson 2014, pp. 21-23). While some biomass may be lost, these adaptations help individuals to survive disturbance events and may even facilitate dispersal (Catford and Jansson 2014, pp. 23-24, 26). However, flood scouring during more severe events can resuspend underlying sediment, thereby reducing plant anchoring ability while augmenting shear stress and generating a cascade of effects (sediment attrition and bank failure), which greatly inhibits plant establishment (Riggsbee et al. 2008, entire). Such erosion trends have been documented following several catastrophic storms in coastal North Carolina and are suggested to result from increased impervious surfaces at roadsides, bridges, boat ramps, and other human access points (Phillips 2022a, entire; J. Bolin 2026, pers. comm.).

While Barni et al. (2013, p. 12) noted that submerged *Isoëtes* are sensitive to fast water flow and scour action, with the sudden inflow of turbid water heavily affecting important phases in species demography—i.e., germination, sporeling establishment and initial growth (Szmeja 1994, entire), amphibious *Isoëtes* have been regularly observed in locales with significant seasonal flood scouring and erosion (e.g., Louisiana quillwort (*I. louisianensis*); Georgia quillwort (*I. georgiana*) and its hybrids), suggesting that some semi-aquatic species are well-suited to these processes (Brunton and Britton 1996, p. 110-111; Larke 1997, p. 2; Madsen et al. 2001, entire; Zhu et al. 2012, p. 2; Brunton and Sokoloff 2020, p. 40). In some blackwater habitats, subsurface networks of tree rootlets and surficial moss patches appear to assist in anchoring plants during intense scouring events (Brunton and Britton 1996, p. 110). Wintergreen quillwort (*I. hyemalis*), another North Carolina blackwater stream taxon, persisted despite severe windstorm damage in 1992 that eliminated most forest cover and increased siltation (Brunton et al. 1994, p. 19; Brunton et al. 2025, p. 40).

In fact, evidence suggests that the formation of new *I. louisianensis* colonies (“populations”) is enhanced by periodic flooding that scours out new depressions along blackwater stream channels and floodplains (Hoot et al. 2000, p. 9). Because organic matter and sediment accumulation can have a smothering effect on existing plants, flood scouring has the potential to maintain suitable habitat by flushing vegetative material and fine sediments downstream to expose bare soil where quillwort sporelings can root and establish (Leonard 1998, p. 8; Leonard 2011, pp. 3, 8-9). Like thin-wall quillwort, Louisiana quillwort grows both in primary scour channel(s) and in floodplain areas that sustain some amount of soil moisture (Sorrie and Leonard 1999, p. 896; Lyman 2017, pp. 3-4).

While we do not have data to demonstrate impacts of high water and flood scour events on population viability, recent hurricanes may have affected populations of thin-wall quillwort through river scour. Data from 2021 surveys suggest that at least one population of thin-wall quillwort has been impacted by

flood scouring (Schaner et al. 2021, p. 12). The type-locality at the White Oak River Campground (Maysville site) had 1,000+ plants when it was first discovered in 1995, yet a 1998 survey demonstrated that the site had been heavily scoured by storm waters and waterborne debris after hurricanes in 1996 and 1997 (Brunton and Britton 1998, p. 268). The destruction of many thin-wall quillwort plants was noted, perhaps due to the resulting erosion (washing away) of the relatively thin soil layer overlying shellstone bedrock (Brunton and Britton 1998, p. 268). In 2015, surveyors found only a few feeble clumps of thin-wall quillwort clinging to moss and roots on a nearby vertical cut-bank, and a return trip in 2021 yielded no plants (Schaner et al. 2021, p. 13).

B. Flood Debris and Sedimentation (Aggradation)

Hurricanes and similar severe storm events have the potential to act as major disturbance agents to stream ecosystems in blackwater systems (Mallin and Corbett 2006, entire). Accompanying high winds denude riparian vegetation, depositing material into stream channels, and intense rainfall increases runoff to local waterways and flushes litter from dry surfaces, resulting in elevated surface water and significant loading of organic matter and sediment to nearby streams and rivers (Schafer et al. 2020, p. 1; Strickland et al. 2024, p. 824). Such debris flows, whether inputs are of an organic or inorganic source, invariably impact receiving water bodies and linked organismal dynamics (Meyer et al. 1997, entire; Mallin and Corbett 2006, entire).

While quillworts seem to be able to withstand periodic flooding with its attendant debris and siltation, they are rarely found in waterways where such conditions are persistent (Hoot et al. 2000, p. 9). And, although in-stream snags and other woody material form debris dams that affect detrital dynamics and provide important habitat for benthic macroinvertebrates and fish, high water level and rapid currents increase sediment resuspension, which may affect plant survival via abrasion (mechanical injury) and decreased light (Madsen et al. 2001, entire; Nilsson and Svedmark 2002, p. 471; Paller et al. 2010, p. 2; Barni et al. 2013, p. 12). Under these conditions, excessive litter and sediment accumulation may have a marked effect on *Isoetes* populations by negatively influencing regeneration dynamics (Leonard 1998, pp. 7-8; Hoot et al. 2000, pp. 9-10; Rhazi et al. 2004, pp. 681-682; Chappuis et al. 2015, entire; Vander Stelt et al. 2017, pp. 4-5). This may be especially true for taxa that occur in blackwater systems whose rivers, although rich in organic material, typically carry little sediment (Meyer 1990, p. 643; Rapp et al. 2001, p. 247; Ogg et al. 2013, p. 394).

Although we do not have data demonstrating effects of storm-associated litter and sedimentation on thin-wall quillwort populations, authors have noted their local influence following 1996 and 1997 hurricanes (Brunton and Britton 1998, p. 268; Schaner et al. 2021, pp. 9, 13). In addition, other amphibious species, such as limestone quillwort (*I. butleri*), reportedly respond poorly to significant litter accretion, with spore burial under litter believed to reduced sporeling survival and dampening temperature and moisture fluctuations at the soil surface interfering with signals needed to trigger root and leaf growth (Karrfalt 1977, p. 2; Vander Stelt et al. 2017, entire). Similarly, substantial silt and leaf pack sedimentation has been noted to have grave effects on Louisiana quillwort (Leonard 2011, pp. 3, 8, 28-29).

3.2.3 Drying/Desiccation

A. Prolonged Drought

Plants commonly found in blackwater swamp habitats are largely adapted to alternating wet and dry phases and to large interannual fluxes in hydroperiod (seasonal timing, duration, and frequency of flooding) (Wharton and Brinson 1978, pp. 36-37; Thorp et al. 1985, p. 56; Burke et al. 2003, p. 1000; Todd et al. 2009, p. 1494). The seasonal change in blackwater discharge and extent is highly precipitation-dependent, with floodplain inundation typically lasting from winter to early spring (in

North Carolina, February to early March), followed by long dry periods during summer when the stream channel often completely dries (Wharton et al. 1982, p. 21; Burke et al. 2003, p. 989; Todd et al. 2010, p. 2; Flotemersch 2023, p. 672; Schafale 2024, p. 676).

Isoëtes as a whole are consistently deemed desiccation-tolerant and some quillwort taxa—including federally endangered Louisiana quillwort and *I. taiwanensis*—have demonstrated the ability to survive extended wet-dry cycles through the shedding of aboveground structures (sporophylls) while underground structures (corm) remain extant but dormant, and others have remained wholly viable following severe drought stress (Rhazi et al. 2009, p. 123; Huang et al. 2018, entire; Alejo-Jacuinde and Herrera-Estrella 2022, pp. 3-4; Gao et al. 2023, p. 2; Montiel-Molina et al. 2023, p. 2). Under such conditions, spore maturation is likely accelerated (J. Bolin 2026, pers. comm.). Furthermore, some research indicates that spores of some quillworts (i.e., *I. cangae*, a wholly aquatic species) can survive for 1 year of desiccation at room temperature (~25° C), despite suffering a decrease in spore viability (Cavalheiro-Filho et al. 2023, p. 2; Lopes et al. 2023, p. 13).

Mature *Isoëtes* spores are reported to develop and disperse before pronounced reductions in soil humidity (Taylor and Hickey 1992, p. 621). For these amphibious taxa, spore dispersion is more complicated than for aquatics, being limited by the rainy season and other factors, such as wind, thereby affecting the genetic structure of populations and direction of gene flow (Troia 2016, entire; Ma et al. 2019, entire). If a resident stream dries out for too long, desiccation may occur before the young are safely established (Hoot et al. 2000, p. 9).

Like other *Isoëtes*, thin-wall quillwort is alleged to tolerate short periods of drought, but its precise tolerance and mechanisms thereof remain unknown (Pfeiffer 1922, p. 91). In cases of sufficient soil moisture, it remains capable of completing sporulation, but in dry(er) conditions, plants will go dormant (and not sporulate) (J. Bolin 2026, pers. comm.). As such, it has been speculated that prolonged drought, especially when associated with climate change, may directly harm semi-aquatic quillwort species, which rely heavily on water for each phase of their life cycle and spore dispersal and must therefore be prepared to survive periods when standing water and soil saturation is unavailable (Freund 2022, p. 21). For example, when exposed to increasing temperatures, quillwort microspores typically germinate earlier and more successfully than megaspores (Čtvrtlíková 2012, p. 141). In this manner, high temperatures associated with drought conditions have the potential to yield a marked desynchronization of mega- and microspore germination, making them unavailable for concurrent fertilization (Čtvrtlíková 2012, p. 150).

Figure 3.2.3.A Drought trends recorded across the range of thin-wall quillwort, 2000-2026 (USDM 2026a, unpaginated) (*color legend in Table 3.2.3.A below*).

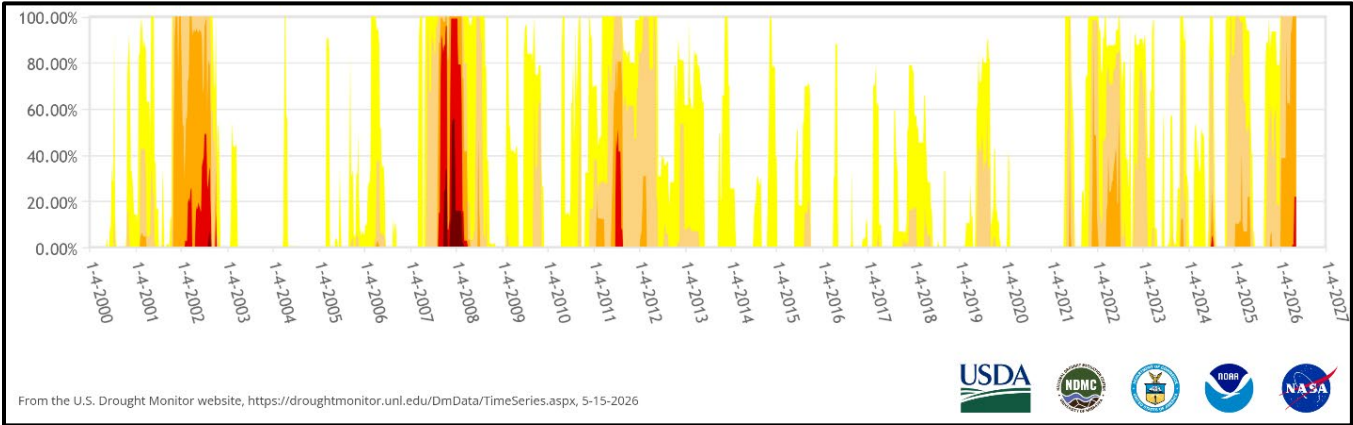


Table 3.2.3.A Drought severity classification (USDM 2026a, unpaginated).

Category	Description	Possible Impacts
D0	Abnormally Dry	Going into drought: • Short-term dryness slowing planting, growth of crops or pastures Coming out of drought: • Some lingering water deficits • Pastures or crops not fully recovered
D1	Moderate Drought	• Some damage to crops, pastures • Streams, reservoirs, or wells low, some water shortages developing or imminent • Voluntary water-use restrictions requested
D2	Severe Drought	• Crop or pasture losses likely • Water shortages common • Water restrictions imposed
D3	Extreme Drought	• Major crop/pasture losses • Widespread water shortages or restrictions
D4	Exceptional Drought	• Exceptional and widespread crop/pasture losses • Shortages of water in reservoirs, streams, and wells creating water emergencies

3.2.4 Climate Change

Climate change is increasing the frequency of extreme storms and persistent droughts, which influences water resource availability, hydrological connectivity, and flood-associated disturbances (Gu et al. 2023, entire; Uehling and Schreck III 2024, entire; Bernal et al. 2025, p. 11). Progressively, these large floods occur over a short time period, with increasing solutes and sediments pulsed into recipient water bodies, and increased water velocity and turbidity potentially hampering stream metabolic activity (Bernal et al. 2025, entire).

Although climate impacts on ecosystems are acknowledged, uncertainty remains about future conditions in the Southeast, particularly North Carolina, and how species will respond. In the 2020 North Carolina Climate Science Report, Kunkel et al. (2020, pp. 5-8, 18-22) indicated the likelihood of increasing trends in temperature, heavy storm event frequency, and drought severity across the state, which are projected to increase into the future. Climate trends of the Coastal Plain, specifically, where thin-wall quillwort occurs, are expected to mirror those for the entire state (Kunkel et al. 2020, pp. 96-118). Furthermore, future tropical events along the North Carolina coast are predicted to generate higher rainfall rates, driving substantial increases in flood extent and depth (compound flooding) (Grimley et al. 2026, entire).

The greatest threat from climate change may arise from synergistic effects or compound events (Kunkel et al. 2020, pp. 187-198). These are observed and/or forecasted changes linking multiple climate factors, which act as “threat multipliers” that increase the risk and severity of more localized, imminent stressors, such as land use (Mulholland et al. 1997, entire; Kunkel et al. 2020, p. 9; Goodman and Baudau 2023, entire). For example, in coastal North Carolina the concentrated impacts of recent tropical storms and associated floods are exacerbated by climate change, as well as by urban and agricultural land use (Paerl et al. 2019, entire). There are reports positing specific *Isoetes* species will be impacted by climate change via increased year-round temperatures and reduced stream flow (Maslovat et al. 2021, p. 302). Others suggest that geographically-restricted species, such as thin-wall quillwort, are substantially more vulnerable to rapid environmental change, like climate change, than more broadly distributed amphibious species (D. Wickell 2026, pers. comm.). Consequently, despite geological persistence following major mass extinctions associated with prehistoric climate change(s), which indicate that

quillworts may persist through further environmental change, thin-wall quillwort's capacity to withstand projected climate change cannot be stated with certainty (Čtvrtlíková et al. 2023, pp. 2, 10).

3.2.5 Implications of Hydrological Alteration on Thin-wall Quillwort

There is no clear indication as to how thin-wall quillwort will respond to the effects described above and there are no data directly linking the influence of hydrological modification on thin-wall quillwort. Ultimately, its survival depends on the timing, frequency, depth, and duration of events (Suir and Berkowitz 2021, entire; Burandt et al. 2023, p. 2). We posit that compound storm events and resultant flooding, as well as severe drought, may decrease the amount of suitable habitat for thin-wall quillwort. As such, we included this factor in our current and future conditions analyses.

3.3 Water Quality

Blackwater rivers and streams of the southeastern Coastal Plain are generally characterized by low pH (acidic), dissolved oxygen, conductivity, and nutrients, as well as high dissolved organic carbon (DOC) and tannins (i.e., organic acids derived from plant material) (Flotemersch 2023, pp. 667, 673; Flotemersch et al. 2024, p. 2; Blocksom et al. 2025, entire). However, eutrophication (enhanced nutrient availability) is increasing in frequency and intensity due to direct and indirect—primarily anthropogenic—means (O'Hare et al. 2018, p. 2; Mallin and Cahoon 2020, entire; Pannard et al. 2024, entire). Such nutrient augmentation in blackwater systems, including both the Black and Northeast Cape Fear rivers, has been identified as driving microbial growth and algal (phytoplankton) blooms, which leads to oxygen depletion and hypoxia (i.e., increased biochemical oxygen demand), thereby enhancing water body acidification (Paerl et al. 2001a/b, entire; Mallin et al. 2006, entire).

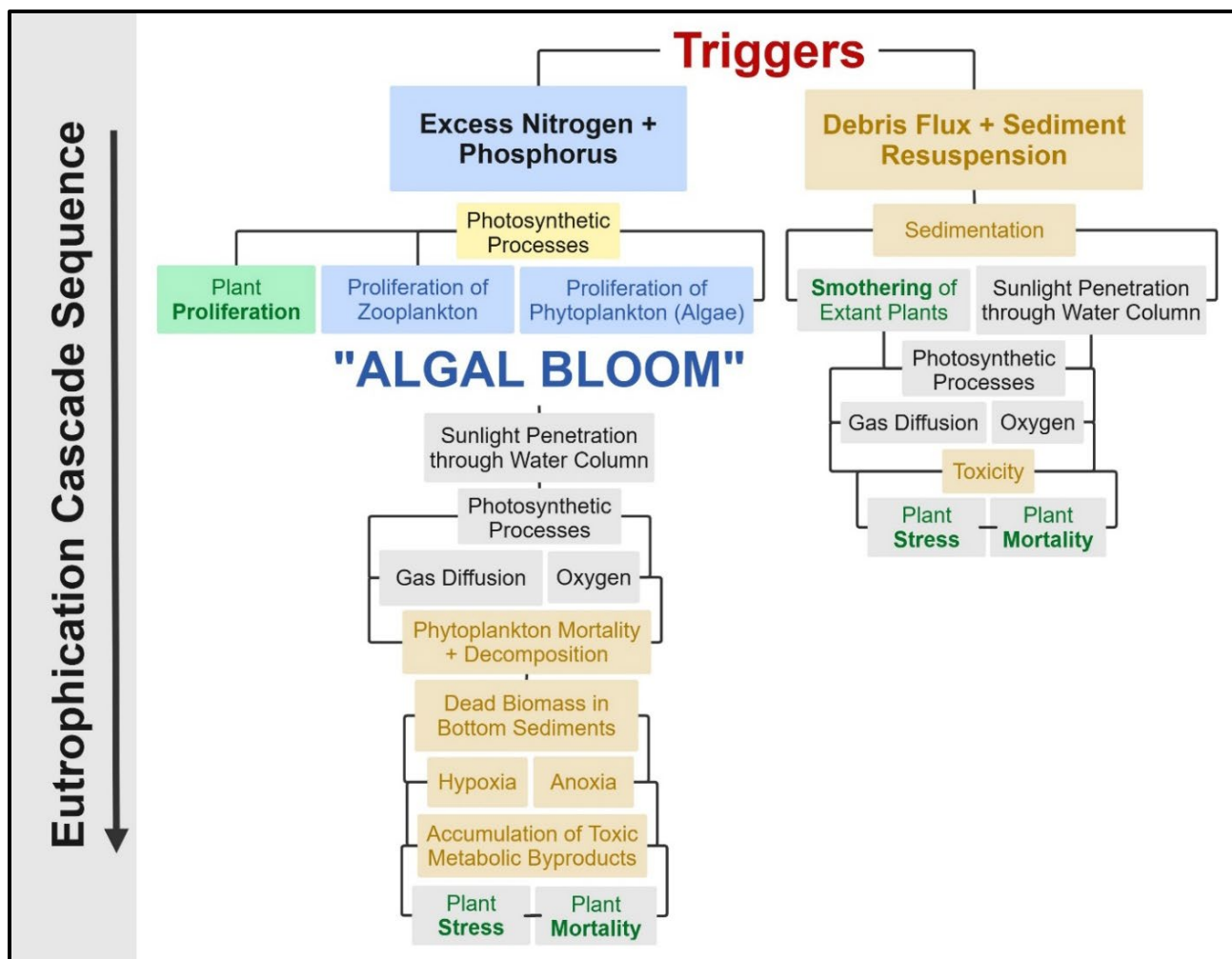


Figure 3.3 Generalized eutrophication cascade sequence triggered by Concentrated Animal Feeding Operations (CAFOs). Colored boxes (blue, tan, yellow) are those influences that are amplified by preceding factor(s) in the cascade sequence. Gray boxes with black font indicate an influence diminished by the antecedent factor(s). Green and gray boxes with bold green font represent the end result for plants under these conditions.

3.3.1 Concentrated Animal Feeding Operations (CAFOs)

Industrialized agriculture is a major cause of eutrophication in thin-wall quillwort watersheds; North Carolina hosts one of the highest concentrations of swine and poultry CAFOs in the United States (Mallin et al. 2004, p. 823; Martin et al. 2018, p. 888; Brown et al. 2020, entire). These facilities produce vast amounts of manure, causing unintended loading of nutrients and other pollutants to receiving water bodies and deliberately sprayed on adjoining fields (Mallin et al. 2002, pp. 999-1000, 1005). Nearby streams and rivers, including the Black River that houses one thin-wall quillwort site, have been documented to receive heavy nutrient loads from CAFOs, chronically through surface and subsurface runoff and more acutely via lagoon breaches, leaks/spills, and floodwater inundation (Mallin et al. 2002, entire; Mallin et al. 2004, pp. 823-824; Martin et al. 2018, p. 892). The Cape Fear River watershed, within which the Black River occurs, has the greatest concentration of swine CAFOs globally (Brown et al. 2020, p. 2). Not surprisingly, agricultural watersheds in eastern North Carolina having CAFOs have been associated with impaired groundwater and surface water quality, which is significantly enhanced in summer months (Harden 2015, entire, especially pp. 50-51; Mallin and McIver 2018, entire). The degradation produced by CAFOs has the potential to be further amplified by catastrophic storms, which

may generate severe flood disturbances in progressively deteriorating systems (Schafer et al. 2020, entire; Neville et al. 2021, entire; Moulton et al. 2024, entire).

Reports have suggested that the gradual decline of the thin-wall quillwort type-locality population may have been influenced by water pollution from upstream hog manure lagoons that regularly breach following high rain events (D. Brunton 2021, pers. comm.). More calamitous animal-waste spills have happened as well; the most notable being in Onslow County, which overflowed in 1995, dumping nearly 26 million gallons of hog waste into nearby rivers (Burkholder et al. 1997, entire; Mallin 2000, entire). Following that catastrophe, innumerable smaller spills and hurricane-induced breaches (Hurricanes Fran, Bonnie, Dennis, Irene, etc.) have occurred in all counties occupied by thin-wall quillwort, but we have no data demonstrating its response (Mallin et al. 1998, pp. 1-2; Schneider et al. 2025, p. 347). Some research indicates, however, that resident macrophytes are the likely primary recipients of excess CAFO nutrients (Chaffin et al. 2025, p. 510).

3.3.2 Urbanization

Similarly, coastal urbanization increases impervious surfaces and thus enhances movement of organic carbon sources (e.g., yard and animal waste, fertilizers) as well as hydrocarbons, heavy metals, and emerging organic contaminants into waterways, which may lead to eutrophication, algal blooms, and low dissolved oxygen (Wahl et al. 1997, entire; Mallin et al. 2004, pp. 824, 834-835; Gold et al. 2020, p. 23; Hopper et al. 2025, pp. 1290-1291). In this manner, alteration of the structure and function of Coastal Plain blackwater streams via land use practices (e.g., roadways and associated infrastructure, residential development, golf courses) may result in adverse effects on fish and invertebrate populations, thereby deleteriously impacting aquatic food webs (Mallin et al. 2009, p. 487). This trophic degradation may be enhanced by storm events, which more readily deliver nonpoint source substances from land to receiving waters (McCabe et al. 2021, entire).

Because quillworts are recognized as indicators of aquatic ecosystem health, impaired water quality, such as eutrophication and acidification, has the potential to affect submerged species that are either sensitive to high nutrient concentrations and/or facing population declines (e.g., *I. lacustris*, *I. malinverniana*) (Barni et al. 2013, entire; Vöge 2015, entire; Salgado et al. 2018, pp. 9-10; Klimaszuk et al. 2020, entire; D. Brunton 2021, pers. comm.; Azzella et al. 2024, p. 1422). For example, increased acidification has been linked with recruitment failure in *I. lacustris* (Čtvrtlíková et al. 2023, entire). Little is known about the response of amphibious quillworts to analogous changes in water quality; however, it has been posited that they may experience comparable effects (D. Brunton 2021, pers. comm.).

Despite the severity of potential effects, urbanization is relatively low within the watersheds where thin-wall quillwort occurs and is uncommon in the immediate area of species occurrence (Appendix E). Presently, two populations may be in the general vicinity of residential infrastructure, although their influence on water quality (and thus the species) is not definite. While these riparian blackwater swamp habitats are usually nonconducive to construction or habitation, one thin-wall quillwort population has sustained substantial impacts from activities related to a private campground, which has a demonstrable development footprint even if not considered true urbanization (see Section 3.5.1).

3.3.3 Implications of Water Quality on Thin-wall Quillwort

There is an extensive body of literature demonstrating the deleterious influence of diminished water quality on various macrophyte communities, wherein it has the potential to play a significant role in aquatic system health. While we have no direct evidence of urbanization, and therefore diminished water quality, acting on thin-wall quillwort populations across its known range, the influence of reduced water

quality as a result of CAFOs in the watersheds that it occupies is clearly problematic to overall system health. We are, therefore, carrying the factor forward to our current and future conditions analyses.

3.4 Interspecific Vegetation Dynamics

Most *Isoetes* species are regarded as stress-tolerators rather than successful competitors (Rhazi et al. 2009, p. 122; Vander Stelt et al. 2017, p. 17; Looy et al. 2021, p. 13). Through this lens, quillworts would be expected to suffer some manner of declines due to interactions with aggressive (native) weeds and/or exotic invasive plants capable of commandeering habitat resources (Marble 2018, entire). Such competitive exclusion is likely to be mediated by hydrology and enhanced by eutrophication common in North Carolina's blackwater systems (Barni et al. 2013, p. 11; O'Hare et al. 2018, entire). However, more information is greatly needed to understand to what extent thin-wall quillwort is susceptible to such behavior.

3.4.1 Woody/Herbaceous Encroachment

Deemed poor competitors, *Isoetes* are often relegated to sub-optimal habitats due to resource competition with other plant species (Looy et al. 2021, entire). Primarily, the increasing density/cover/biomass of shrubs and trees can be observed in habitats with modified land use practices (e.g., decreased fire/grazing) where aquatic and amphibious quillworts occur (i.e., aquatic: Western Lake quillwort (*I. setacea*); amphibious: limestone quillwort (*I. butleri*), *I. durieui*, Columbia quillwort (*I. minima*) (Rhazi et al. 2004, entire; Vander Stelt et al. 2017, entire, particularly p. 17; Maslovat et al. 2021, p. 302). Such competitive exclusion occurs incrementally by means of resource appropriation by woody species, thereby influencing distribution and potentially reducing quillwort biomass and spore production through shading and other means (Rhazi et al. 2004, p. 682). Similarly, competition with herbaceous clonal plant species has the demonstrated ability to displace amphibious Western Lake quillwort in some ephemeral wetland settings (Rhazi et al. 2009, entire).

3.4.2 Exotic Invasive Species

Non-native invasive plants compete overwhelmingly with native species for space, light, water, and nutrients, with changes in community composition stemming from an array of direct displacement processes, soil microbial alteration, and/or disturbance regime disruption (Craine and Dybzinski 2013, entire; Gaertner et al. 2014, entire; Gioria et al. 2023, entire; Nunez-Mir and McCary 2024, entire). Globally, several native quillwort taxa are documented as being under threat by such invasions—e.g., aquatic: *I. lacustris* by numerous, non-specialized species; alpine quillwort (*I. alpinus*) by various submerged species; amphibious: narrow quillwort (*I. sinensis*) by multiple floating and emergent taxa, Columbia quillwort (*I. minima*) by spotted knapweed (*Centaurea stoebe*)—with some research indicating that invasives cause premature drying of seepages, thereby reducing native reproductive success and rendering habitat unsuitable (Kelly and Hawes 2005, entire; Wang et al. 2005, pp. 4069-4070, 4075-4077; Spierenburg et al. 2010, entire; Maslovat et al. 2021, p. 302; Ronowski et al. 2023, pp. 6, 8-10). However, conversely, there are reports of quillworts being the most abundant species at sites dominated by invasive water hyacinth and, unexpectedly, occurring exclusively in infested water bodies (Mengitsu et al. 2017, pp. 672-675).

For thin-wall quillwort, multiple (three) subpopulations were found to coexist with the non-native invasive (herbaceous) alligator-weed (*Alternanthera philoxeroides*) during 2021 status survey efforts (Schaner et al. 2021, pp. 14, 17-18). Portions of these subpopulations contained approximately 10 to 40% cover by alligator-weed, appearing to affect 145 clumps of thin-wall quillwort (Schaner et al. 2021, p. 14). Native to South America, alligator-weed is an aquatic plant, dispersing via stem fragments, that forms large floating mats that may smother co-occurring vegetation on riverbanks and in interior microhabitats (Schaner et al. 2021, p. 10; Harms et al. 2023, entire; Wu et al. 2024, entire; J. Bolin 2026,

pers. comm.). On the Black River, it seems that alligator-weed occupies the same riverbank habitat niche as does thin-wall quillwort, thus potentially limiting available habitat for dispersal, in addition to posing a threat by outcompeting existing populations. Interestingly, alligator-weed has also been noted to impact other quillwort taxa outside of its native range (i.e., narrow quillwort, *I. sinensis*) (Wang et al. 2005, pp. 4075-4077).

3.4.3 Implications of Interspecific Vegetation Dynamics on Thin-wall Quillwort

While competing (woody and herbaceous encroachment) and exotic invasive vegetation may pose a threat to quillwort flora globally, only one plant species occurs across thin-wall quillwort's known range (at fairly low abundance) that demonstrates some capacity to impact thin-wall quillwort through its influence on habitat. However, because we have no clear evidence of its effect on individuals or populations, we did not carry the factor forward to our current and future conditions analyses.

3.5 Habitat Modification and Loss

3.5.1 Trampling/Erosion

Among anthropogenic impacts, the creation of swimming, fishing, and boat launch areas and associated shoreline degradation resulting from recreation is recognized as damaging for *Isoetes* within shallow areas (Mochalova et al. 2015, entire; Azzella et al. 2024, p. 1422; Keskin and Harahan 2025, p. 190). Through these and other related activities, soil can be dislodged (generating erosion), plants trampled, natural hydrology can be altered, and vehicle tires and individual's shoes can introduce/spread invasive plants (Maslovat et al. 2021, p. 302; Mäemets 2023, pp. 15, 17). For example, some amphibious quillwort populations (*I. minima*, *I. asiatica*) have been impacted by the direct physical disturbance of swimmers and picknickers using occupied shoreline/beach habitat (Mochalova et al. 2015, p. 110; Keskin and Karahan 2025, p. 190). However, in small amounts, some magnitude of disturbance associated with recreation may create habitat by dispersing spores, decreasing competition from vascular plants, and changing hydrology in ways that transport megaspores to new habitats (Mochalova et al. 2015, p. 111; Maslovat et al. 2021, p. 302). In one case, *I. velata* was positively associated with experimental disturbance treatments generated by livestock (Van den Broeck et al. 2019, p. 8).

There is some indication that physical damage and destruction of riverbank habitat may pose a threat to thin-wall quillwort. The White Oak River Campground site is directly impacted by habitat modification (foot traffic, flooding, and erosion), with the floodplain of the river having been heavily altered, with only a few canopy trees remaining on the riverbank and the rest maintained as a mowed lawn (Schaner et al. 2021, pp. 12-13, 19). Increasing observations of this form indicate that research is needed to understand the true scope and severity of these threats to thin-wall quillwort populations.

3.5.2 Construction and Development

The disappearance or reduction of several fully aquatic species of *Isoetes* due to infrastructure construction (dam, road) has been posited, but few details exist (Gacia and Ballesteros 1998, entire; Pietsch 1991, p. 101; Wang et al. 2005, p. 4069, 4074; Brunton et al. 2020, pp. 391, 404). While even less literature exists describing the response of amphibious quillworts to such anthropogenic activities, it is likely that they are subject to similar degradation, potentially generating irreversible destruction of habitat and loss of ecosystem function and biodiversity (Bagella and Caria 2013, p. 286). Nonetheless, such artificial barriers have the potential to critically impact dispersal and population connectivity (Jones et al. 2020, entire).

Schaner et al. (2021, p. 14) noted that most known populations of thin-wall quillwort occur adjacent to roadway bridges. At one site, no plants were observed in 2021 in close proximity to the bridge structure, which had been replaced in 2014 (NCNHP 2026a, pp. 5-6). At another site, thin-wall quillwort plants

were located in a different location from where they had been observed prior to disturbances from recent bridge reconstruction (NCNHP 2026a, p. 10). There has also been a suggestion that use of heavy equipment in suitable habitat adjacent to bridges for removal of large woody debris following severe storm events is a cause for concern (J. Schaner 2026, pers. comm.). Nonetheless, there is also evidence of other amphibious, blackwater stream species (i.e., Mississippi quillwort, *I. mississippiensis*) readily revegetating new roadside ditch and sandbar habitat despite severe road crossing disturbance (Schafran et al. 2016, p. 100).

Because development pressure across the known range of thin-wall quillwort is intensifying, populations may eventually be impacted, either through the direct removal of natural forest riparian buffers or indirectly via overall fragmentation of natural areas. Brunswick, Onslow, and Pender counties, where the species is documented to occur, have the fastest rates of human population growth in the state of North Carolina, although this growth (as interpreted through increasing numbers of residential building permits) remains concentrated along the coast (Table 3.5.2; Halls and Magolan 2019, entire, especially pp. 1-2; Hino et al. 2024, p. 56). Of course, as sea levels rise and coastal environments become increasingly inundated, new development may occur in more inland, undeveloped areas of these same counties (Barnard et al. 2024, p. 109-iii).

Table 3.5.2 Change in human population in North Carolina counties occupied by thin-wall quillwort from 2010 to 2024 (NCOSMB 2022, unpaginated; NCSOMB 2025*, unpaginated; NCOSMB 2026, unpaginated).

NC County	Metropolitan Area	2010 Population	2024 Population	Population Change 2010-2024 (#)	Population Change 2010-2024 (%)	% Remaining Non-Municipal Area 2024*
Bladen	--	35,038	29,393	-5,645	-16.1	75.7
Brunswick	Wilmington	107,429	169,448	+62,019	+57.7	46.2
Jones	--	10,163	9,317	-846	-8.3	84.5
Onslow	Jacksonville	177,801	215,481	+37,680	+21.2	59.8
Pender	Surf City	52,195	68,737	+8,130	+15.6	85.7
Sampson	--	63,471	59,960	-3,511	-5.5	80

3.5.3 Implications of Habitat Modification and Loss on Thin-wall Quillwort

Urbanization in the immediate vicinity (within occupied riverine swamp habitats) of thin-wall quillwort occurrence is not significant nor is it anticipated to increase significantly in the future. However, the localized threats of recreation (and the likely effect of bridge construction) is having a direct impact on population persistence, therefore, we carried it forward in our current and future conditions analyses.

3.6 Conservation Measures

Thin-wall quillwort was first described in 1998 and, given the cryptic nature of quillworts and their superficial similarity to clumps of grasses or rushes, members of the genus *Isoetes* are often overlooked (Brunton and Britton 1998, entire). Most of the known populations occur on private land, therefore, offer no protective safeguards (Schaner et al. 2021, p. 12). Thin-wall quillwort was listed as state endangered in 2009 in North Carolina, but it has since been downlisted to state threatened, and receives limited protection from take or tampering by the North Carolina Plant Protection and Conservation Act (LeBlond

and Brunton 2010, p. 501; NCDACS 2021, p. 9; Walker et al. 2024, pp. 23, 54).

3.6.1 Site Protection

Three populations of thin-wall quillwort occur entirely on private property with no formal protection (Schaner et al. 2021, p. 12). However, one population (Black River) is composed of three subpopulations, of which two are protected (ownership and easement by The Nature Conservancy) and one is wholly unprotected (Schaner et al. 2021, p. 17). These two protected subpopulations contain 1,829 clumps of thin-wall quillwort, or 56% of all known clumps (Schaner et al. 2021, pp. 12, 17). An additional population, Sandy Run Swamp, officially occurs on private land but is within 55 feet of Sandy Run State Natural Area, which is managed by North Carolina State Parks (Schaner et al. 2021, pp. 12-13, 18). Due to the buffering methods used to delineate populations, a portion of this population appears on the protected area. Such proximity suggests that future survey efforts should focus on locating thin-wall quillwort on North Carolina State Park land.

3.6.2 Federal Lands Management

Thin-wall quillwort is listed as a rare species of emergent riverbanks on Croatan National Forest in the Croatan National Forest Plan (CNF Plan), although to date there are no documented occurrences (USFS 2002a, pp. 62, 67; USFS 2002b, pp. 480, 485). In theory, the CNF Plan affords habitat protections via maintenance and improvement of aquatic habitats and alignment with relevant best management practices (BMPs), although these measures may not be sufficient to safeguard thin-wall quillwort (USFS 2002b, p. 480). Specifically, the species is classified as occurring in a Streams, Lakes, and Riparian system in the CNF Plan, which is a habitat that was excepted from the monitoring program (USFS 2002b, p. 480; USFS 2016, entire).

And while some potentially suitable stream and riparian habitats may be inventoried as special interest natural areas on the Forest, such designation does not necessarily afford monitoring measures, although it may mean those lands are either withdrawn from or deemed as generally inappropriate for timber production and, therefore, sustain some manner of indirect protections (USFS 2002a, p. 99).

In addition, the Marine Corps Base at Camp Lejeune (MCB-CL) recognizes thin-wall quillwort as a Priority 1 species-at-risk that potentially occurs on emergent riverbanks of the Installation (JV-GMI-AECOM 2015, pp. 4-93, 4-97, 5). Such designation supports the identification and monitoring of known occurrences, as well as (in some cases) avoidance and minimization efforts and ecosystem management practices that support its conservation and germane habitat management (JV-GMI-AECOM 2015, pp. 4-97, 10). Implementation of rare plant species surveys over the years by MCB-CL personnel have, thus far, not revealed any thin-wall quillwort occurrences on the Installation (C. Ten Brink 2026, pers. comm.).

3.6.3 State Laws

Thin-wall quillwort is considered critically imperiled globally (G1) and in North Carolina (S1), indicating it has a very high risk of extinction or elimination throughout its global range and in North Carolina (Walker et al. 2024, pp. 23, 54; NatureServe 2026, unpaginated). These designations, however, do not afford legal protections to the species.

Further, thin-wall quillwort is state listed as threatened in North Carolina (Walker et al. 2024, pp. 23, 54). The threatened designation is defined as “any resident species of plant which is likely to become an endangered species within the foreseeable future throughout all or a significant portion of its range” (NCLEG 2025, p. 1). This law provides limited protection of thin-wall quillwort from unauthorized collection (“uproot, dig, take or otherwise disturb or remove for any purpose” from another’s private property without the owner’s written permission) and trade (“sell, barter, trade, exchange, export”, etc.)

(NCLEG 2025, pp. 5-6). However, the statute does not protect the species or its habitat from destruction resulting from development projects or otherwise legal activities. Additional legislation also authorizes the state to establish nature preserves for protected species and their habitats; however, no such preserve has been created for thin-wall quillwort (NCNHP 2015, entire). Thin-wall quillwort was added to the Species of Greatest Conservation Need list in the 2025 North Carolina State Wildlife Action Plan (NCWRC 2025, pp. 1183, 1263).

3.6.4 Other

No quillworts are listed on the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) lists. The International Union for Conservation of Nature (IUCN) Red List categorized thin-wall quillwort as an endangered entity in 2024 (Treher 2024, entire). Lastly, there is no indication that members of the genus *Isoetes* are under threat by the horticultural or gardening trade in the Southeast.

3.7 Summary of Influencing Factors

Our evaluation of the potential factors affecting thin-wall quillwort indicated that sustained catastrophic hydrological alteration at the regional level (i.e., repeated compound flood events, long-term drought) together with regional water quality stressors (i.e., CAFOS) and localized habitat modification/loss stressors (i.e., recreation) pose the greatest risks to overall species viability. These factors are further analyzed in both current (Chapter 4) and future conditions (Chapter 5).

CHAPTER 4: CURRENT CONDITION

4.1 Delineating Analysis Units

Detailed information regarding thin-wall quillwort genetics and population structure are not available; however, we considered how the spatial arrangement of populations is generally linear due to plant dispersal most readily following stream courses, with spacing of individuals and populations demonstrating patchy establishment patterns. Consequently, for the purposes of this assessment, we based our thin-wall quillwort unit delimitation on field-delineated sub-element occurrences (sub-EOs) provided by the North Carolina Natural Heritage Program (NCNHP), which followed NatureServe's habitat-based criteria for plants occurring along a riparian corridor (inter-population distance of ≤ 10 km) (NatureServe 2020, pp. 9-10). To generate our analytical units (AUs) (Table 4.1), we buffered each of nine sub-EO points by 1 km upstream and downstream (to account for potential downstream dispersal), as well as laterally, thus creating a 1 km riparian buffer. Any overlapping buffered occurrences were joined as a single analysis unit. Most of the resulting AUs were consistent with NCNHP sub-EO delineation. The final seven delimited AUs were used in all subsequent analyses, including our evaluation of resiliency condition metrics (see Section 4.3).

We delineated representation units at the level of HUC-10 watersheds (Table 4.1). While the species is known to inhabit four different watersheds of North Carolina's Coastal Plain and is likely primarily a short-distance disperser (dispersal limited), occupied habitats are structurally similar throughout its range (i.e., riverine blackwater swamps with abundant hardwoods), with no glaring fundamental ecological differences. These representation units corresponded to EOs delineated by NCNHP. For the purposes of this assessment, analysis units were deemed equivalent to populations; thus, terms may be used interchangeably.

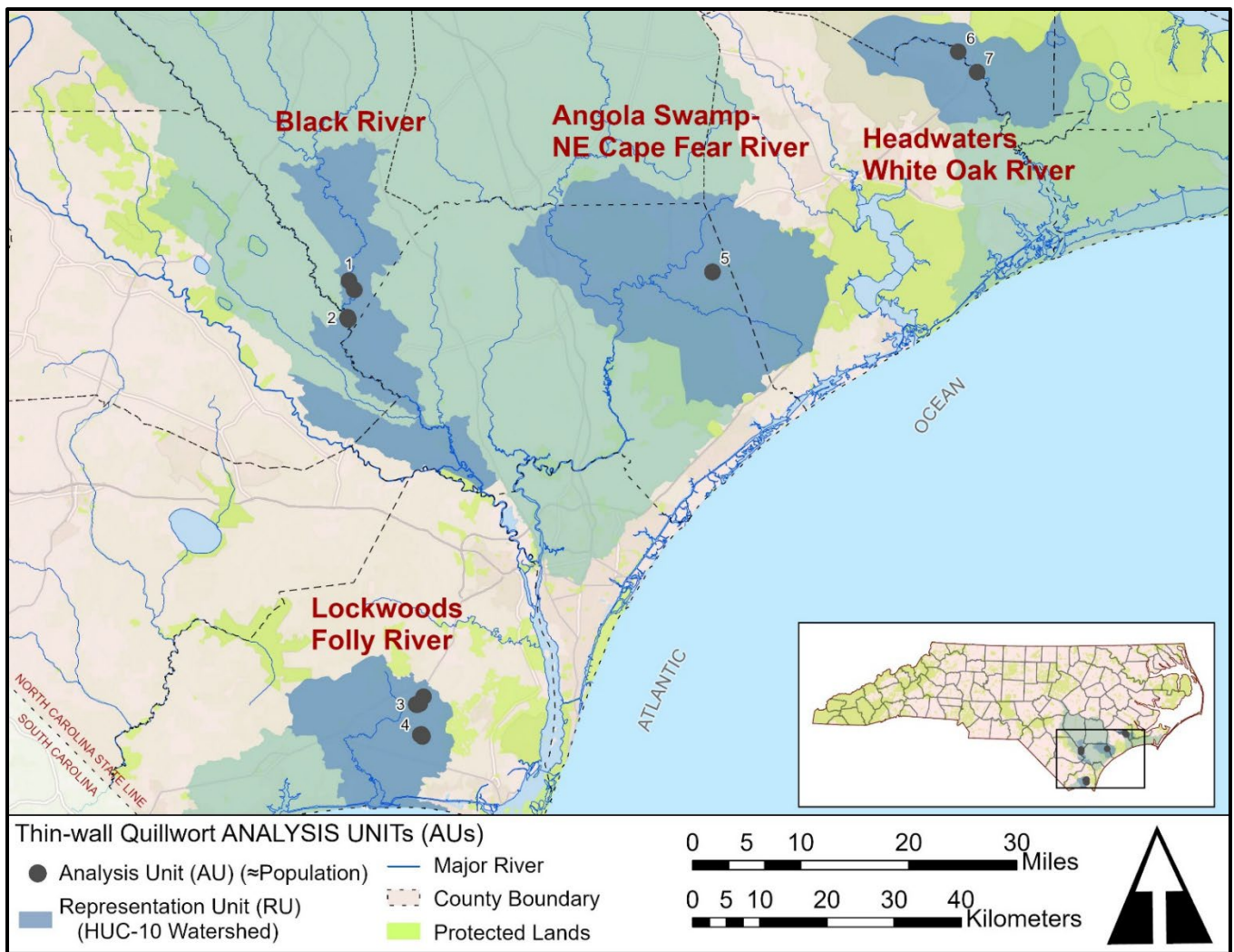


Figure 4.1 Thin-wall quillwort analysis units (AUs≈populations) and representation units (RUs=HUC-10 watersheds). HUC-10 watershed names in red.

Table 4.1 First and last collection date within each analysis unit (AU) and representation unit (RU), plus HUC-10 watershed and HUC-6 river basin names (NCNHP 2026a, entire).

AU	RU	HUC-10 Name	HUC-10 Code	HUC-6 Name	Date of First Collection	Date of Last Collection
1	1	Black River (Lower)	0303000608	Cape Fear	2006	2021
2	1	Black River (Lower)	0303000608	Cape Fear	2021	2021
3	2	Lockwoods Folly River	0304020801	Lumber	1950	2026
4	2	Lockwoods Folly River	0304020801	Lumber	1998	2025
5	3	Angola Swamp-NE Cape Fear River (Lower)	0303000706	Cape Fear	2010	2021
6	4	Headwaters White Oak River	0302030101	White Oak	2005	2022
7	4	Headwaters White Oak River	0302030101	White Oak	1951	2022

4.2 Methods for Estimating Current Condition

To assess the current biological status of thin-wall quillwort across its known range, we used best available information, which included peer-reviewed scientific literature, the NCNHP (Schaner et al.) 2021 status survey data and report, and unpublished communications from species experts. Following the SSA framework, we conducted current condition analyses and determined resiliency for each AU and then evaluated redundancy and representation at the range-wide scale. For the purposes of this assessment, we consider a resilient thin-wall quillwort population as conforming to the following factors:

- has abundant suitable habitat for successful growth and development of mature plants and establishment of dispersing propagules
- has water quality adequate for ongoing survival
- maintains enough individuals to support effective reproduction and gene flow
- has sufficient connectivity between populations to facilitate successful dispersal and gene flow
- can recover from stochastic events or disturbances (e.g., drought, flooding, etc.) over time

4.3 Current Condition Metrics

4.3.1 Habitat Factor

Water Quality

While mature *Isoetes* may demonstrate some degree of tolerance of naturally acidic conditions, immature stages are often documented as sensitive to an assortment of water chemistry variables, particularly nutrient loading (Čtvrtlíková et al. 2020, p. 9; Čtvrtlíková et al. 2023, entire). To assess water quality of thin-wall quillwort habitats and how the species may respond, we used three data sources: 1) the Assessment, Total Maximum Daily Load (TMDL) Tracking and Implementation System (ATTAINS); 2) the North Carolina Department of Environmental Quality database of permitted animal swine facilities (NCDEQ 2024, unpaginated; USEPA 2026, unpaginated); and 3) the Environmental Working Group (EWG) Animal Facilities in North Carolina in 2022 webpage (EWG 2022, unpaginated). The ATTAINS dataset is a centralized database that collates water quality data to help agencies identify prioritized waters, prioritize restoration efforts, and track progress toward meeting Clean Water Act goals (USEPA 2025, unpaginated). It combines HUC-12 watershed-level information, designating a watershed as being in good condition or impaired.

First, using ATTAINS, we evaluated the general conditions within each occupied HUC-12 watershed for each population, which is the scale available for this dataset. Results from this dataset indicate that water quality is designated as “good quality” in all HUC-12 watersheds where thin-wall quillwort populations occur. Since results are the same across all units, we do not present them further in our analysis. We then used the NCDEQ animal facility database to quantify the number and relative proximity of permitted CAFOs within each occupied HUC-12 and HUC-10 watershed. This dataset includes only permitted swine operations. To consider poultry farms, we used the Environmental Working Group database of poultry facilities estimated with satellite imagery (EWG 2022, unpaginated). Since we do not have access to the point data, we were able to estimate the number of poultry operations within each HUC-10 watershed. We assessed both scales to evaluate potential immediate risk that is closer to AUs (HUC-12 scale) as well as possible larger-scale risk throughout the greater watershed (HUC-10 scale). As part of this, we also considered the proximity of CAFOs upstream from the analytical unit to evaluate the likelihood that the CAFOs would present an actual risk to each population. In some cases, CAFOs were located on tributaries that would not affect the AU. Combining both number of CAFOs within the HUC-12/HUC-10 as well as upstream location, we identified three risk categories to provide a measure of exposure risk for each analysis unit (Table 4.3.1.2).

Table 4.3.1 Scoring rubric for habitat suitability metric {water quality} assessed in current condition.

Water Quality	Rank (Score)
Many (>20) CAFOs in HUC-12 and HUC-10, located upstream	Low (1)
Few (<20) CAFOs in HUC-12 or HUC-10, OR Many CAFOs in HUC-10, BUT not upstream	Moderate (2)
No CAFOs in HUC-12, Few CAFOs in HUC-10, AND/OR No CAFOs upstream	High (3)

4.3.2 Demographic Factors

Population Size

Each thin-wall quillwort analysis unit was assigned to one of three population size categories: Low (<100 plants), Moderate (100 to 1000 plants), and High (>1000 plants). Expert elicitations indicated that less than 100 plants should be considered “low condition” due to their vulnerability to stochastic events. Over 1000 plants was identified as “high condition.” These values are consistent with those presented by Dalapicolla et al. (2021, p. 10123; *I. cangae*) and Frankham et al. (2014, entire), who define 50-100 as the minimum number of individuals within a population to avoid inbreeding and 500-1000 as the minimum to maintain species evolutionary potential.

Table 4.3.2.1 Scoring rubric for demography metric {population size} assessed in current condition.

Population Size (Range)	Rank (Score)
<100 plants (“Tens”)	Low (1)
100-1000 plants (“Hundreds”)	Moderate (2)
>1000 plants (“Thousands”)	High (3)

Population Connectivity

To evaluate thin-wall quillwort population connectivity, we used the National Aquatic Connectivity Collaborative (NACC) Barrier Inventory and Prioritization Tool to assess the presence of known large dams between populations that may serve as obstructions to dispersal (NACC 2026, unpaginated). Although we lack direct evidence, we assume that major barriers such as dams could impact stream flow and width, hindering downstream dispersal of thin-wall quillwort. For each AU, we evaluated its overall connectivity potential through two metrics: barriers and proximity (Table 4.3.2.2). To evaluate barriers, we noted any major barriers to dispersal (dams) 2 km upstream or downstream of the AU. We designated an AU as impaired if it has a barrier present or intact if a barrier was absent. To evaluate proximity, we measured the distance to the nearest AU and noted if the nearby unit was upstream or downstream. We assigned a value of 1 if no barriers were present and a value of 0.5 if barriers were present.

While upstream propagule dispersal has been reported for many riparian species, we lack evidence of its occurrence in thin-wall quillwort and thus assume downstream dispersal is more likely (Wubs et al. 2016, entire). We assigned each AU to one of three proximity categories and provided a numerical value for each:

- Isolated (no upstream/downstream AU(s), therefore, no rescue probability),
- Remote (downstream AU(s) only, therefore, limited rescue probability),
- Proximate (upstream AU(s) present, therefore, greater probability of rescue).

To provide an overall ranking for connectivity, we combined the barrier and proximity scores for an overall connectivity score (Table 4.3.2.3). Overall connectivity scores can range from 1.5 to 3. An AU with a score of 1.5 does not meet either criterion for connectivity, and is thus considered in low condition for connectivity. An AU with an overall score of 2 to 2.5 is impaired in either barriers or proximity, and is considered to be in moderate condition. An AU with an overall score of 3 meets all criteria for connectivity and is considered to be in high condition for connectivity.

Table 4.3.2.2 Scoring rubric for demography metric {population connectivity} assessed in current condition.

Population Connectivity			
Barriers	Rank (Score)	Proximity	Rank (Score)
Barrier(s) present	Impaired (0.5)	No AU upstream OR downstream	Isolated (1)
No barrier(s) present	Intact (1)	AU(s) downstream	Remote (1.5)
		AU(s) upstream	Proximate (2)

Table 4.3.2.3 Overall connectivity scores and designation.

Combined Connectivity Score	Connectivity Ranking
1.5	Low
2 to 2.5	Moderate
3	High

4.3.3 Current Resiliency Ranking

To determine current resiliency, we evaluated each analytical unit using the demographic and habitat metrics described above. We combined scores from each of these metrics to assign an overall resiliency score based on the ability of the unit to meet its population-scale needs. The combined scores of water quality, number of plants, and population connectivity may range from 4.5 to 10 overall (Table 4.3.3).

To incorporate the impacts of local threats, we reduced the resiliency score for each significant, independent threat present at a site. We provide descriptions of local threats present at each site in the AU summary in the following sections, which include active mining/quarry operations, presence of invasive plant species, close proximity to residential development, and recreation access. We considered independent threats as imposing independent impacts on individuals within the delineated analysis unit, and deemed each to have an overall negative effect on resiliency. For each threat present within an AU, we subtracted a score of 0.5 from the combined resiliency total. For example, if an AU was in close proximity to residential development and invasive plants were reported within the AU, resiliency would be reduced by a value of 1.

Land ownership type is described for each population but is not evaluated in resiliency.

We assigned an overall resiliency designation based on the final score within each AU. We assigned a resiliency of “low” to an AU with a score of less than 5. Analysis units with these scores are those impaired in multiple demographic or habitat categories: the ability of these units to meet the unit needs is reduced, and the ability of the unit to withstand stochastic disturbances is reduced. We assigned a resiliency of “moderate” to an AU with an overall score of 5 to 7: these analysis units are typically impaired in either a demographic or habitat category or have few local threats present. The needs of these units are nearly met but are somewhat impaired. We assume that these units are able to withstand

stochastic events, but their ability to do so is somewhat reduced from an ideal condition. We assigned a resiliency of “high” to an AU with a score greater than 7: these analysis units typically meet the high condition designation of multiple metric categories and have no local threats present.

Table 4.3.3 Combined resiliency scores and designation.

Combined Resiliency Score	Overall Resiliency Ranking
<5	Low
5-7	Moderate
>7	High

4.3.4 Current Resiliency Summary

Most (5 of 7) analysis units are considered highly and moderately resilient and two AUs have low resiliency (Figure 4.3.4; Table 4.3.4). The most resilient populations are those with a greater number of individuals and persist in landscapes having fewer CAFOs (thus enhanced water quality) and fewer barriers to dispersal. Across its range, three populations (43% of total) occur on lands at least partially managed for conservation and the remaining four populations (57%) occur on private lands with no known protection (Table 4.3.4).

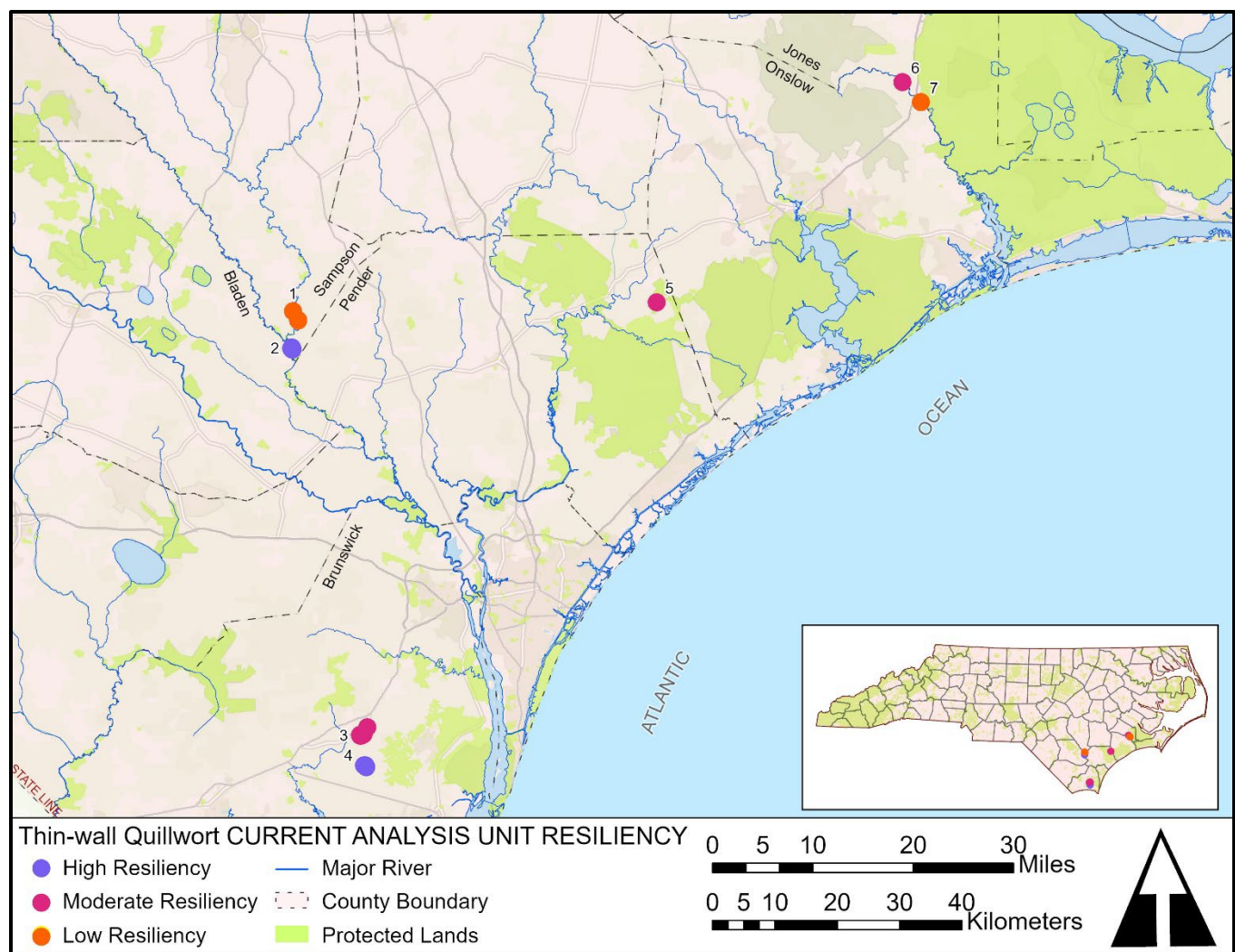


Figure 4.3.4 Map of thin-wall quillwort current analysis unit resiliency.

Table 4.3.4 Summary of thin-wall quillwort current analysis unit resiliency metric results.

AU	METRICS			OTHER FACTORS		RESILIENCY
	HABITAT	DEMOGRAPHY				
	Water Quality (# CAFOs + Proximity)	Population Size	Population Connectivity (Barriers + Proximity)	Local Threats	Land Ownership	
1	Low	1-100+	Intact + Remote	Active sand quarry, recreation access	Private + TNC easement	Low
2	Moderate	1,828	Intact + Proximate	Invasive plant presence	TNC ownership	High
3	High	140-275	Intact + Remote	Residential development, bridge	Private	Moderate
4	High	1,275-1,400+	Intact + Proximate	Residential development, bridge	Private	High
5	Moderate	97	Intact + Isolated	NA	Private + State ownership	Moderate
6	High	177	Impacted + Remote	NA	Private	Moderate
7	High	0-5	Impacted + Proximate	Campground: recreation access (trampling, erosion), mowing	Private	Low

Table 4.3.5 Scoring for all metrics used to evaluate thin-wall quillwort analysis units resiliency.

AU	METRICS			OTHER FACTORS	RESILIENCY SCORE	RESILIENCY RANK
	HABITAT	DEMOGRAPHY				
	Water Quality	Population Size	Population Connectivity	Local Threats		
1	1	1	2.5	-1	3.5	Low
2	2	3	3	-0.5	7.5	High
3	3	2	2.5	-1	6.5	Moderate
4	3	3	3	-1	8	High
5	2	1	2	0	5	Moderate
6	3	2	1.5	0	6.5	Moderate
7	3	0	2	-0.5	4.5	Low

Analysis Unit 1

Analysis units 1 and 2 are both located in the Lower Black River watershed within the Cape Fear River Basin. Analysis unit 1 occurs in the Kings-Branch Black River sub-watershed, and the southern half of this AU is located within a Nature Conservancy (TNC) conservation easement managed by the North Carolina Coastal Land Trust. Recent population estimates indicate 1-100 individual plants at this location. While located 8 km upstream from AU 2, with no barriers identified between the two AUs, this AU has the greatest risk of experiencing water quality impacts due to potential spills from nearby CAFOs, with 15 swine operations located within the HUC-12 unit. Within the HUC-10 unit are 36 swine operations and ~50 poultry operations.  is also subject to some magnitude of development threat due to its location next to an active sand quarry and areas of residential development. Additionally, this AU is situated near a recreational access point for the Black River. Due to the relatively low number of individuals and low water quality, we assigned LOW resiliency to population 1.

Analysis Unit 2

Analysis unit 2 occurs in the Lake Creek-South River sub-watershed 8 km downstream from AU 1. The stretch between these two AUs is unimpeded, indicating intact connectivity with potential for demographic rescue. Analysis unit 2 has the highest number of recorded plants of all AUs across the species range, with an estimated 1,828 plants recorded during 2021 surveys. Within the HUC-12 sub-watershed, no CAFOs are present; however, 36 swine operations and ~50 poultry operations are located within the larger HUC-10 watershed, indicating potential risk to water quality. Recent surveys have reported the presence of invasive exotic alligator-weed at the site, but there is no evidence of population-scale impacts due to its presence. Given the large number of plants and moderate water quality, we assigned AU 2 as having HIGH overall resiliency.

Analysis Unit 3

Analysis units 3 and 4 are located in the Lockwoods Folly River watershed within the Lumber River Basin. Both AUs are located on private lands in the Middle Swamp sub-watershed. Analysis unit 3 is 6.8 km upstream from AU 4 on an unimpeded river stretch, indicating intact connectivity between these populations. There are no CAFOs located in the HUC-12 sub-watershed and two swine operations located within the larger HUC-10 watershed, indicating reasonably high water quality for both populations. Between 2016-2026, 140 to 275 plants were estimated at AU 3; however, one of the sub-EOs comprising this analysis unit has not been observed since 2015 (Schaner et al. 2021, p. 14; NCNHP 2026a, p. 3). This AU is located near residential areas, which increases the impervious surface cover within the AU area, increasing the risk of runoff from nearby development. North Carolina Natural Heritage Program biologists have reported possible impacts to this population and the surrounding area due to human development, including trash, vegetation overgrowth, and disturbance (NCNHP 2026a, p. 3). A portion of this unit is located in close proximity to a bridge, where neighboring plants have been documented in a different location following bridge reconstruction (NCNHP 2026a, p. 10). A nearby proposed housing development may result in impacts to the population directly via habitat conversion or indirectly due to runoff or sedimentation  (NCNHP 2026b, p. 2). Due to the moderate number of individuals and high water quality, we assigned AU 3 an overall MODERATE resiliency.

Analysis Unit 4

Analysis unit 4 is the second largest site within the species range, with 1275-1400 plants estimated between 2016-2026. This AU is located in proximity to residential areas and has some impervious surface within the AU area, but a low amount (2%). This unit is located in close proximity to a bridge, where its replacement in 2014 was reported to have altered adjacent habitat (Schaner et al. 2021, p. 14;

NCNHP 2026a, pp. 5-6). Given the AU size and high water quality in this area, we assigned an overall HIGH resiliency to AU 4.

Analysis Unit 5

Analysis unit 5 is in the Headwaters Sandy Run Swamp sub-watershed within the Angola Swamp-Northeast Cape Fear watershed. This is the only AU detected within this watershed. Due to the buffer used to delineate analytical units, this AU is partially located within the Sandy Run Savannas State Natural Area managed by North Carolina Division of Parks and Recreation, so is not wholly protected. Recent surveys estimate 97 plants at this AU. Three CAFOs are located 5 km upstream within the HUC-12 watershed. And while there are 30 swine CAFOs located within the larger HUC-10 watershed, these are located on tributaries of Angola Creek, which are not likely to affect this AU. Additionally, there are 20-25 poultry operations within the HUC-10 watershed, but we do not have location information to determine connectivity. Being the only detection at this site, this AU appears isolated; however, there are no barriers upstream or downstream that would serve as obstructions to dispersal. Given the number of plants and moderate water quality, we assigned AU 5 to MODERATE overall resiliency.

Analysis Unit 6

Analysis units 6 and 7 are both located in the Headwaters White Oak River watershed within the White Oak River Basin. Neither AU contains CAFOs within their HUC-12 sub watershed and two swine operations and one poultry operation are located within the larger HUC-10 watershed. Analysis unit 6 is located within the Town of Maysville-White Oak River sub-watershed on private land. Its proximity to a bridge and roadside may expose resident plants to some level of impacts, but no observations have yet to be documented. Recent surveys estimate 177 plants at this AU. There are no AUs detected upstream of this AU, so it may be relatively isolated. Given the moderate number of plants and high water quality, we assigned this AU an overall resiliency of MODERATE.

Analysis Unit 7

Analysis unit 7 is located in the Brick Kiln Branch-White Oak River sub-watershed in the White Oak River Campground. Although this AU is 6 km downstream from AU 6, a lake is located between these two locations, presenting a likely barrier to dispersal for the species. Very few plants have been detected at this location, with past surveys indicating that the AU is likely declining. State researchers attribute these declines to trampling due to heavy recreational use from the nearby campground as well as flooding and scouring due to hurricane events. Due to these factors, we assigned an overall LOW resiliency to AU 7.

4.4 Representation

We do not have explicit information regarding many attributes of thin-wall quillwort, including genetic diversity within and among all thin-wall quillwort populations across its range. Therefore, we evaluated representation using the best available information for *Isoetes*, including chiefly-amphibious taxa, and summarized attributes of adaptive capacity following Service guidance (2026, unpaginated) and Thurman et al. (2020, entire). We summarized the traits based on three primary mechanisms that contribute to adaptive capacity: 1) dispersal, 2) evolutionary potential, and 3) phenotypic plasticity (Table 4.4). Per DeWan et al. (2010, p. 15), these life history traits can influence species vulnerability to climate change (Figure 4.4).

Table 4.4 Three primary mechanisms (and associated factors) contributing to adaptive capacity (Service 2026, unpaginated).

Dispersal	Evolutionary Potential	Phenotypic Plasticity
-----------	------------------------	-----------------------

Habitat Specialization	Population Size	
Dispersal Distance	Genetic Diversity	Phenological Shifts
Climatic Niche Breadth	Extent of Occurrence	Morphological Changes
Physiological Tolerances	Physiological Tolerances	Physiological Tolerances

Species least at risk	Species most at risk
• Physiological tolerance to broad range of factors such as temperatures, water availability and fire • High degree of phenotypic plasticity • High degree of genetic variability • Short generation times (rapid life cycles) and short time to sexual maturity • High fecundity • 'Generalist' requirements for food, nesting sites, etc. • Good dispersal capability • Broad geographic ranges	• Narrow range of physiological tolerance to factors such as temperature, water availability and fire • Low genetic variability • Long generation times and long time to sexual maturity • Specialized requirements for other species (e.g. for a disperser, prey species, pollinator or photosynthetic symbiont) or for a particular habitat that may itself be restricted (e.g. a particular soil type) • Poor dispersers • Narrow geographic ranges

Figure 4.4 Climate vulnerability based on species traits (DeWan et al. 2010, p. 15).

1) *Dispersal*

Plant traits that contribute to dispersal are habitat specialization, dispersal distance, climatic niche breadth, and physiological tolerances. Thin-wall quillwort is generally considered a **habitat generalist** (and found across a matrix of swamp riverine habitat types). Dispersal distance for thin-wall quillwort is moderate—being a semi-aquatic organism, it routinely disperses in water currents and is purportedly transported by wildlife. In this manner, its dispersal/movement capability may not be truly restricted, although some unknowns remain. While its ability to shed leaves and enter dormancy in response to desiccation and drought conveys a functional adaptation to a changing climate (climatic niche breadth), some habitat across its range may be vulnerable (via decreased suitability) and have far reaching consequences for polyploids (Kurki-Fox et al. 2019, entire; Babin and Bell 2023, entire). In sum, our evaluation suggests that thin-wall quillwort ranks moderate for traits that contribute to dispersal (thus contributing to moderate adaptive capacity) (Table 4.4.1).

2) *Evolutionary Potential*

Traits that contribute to evolutionary potential are population size, genetic diversity, extent of occurrence, and physiological tolerances. Population sizes for thin-wall quillwort vary between low to moderately low, indicating that most have been tracked with fewer than 2500 mature individuals and some with less than 250. Genetic diversity is unknown; however, polyploid *Isoetes* species, like thin-wall quillwort, are suggested to have reduced diversity relative to diploid species (although enhanced genetic variation) (Kim et al. 2009, pp. 496, 499-503; Suissa et al. 2022, p. 11). The known extent of occurrence is moderate for this endemic species; however, due to the high degree of contiguous riverine swamp habitat and limited landscape modification, we presume that suitable habitat (and thus occurrence) is widespread and likely greater than currently documented. The physiological tolerance(s)

of thin-wall quillwort is not wholly known; however, its conceivable ability to photosynthesize using CAM, a process exploited by many amphibious quillwort taxa, could facilitate greater tolerance of underwater submergence while its polyploid nature may engender it with augmented genetic variation, which could potentially confer increased tolerance to a broad spectrum of ecological conditions (Keeley 1982, entire; Keeley 1998, p. 122; Liu et al. 2004, pp. 1243-1244; Husband et al. 2013, p. 270; Yang and Liu 2015, p. 6; Troia et al. 2016, entire; Larsén and Rydin 2016, p. 169; Suissa and Green 2020, entire, esp. p. 140; Lopes et al. 2023, pp. 12, 14; Koga et al. 2024, p. 593; Xu et al. 2026, pp. 1003, 1005). While, the evolutionary consequences of these traits are context dependent, CAM has been posited to have been a critical survival strategy used by ancestors of extant *Isoetes* during prehistoric extinction events (Xu et al. 2026, entire). Based on our investigation, thin-wall quillwort ranks moderate for traits that contribute to evolutionary potential (therefore, moderate adaptive capacity) (Table 4.4.1).

3) Phenotypic Plasticity

Traits that contribute to phenotypic plasticity include physiological tolerances, morphological changes, and phenological shifts. As described above, physiological tolerances are (assumed) high. Detailed phenological shifts are unknown, and we do not know about specific morphological changes that thin-wall quillwort may undergo to facilitate persistence under novel or stressful conditions. However, evidence demonstrates that many emergent (i.e., amphibious) *Isoetes* display phenotypic plasticity, e.g., adapting to aerial environments by developing stomata and switching to a terrestrial photosynthetic pathway (C₃) (Keeley 1998, p. 122; Cavalheiro-Filho et al. 2021, entire; Bjorn et al. 2022, pp. 3-4; Tan and Chen 2023, entire). Quillworts, generally, are iteroparous (multiple reproductive events in their lifetime), and adults are highly fecund, with individuals observed to produce anywhere from 150,000 to 1,000,000 (avg. 300,000 to 400,000) microspores and up to hundreds (50-300) of megaspores during one sporulation event/season (Smith 1900, p. 251; Husk et al. 2000, p. 5; Wang et al. 2005, p. 4069, 4076; Vöge 2006, p. 232; Blackmore et al. 2012, p. 85; Nunes et al. 2018, p. 13; Caldeira et al. 2019, entire; Larsén et al. 2025, p. 1). Although information is incomplete, some quillwort species populations demonstrate a top-heavy age structure, exhibiting a higher distribution of mature individuals, with these individuals becoming sexually mature somewhat rapidly (within one to five years) (Pfeiffer 1922, p. 91; Gacia and Ballesteros 1994, p. 86; Vöge 2003; Vöge 2006, entire; Čtvrtlíková et al. 2009, pp. 142, 144; Čtvrtlíková et al. 2023, pp. 4, 11). In addition, *Isoetes* readily navigate the challenging domains of hydrological drought and elevated temperature via shedding of aboveground biomass (leaves) while the underground corm remains extant but dormant (Rhazi et al. 2009, p. 123; Huang et al. 2018, entire; Alejo-Jacuinde and Herrera-Estrella 2022, pp. 3-4; Gao et al. 2023, p. 2; Montiel-Molina et al. 2023, p. 2). Whether they are inherently physiologically capable of withstanding such stress, their adaptive capacity is enhanced by this behavior. Collectively, these traits contribute to a high adaptive capacity (Table 4.4.1).

While thin-wall quillwort populations are known to occur in two ecoregions of North Carolina that span both the Inner and Outer Coastal Plain, we do not know whether the species was historically distributed throughout. At present, two highly resilient and two moderately resilient populations, as well as a single low resiliency population, occur across the southern portion of the Outer Coastal Plain. One high and one moderate resiliency population occur in the southernmost Inner Coastal Plain.

4.4.1 Current Representation

In summary, thin-wall quillwort exhibits varying levels of adaptive capacity depending on the attribute under consideration, with high levels of adaptive capacity attributable to its fecundity and habitat specialization (as well as physiological tolerances and reproductive phenology), moderate-high levels attributable to its life span, moderate levels attributable to climatic niche breadth and dispersal distance,

and moderately low levels attributable to its extent of occurrence and population size. Genetic diversity is unknown. The species’ occupancy within a modest range of the North Carolina Coastal Plain indicates no apparent reduced representation, suggesting it may be capable of either/both moving across the landscape outside of its required habitat type to other suitable conditions (“shift in space”) and/or “persist in place” (Thurman et al. 2020, pp. 521-522).

Table 4.4.1 Representation (adaptive capacity) summary for thin-wall quillwort.

Representation Mechanism	Rank	REPRESENTATION
Dispersal	Moderate	Moderate
Evolutionary Potential	Moderate	
Phenotypic Plasticity	High	

4.5 Current Redundancy

Thin-wall quillwort redundancy is characterized by having multiple resilient AUs across the species’ known range. Considering these AUs occur within two ecoregions, spanning both the Inner and Outer Coastal Plain of North Carolina, as well as across three contiguous river basins (and four HUC-10 watersheds), this species may be able to withstand catastrophic events such as widespread storm-induced flooding or other disturbances; thus, we believe the redundancy for thin-wall quillwort is moderately high. 

4.6 Summary of Current Condition

 The viability of thin-wall quillwort is estimated to be stable across its narrow range. Our current condition analysis indicates that the 3Rs—resiliency, representation, and redundancy—are sufficient to sustain the species. Currently, most analysis units are moderately (43%) and highly (28.5%) resilient—save two low resiliency AUs (28.5%)—and are distributed throughout the species’ known range such that the species is expected to withstand stochastic and catastrophic events. Over two-thirds of all known plants (3,228 of 3,882) occur at the two largest AUs, with the largest (1,828) occurring on lands owned by The Nature Conservancy, which is protected from future development and, therefore, is expected to remain highly resilient. Further, the species possesses traits that lend themselves to moderate adaptive capacity. And while a narrow range endemic, it occurs across a moderate expanse of the North Carolina Coastal Plain in two ecoregions and experts indicate that with increased sampling more populations are likely to be discovered.

CHAPTER 5: FUTURE CONDITIONS

In previous chapters, we considered thin-wall quillwort life history characteristics, reviewed the factors that may be driving historical, current, and future conditions of the species, identified the habitat and demographic requisites needed for viability, and estimated the current condition of those needs through the lens of the 3Rs (Chapters 2-4). In this chapter, we project future habitat and AU conditions across thin-wall quillwort’s range based on a range of plausible future scenarios.

5.1 Future Scenario Methods

As outlined in Chapter 3, the primary factors influencing the viability of thin-wall quillwort appear to include significant changes in hydrology, diminished water quality, and habitat modification and loss resulting from recreation and development. We focused our future conditions analysis on how the species may be impacted by changes in available habitat, water quality, drought, and severe storms in the future. Although we are unable to project certain influences into the future, we have identified appropriate surrogate variables to represent change in intensity and magnitude.

We evaluated future conditions under two plausible scenarios: scenario A, a lower impacts scenario, and scenario B, a higher impacts scenario. A summary of all metrics, datasets, and scenario inputs is summarized in Table 5.1. Detailed methods for each scenario are described in the following sections. We present future condition results for 2050, 2070, and 2100 under these two scenarios.

Table 5.1 Summary of future condition analysis, including influences, datasets, metrics, measurement scales, and model output scenarios selected for scenarios A and B.

Influence	Dataset	Metric	Scale	Scenario A Lower Impact	Scenario B Higher Impact
Habitat Loss	FUTure Urban-Regional Environment Simulation (FUTURES)	Percent of area converted to development	AU	Low	High
	NOAA SLR	Area inundated	AU	Intermediate	High
Water Quality	Integrated Climate and Land Use (ICLUS)	Percent of area converted to unsuitable land use	HUC-12	SSP-2	SSP-5
	NOAA SLR	Area inundated	AU	Intermediate	High
Drought	MACAv2METDATA	Standard Precipitation Evapotranspiration Index (SPEI)	Range-wide	RCP4.5	RCP8.5
Hurricane Risk	Lopez et al. 2024	Projected hurricane frequency and intensity	Range-wide	No scenario	No scenario

5.2 Metrics and Methods for Estimating Future Conditions

5.2.1 Available Habitat

In order to predict future changes in immediate available habitat, we evaluated future habitat loss at the population scale using outputs from the U.S. Geological Survey (USGS) FUTure Urban-Regional Environment Simulation (FUTURES) model (Meentemeyer et al. 2013, entire). Specifically, we used the probabilistic projections of urbanization by decade from FUTURES version 2.0 (Petrasova et al. 2023, unpaginated). These projections provide probabilities of new development ranging from 0 to 1, with higher values indicating areas more likely to be developed. We used FUTURES projections from 2050, 2070, and 2100 under a low and high development scenario. We calculated the percentage of area within each AU projected to be developed under these two scenarios. We assume that development at the population scale immediately surrounding the known occurrence records would result in direct habitat loss and the potential loss of individual plants due to development activities and land cover conversion.

Across all analysis units, no change is anticipated in the low development scenario, and minor changes are anticipated in the high development scenario (Table 5.2.1.1).

Table 5.2.1.1 Percent increase of developed area within analysis units under low and high development scenarios at 2050, 2070, and 2100.

AU	Low Development			High Development		
	2050	2070	2100	2050	2070	2100
1	0	0	0	0.02	0.02	0.15
2	0	0	0	0.02	0.02	0.04
3	0	0	0	1.73	1.92	2
4	0	0	0	1.62	2.27	2.69
5	0	0	0	0.17	0.18	0.24
6	0	0	0	0.04	0.04	0.08
7	0	0	0	0.95	1	1.99

For AUs that may be affected by tidally influenced systems, we describe potential habitat loss due to inundation using the National Oceanographic and Atmospheric Administration (NOAA) sea level rise dataset (Sweet et al. 2022, entire; NOAA 2026, entire). We used the intermediate and high scenarios, which correspond to predictions from two plausible global emission scenarios, SSP2-RCP4.5 and SSP5-RCP8.5 (Sweet et al. 2022, p. 27). Local predictions for these two scenarios are available from NOAA, which provides estimates of sea level rise for stations at decadal time steps in the future to year 2100 (Sweet et al. 2022, entire; NOAA 2026, entire). The amount of sea level rise projected under each scenario in 2050, 2070, and 2100 was calculated using the local scenario projections from local tidal gauge ID 1005602820 (Sweet et al. 2022, entire). Due to relative uncertainty regarding the rate of inundation and the species' ability to expand to newly inundated areas, the risk of habitat loss due to inundation is unclear. We describe it due to this uncertainty but do not consider it a major risk to the species.

Only three AUs are expected to experience some degree of inundation in the future (Table 5.2.1.2). Analysis units 1 and 2 located in the Lower Black River watershed in the Cape Fear River basin are projected to undergo minor inundation under the high scenario out to 2100. Analysis unit 7 may experience minor inundation under the high scenario by 2100 with less than 35 acres of the delineated population area inundated.

Table 5.2.1.2 Number of acres inundated within each analysis unit at 2050, 2070, and 2100 under intermediate (RCP 4.5) and high (RCP 8.5) scenarios.

AU	Intermediate (RCP4.5)			High (RCP8.5)		
	2050	2070	2100	2050	2070	2100
1	57	57	57	57	57	57
2	35	35	35	35	35	35
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0

6	0	0	0	0	0	0
7	0	0	0	0	0	35

5.2.2 Water Quality

To evaluate deviations in water quality, we evaluated land cover change within each HUC-12 watershed as well as at the AU scale. In current condition we used water quality data as well as the number of animal facilities within each watershed. However, because we are unable to project changes in the number of animal operations in the future and assume that the impacts and potential risks from these operations continue forward, we used land cover change and conversion as a surrogate variable to consider additional changes to future water quality. Future land use and land cover change was measured using the U.S. EPA Integrated Climate and Land-Use Scenarios (ICLUS) dataset. The ICLUS model projects scenarios of housing density and derived impervious surface cover for the United States for each decade up to 2100 (USEPA 2009, pp. 1-1–1-2). ICLUS scenarios use a combination of SSP (Shared Socioeconomic Pathways) and RCP (Representative Concentration Pathways) scenarios. The SSPs incorporate major components of projected development, including social, economic, and demographic information (USEPA 2009, p. 1-1). The RCPs outline pathways for emissions out to 2100 (USEPA 2017, p. xv).

For this analysis, we used the ICLUS version 2 dataset, which includes updated projections for population and land use (USEPA 2017, entire). For scenario A, the lower plausible limit scenario, we used the SSP2-RCP4.5 scenario, which represents a moderate or “middle-of-the-road” perspective compared to the other SSPs and projects up to a 4° C increase at the end of the century (Fricko et al. 2017, entire; Riahi et al. 2017, entire). For scenario B, the upper limit plausible scenario, we used the SSP5-RCP8.5 scenario, a more fossil fuel-based development scenario that represents the upper end of the scenario literature in fossil fuel use, food demand, energy use, and greenhouse gas emissions (Kriegler et al. 2017, entire; Riahi et al. 2017, entire).

Development under the ICLUS model represents residential land uses, which are the major types of development and intensification of land use related to urbanization in the United States. The model also incorporates commercial, residential, and industrial land use, as well as other land cover types. To evaluate watershed scale impacts to water quality, we evaluated the amount of area within each HUC-12 that would be converted to unsuitable land cover in the future using the categories designated in the ICLUS model, including mining, suburban, urban low, urban high, commercial, industrial, institutional, and transportation land cover classes. We assume the higher human density and impervious surface associated with these land cover classes will affect water quality in the future within the HUC-12 watershed.

Across the species range, the amount of land conversion to land cover classes that may affect water quality is relatively low (Table 5.2.2). The greatest increase is anticipated in populations 3 and 4 under the SSP5 scenario, which are located in the same HUC-12 watershed.

Table 5.2.2 Percentage change of unsuitable land cover within each HUC-12 watershed at low (SSP2) and high (SSP5) development scenarios at 2050, 2070, and 2100. Unsuitable land cover types include mining, suburban, urban low, urban high, commercial, industrial, institutional, and transportation cover classes.

AU	Low Development (SSP2)			High Development (SSP5)		
	2050	2070	2100	2050	2070	2100

1	-0.3	-0.3	-0.3	-0.86	-1.07	-1.23
2	0	0	0	0.02	-0.2	-1.12
3	1.87	2.89	3.8	4.97	8.15	13.43
4	1.87	2.89	3.8	4.97	8.15	13.43
5	0.48	1.19	2.67	-0.01	1.13	2.03
6	0.16	0.56	1.27	2.19	3.25	4.2
7	0	-0.33	-0.23	0.16	0.23	1.5

To assess water quality changes associated with salinity, we used the NOAA sea level rise dataset as described in Section 5.2.1 (Sweet et al. 2022, entire; NOAA 2025, entire). Analysis units 1 and 2 are the only AUs likely influenced by saltwater inundation in the future; however, the magnitude and intensity of inundation are not expected to increase under either scenario out to 2100. Although this area may experience inundation, the associated salinity levels and species' tolerance are unknown. Given the low magnitude of change, the likelihood of significant change in salinity affecting the species is unlikely to occur given the low rate of change anticipated in the future.

5.2.3 Drought and Flooding

We use the Standardized Precipitation-Evapotranspiration Index (SPEI) to assess exposure to extended meteorological drought conditions into the future. To measure SPEI, we used the U.C. Merced Climate Toolbox, which uses outputs from the Multivariate Adaptive Constructed Analogs MACAv2-METDATA of the Coupled Model Intercomparison Project (CMIP5), a collection of 20 downscaled global climate models (Hegewisch and Abatzoglou 2026, unpaginated). SPEI is considered one of the more robust metrics for characterizing future drought conditions under climate change. This metric summarizes the departure of the climatic water balance from normal conditions based on both temperature and precipitation, and accounts for the impact of potential evapotranspiration. We used the U.S. Drought Monitor (USDM) standard to translate SPEI values into drought classifications (Table 5.2.3.1; USDM 2026b, unpaginated).

Table 5.2.3.1 U.S. Drought Monitor drought categories associated with SPEI values (USDM 2026b, unpaginated).

Category	Description	% Range	SPEI Values
None	Normal or Wet Conditions	30.01 +	-0.49 +
D0	Abnormally Dry	20.01 to 30.0	-0.5 to -0.79
D1	Moderate Drought	10.01 to 20.0	-0.8 to -1.29
D2	Severe Drought	5.01 to 10.0	-1.3 to -1.59
D3	Extreme Drought	2.01 to 5.0	-1.6 to -1.99
D4	Exceptional Drought	0.0 to 2.0	≤ -2.0

We selected SSP2-RCP4.5 for the lower-bounds scenario, representing an intermediate greenhouse gas emissions scenario with socioeconomic factors following historical trends, and SSP5-RCP8.5 as the upper-bounds scenario, representing a very high greenhouse gas emissions scenario with no additional climate policy (IPCC 2021, p. 54). To account for variation in projections across generalized circulation models (GCMs), we used a mean model estimate to compare change across time. We projected this

metric continuously from 1950 to 2100 under the RCP4.5 and RCP8.5 scenarios across 12-month annual estimates. We compared seasonal outputs to assess potential changes in magnitude; however, differences across all seasons were minimal.

Under both scenarios, the projected SPEI shows minor changes in future drought risk until 2100 (Figure 5.2.3.1; Table 5.2.3.1). Using the U.S. Drought Monitor values to classify drought intensity, most scenarios and timesteps remain under normal or wet conditions. The RCP8.5 scenario at 2100 of -0.5 is considered “abnormally dry” by these standards, but it does not elevate to a drought category.

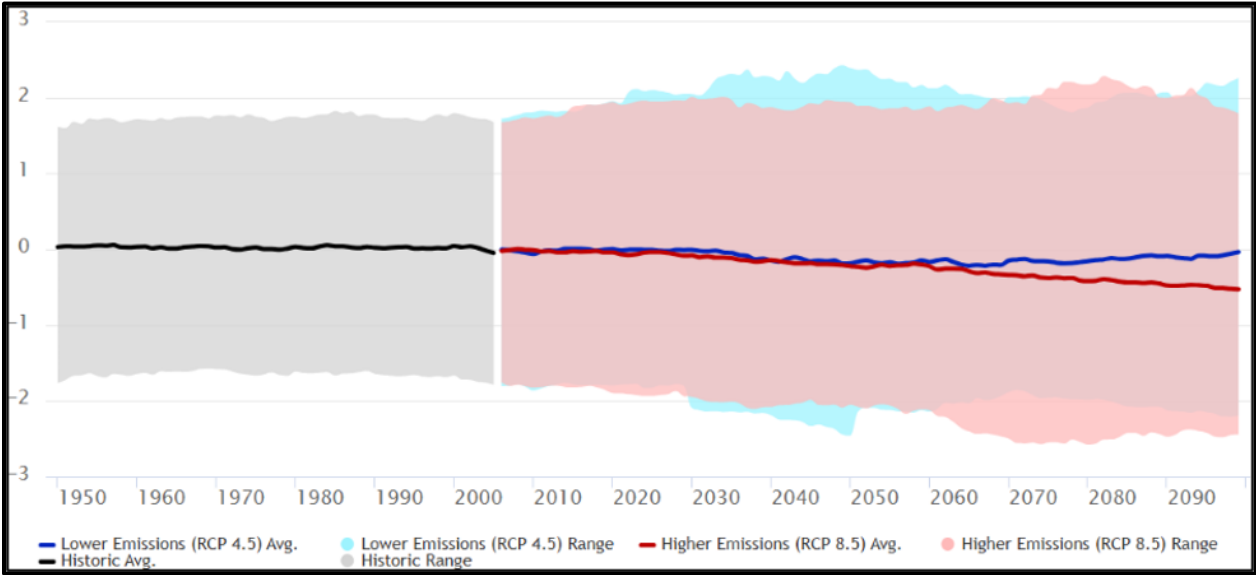


Figure 5.2.3.1 12-month standardized potential evapotranspiration index (SPEI) within the range of thin-wall quillwort between 1950-2100. The black line represents the historical average, while the grey shading represents the 20-model range of estimates during the historical period. The dark blue line represents the model averaged estimate under the RCP4.5 scenario. The light blue shading represents the range of model outputs under the RCP4.5 scenario. The red line represents the model averaged estimate of the RCP8.5 scenario. The pink shading represents the range of model outputs under the RCP8.5 scenario.

Table 5.2.3.2 Twenty global climate model-averaged estimates of SPEI under RCP4.5 and RCP8.5 scenarios at 2050, 2070, and 2100.

	RCP4.5			RCP8.5		
	2050	2070	2100	2050	2070	2100
SPEI	-0.2	-0.1	0	-0.2	-0.3	-0.5

To evaluate risk of future floods, we focus on the frequency and intensity of future hurricane events using results from a study investigating projections of tropical cyclones. In the North Atlantic, models predict a significant rise in interannual variability in tropical cyclone frequency: by mid-century, the variance in accumulated cyclone energy is expected to increase by approximately 36%, signaling both more extreme active seasons and more extreme inactive ones (Lopez et al. 2024, p. 2). This heightened variability stems from greater fluctuations in vertical wind shear and atmospheric stability, linked to ocean temperature changes between the Pacific and Atlantic (Lopez et al. 2024, pp. 6-9). Consequently, future hurricane seasons will likely exhibit less predictability and more extreme swings, with a growing



frequency of very strong individual storms driving the increase in accumulated cyclone energy (Lopez et al. 2024, p. 4).

5.3 Future Resiliency Summary

Under both scenarios in 2050 and 2070 we do not anticipate any changes to resiliency within any analysis units from current condition (Table 5.3). Under both scenarios by 2100, the majority of AUs are not expected to experience changes in resiliency; however, given the threats on the landscape, one AU (AU 7) may possibly be extirpated (Figure 5.3). Additional details on projected influences and species response are provided in detail for each analysis below.

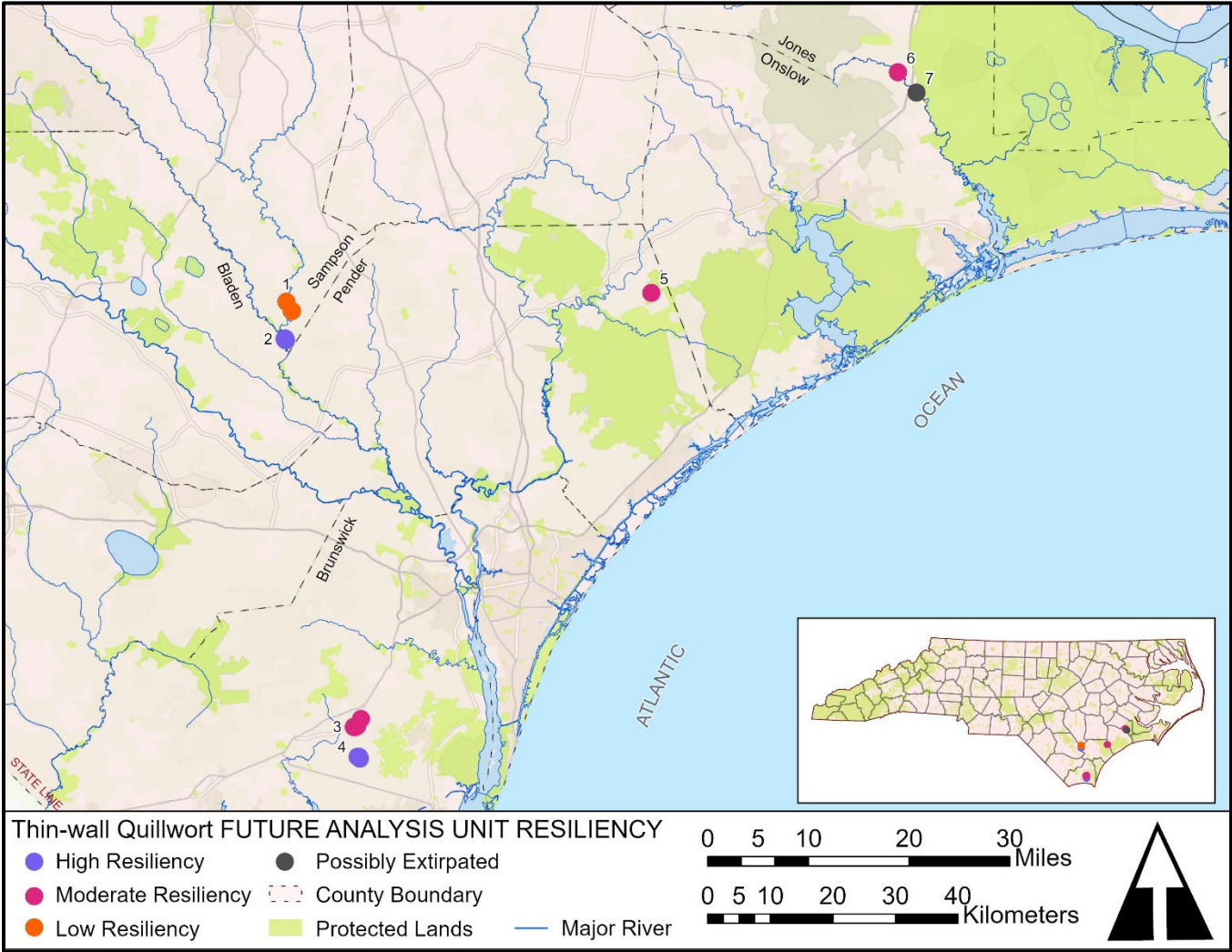


Figure 5.3 Map of projected thin-wall quillwort future analysis unit resiliency at 2100.

Table 5.3 Summary of analysis unit resiliency for current and future timesteps under Scenario A (lower impacts) and Scenario B (higher impacts) in 2100.

AU	Current Resiliency	Local Threats	FUTURE RESILIENCY					
			SCENARIO A: LOWER IMPACTS			SCENARIO B: HIGHER IMPACTS		
			2050	2070	2100	2050	2070	2100
1	Low	Sand quarry MAY▲	Low	Low	Low	Low	Low	Low

2	High	Invasive plant MAY ▲	High	High	High	High	High	High
3	Moderate	Residential footprint MAY ▲	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate
4	High	Residential footprint MAY ▲	High	High	High	High	High	High
5	Moderate	NA	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate
6	Moderate	NA	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate
7	Low	Recreation impacts LIKELY ▲	Low	Low	Possibly Extirpated	Low	Low	Possibly Extirpated

Analysis Units 1 and 2

As stated in Section 4.3.4, AUs 1 and 2 are both located in the Lower Black River (HUC-10) watershed within the Cape Fear River Basin, with AU 2 occurring approximately 8 km downstream from population 1.

For both analysis units, there is little projected land cover change at either the HUC-12 or AU-scale indicating substantial development in either scenario. However, water quality risk due to animal operations (both swine and poultry) will likely persist into the future and thus the likelihood of events associated with flood risk in these operations are also likely to increase over time (i.e., more time = more exposure/opportunities for events to transpire). Additionally, a slight risk of inundation (< 1% for both populations) due to sea level rise may somewhat reduce available habitat and decrease water quality due to augmented salinity. The amount of area inundated is not projected to increase over time, indicating this area is likely tidally influenced and is therefore subject to periodic inundation.

For AU 1, while the southern portion is located within a TNC conservation easement, another segment is situated next to an active sand quarry, which may negatively affect extant individuals via future expansion. Analysis unit 2 currently sustains a moderate quantity of exotic invasive alligator-weed, which may conceivably multiply and spread via increased eutrophication from animal operations and storm events (Pfungsten et al. 2024, entire).

Despite current persistence of thin-wall quillwort at these AUs, its eutrophication and salinity tolerance as well as its response to competitive plant interactions remain unknown, thus we cannot definitively determine what effect these future factors may have. Nonetheless, thin-wall quillwort has continued to persist despite documented flooding events leading to sewage overflow from CAFOs in these areas and presence of invasive species. Therefore, although there is a future risk of these events, we are uncertain that these events would result in meaningful impacts to population viability.

Given the likelihood of ongoing water quality issues, as well as possible effects of salinity and adjacent quarry operation, we maintain an overall LOW resiliency for AU 1. Due to CAFO influence on water quality, as well as possible impacts of plant invasion and negligible salinity inputs, we maintain an overall HIGH resiliency for AU 2. If these influences and subsequent species response result in population-scale impacts, future resiliency could be reduced to MODERATE, but we are uncertain of the species response to invasives and water quality changes.

Analysis Units 3 and 4

Analysis units 3 and 4 occur on private lands in a single HUC-12 sub-watershed (Middle Swamp) separated by a distance of ~6.8 km. Both AUs are projected to be at their highest risk of development under the high scenario in 2070 and 2100 due to current proximity of residential zones, whose expansion may increase encroachment on thin-wall quillwort habitat. Such development would also likely affect water quality due to increased impervious surfaces, although not significantly. While resiliency may be weakened as a result, the magnitude of this decline is not anticipated to generate a substantial reduction in overall resiliency.

Even with the prospect of continued residential development and related impacts to water quality, although at a negligible scale, we maintain a MODERATE resiliency for AU 3 and HIGH resiliency for AU 4.

Analysis Unit 5

Analysis unit 5 is in the Headwaters Sandy Run Swamp sub-watershed (HUC-12) within the Angola Swamp-Northeast Cape Fear watershed and garners some indirect protection from its occurrence on state lands. This AU is projected to sustain only meager changes in land cover through 2100 under both low and high scenarios (1 to 3% at the HUC-12 level) and is not predicted to be impacted by sea level rise or otherwise faces local threats that can be projected into the future.

Given the lack of future change captured in our future conditions analysis, AU 5 will retain a MODERATE overall resiliency.

Analysis Units 6 and 7

Analysis units 6 and 7 are both located in the Headwaters White Oak River HUC-10 watershed within the White Oak River Basin, with AU 7 situated approximately 6 km downstream from AU 6.

Analysis unit 6 is projected to sustain no change under all low development scenarios to 2100 and very little land cover change under high development scenarios (1 to 4%). Further, there are no projected impacts of sea level rise on AU 6. Analysis unit 7 is similarly projected to sustain negligible land cover change through 2100 (1-2%) and while it is projected to have a lower percentage of unsuitable land cover at 2050-2100, its occurrence on a recreational shoreline where trampling has already reduced population size suggests its future is not bright. Furthermore, it is projected to experience some magnitude (albeit miniscule) of sea level rise impacts at 2100.

Given the lack of substantial future change captured in our analysis, we determine that AU 6 will retain MODERATE resiliency. Conversely, because AU 7 remains subject to active recreational use via trampling, it will likely continue to decline and is at risk of local extirpation.

5.4 Future Representation

As noted in Section 4.4, thin-wall quillwort exhibits varying degrees of adaptive capacity across its range of known/investigated attributes (Table 4.4.1; Appendix F). Some of these attributes more appreciably lend themselves to prospective future changes in habitat and climate. For example, thin-wall quillwort's ability to enter dormancy during prolonged dry periods indicates that the species may be resilient to harsh future climate. Similarly, its ostensible ability to photosynthesize and respire during periods of prolonged inundation may put it at a physiological advantage during climate fluxes or following severe floods. Regarding future range dynamics, even as a rare occurrence, long distance dispersal events could have a significant effect on maintaining connectivity, establishing new populations, or preserving thin-wall quillwort lineages following local extinction (D. Wickell 2026, pers. comm.). Since there is only low to moderate habitat and land cover change predicted across the species'

range, the species' future representation, while naturally low as a narrow endemic, is expected to remain comparable to its current moderate levels. Nonetheless, it remains uncertain how thin-wall quillwort may respond to future encroachment by alligator-weed and/or other exotic invasive plant species that occupy the same or adjacent habitats.

5.5 Future Redundancy

In the future, five moderately to highly resilient thin-wall quillwort AUs are expected to be distributed across the species' known range. Drought risk is low across all scenarios and timesteps range-wide. Hurricane risk (intensity) persists into the future, yet the species has persisted despite powerful storm events, and connectivity between most AUs is greatly intact, and thus the ability for demographic rescue for many AUs remains possible. Overall, the number and distribution of resilient AUs in the future are likely to continue to enable the species to withstand and/or recover from catastrophic events.

5.6 Summary of Future Conditions

For our future conditions analyses, we measured how changes in available habitat, water quality, drought, and storm events may affect thin-wall quillwort in the future. Our overall estimate of the species' future viability is moderate to high across most of its geographic range. Future development is unlikely to result in declines and is therefore not anticipated to affect species' viability through 2100. While climate change, as manifested through a composite drought metric (that includes temperature, precipitation, and evapotranspiration), is not projected to substantially affect the species out to 2100, we do not know if the species will respond negatively to the projected increase in storm intensity or to an expected increasing abundance of invasive vegetation. While recreational use may impact a single analysis unit via trampling and erosion, our analyses suggest that thin-wall quillwort as a species is expected to maintain viability through the end of the century.

REFERENCES

- Abeli, T., A. Bacchetta, F. Basilico, P. Berardo, A. Colli, G. Colonna, A.B. Pastor, A. Ruzzoli, M. Tucci, G.A. Vesconi, D. Vitaloni, and M. Tarascio. 2026. Megaspore germination and sporophyte emergence requirements in the critically endangered endemic quillwort *Isoetes malinverniana*. *Aquatic Botany* 203(103965):1-6.
- Alejo-Jacuinde, G., and L. Herrera-Estrella. 2022. Exploring the high variability of vegetative desiccation tolerance in pteridophytes. *Plants* 11(1222):1-22.
- Ali, J., T. Wahl, J. Morim, A. Enriquez, M. Gall, and C.T. Emrich. 2025. Multivariate compound events drive historical floods and associated losses along the U.S. East and Gulf coasts. *Npj Natural Hazards* 2(19):1-10.
- Arrigo, N., and M.S. Barker. 2012. Rarely successful polyploids and their legacy in plant genomes. *Current Opinion in Plant Biology* 15:1-7.
- Arts, G.H.P. 2002. Deterioration of Atlantic soft water macrophyte communities by acidification, eutrophication and alkalization. *Aquatic Botany* 73:373-393.
- Aucott, W.R. 1996. Regional Aquifer-System Analysis-Southeastern Coastal Plain: Hydrology of the Southeastern Coastal Plain Aquifer System in South Carolina and Parts of Georgia and North Carolina. U.S. Geological Survey Professional Paper 1410-E, Denver, CO. 93 pp.
- Azzella, M.M., A.D. Vecchia, T. Abeli, J. Alahuhta, V.B. Amoroso, E. Ballesteros, V. Bertrin, D. Brunton, A.A. Bobrov, C. Caldeira, S. Ceschin, E.V. Chemeris, M. Čtvrtlíková, M. de Winton, E. Gacia, O.G. Grishutkin, D. Hofstra, D. Ivanova, M.O. Ivanova, N.K. Konotop, D.M. Larson, S. Magrini, M. Mjelde, O.A. Mochalova, G. Oliveira, O. Pedersen, J.B. de S. Pereira, C. Ribaud, M. Inmaculada Romero Buján, A. Troia, Y.S. Vinogradova, P.A. Volkova, D. Basílio Zandonadi, N.V. Zueva, and R. Bolpagni. 2024. Global assessment of aquatic *Isoetes* species ecology. *Freshwater Biology* 69:1420-1437.
- Babin, C.H., and C.D. Bell. 2023. The effects of climate change on cytotype distributions of endemic genera in the North American Coastal Plain. *Plant Ecology and Diversity* 16(1-2):45-59.
- Bagella, S., and M.C. Caria. 2013. Sensitivity of ephemeral wetland swards with *Isoetes histrix* Bory to environmental variables: implications for the conservation of Mediterranean temporary ponds. *Aquatic Conservation: Marine and Freshwater Ecosystems* 23:277-290.
- Barnard, P.L., K.M. Befus, J.J. Danielson, A.C. Engelstad, L.H. Erikson, A.C. Foxgrover, M.K. Hayden, D.J. Hoover, T.W.B. Leijnse, C. Massey, R. McCall, N.C. Nadal-Caraballo, K. Nederhoff, A.C. O'Neill, K.A. Parker, M. Shirzaei, L.O. Ohenhen, P.W. Swarzenski, J.A. Thomas, M. van Ormondt, S. Vitousek, K. Vos, N.J. Wood, J.M. Jones, and J.L. Jones. 2024. Projections of multiple climate-related coastal hazards for the US Southeast Atlantic. *Nature Climate Change* 15:101-109.
- Barni, E., C. Minuzzo, F. Gatto, M. Lonati, T. Abeli, C. Amosso, G. Rossi, and C. Siniscalco. 2013. Estimating influence of environmental quality and management of channels on survival of a threatened endemic quillwort. *Aquatic Botany* 107:1-17.
- Baskin, J.M., and C.C. Baskin. 1978. Geographical distribution of *Isoetes butleri* in the southeastern United States. *American Fern Journal* 68(1):7-8.
- Baskin, J.M., and C.C. Baskin. 1979. The role of temperature in the vegetative life cycle of *Isoetes butleri*. *American Fern Journal* 69(4):103-110.

- Bason, C.W., D.E. Kroes, and M.M. Brinson. 2017. The effect of beaver ponds on water quality in rural Coastal Plain streams. *Southeastern Naturalist* 16(4):584-602.
- Bauder, E.T. 2000. Inundation effects on small-scale plant distributions in San Diego, California vernal pools. *Aquatic Ecology* 34:43-61.
- Beever, E.A., J. O'Leary, C. Mengelt, J.M. West, S. Julius, N. Green, D. Magness, L. Petes, B. Stein, A.B. Nicotra, and J.J. Hellmann. 2016. Improving conservation outcomes with a new paradigm for understanding species' fundamental and realized adaptive capacity. *Conservation Letters* 9:131-137.
- Belda, M., E. Holtanova, T. Halenka, and J. Kalvova. 2014. Climate classification revisited: from Köppen to Trewartha. *Climate Research* 59:1-13.
- Bell, M.B., and G.M. Lackmann. 2025. Inland tropical cyclone impacts in a warming climate: semi-idealized simulations of Hurricane Fran (1996 and 2096). *Journal of Climate* 38:6821-6836.
- Benedetti, M.M., M.J. Raber, M.S. Smith, and L.A. Leonard. 2006. Mineralogical indicators of alluvial sediment sources in the Cape Fear River Basin, North Carolina. *Physical Geography* 27(3):258-281.
- Benke, A.C., I. Chaubey, G.M. Ward, and E.L. Dunn. 2000. Flood pulse dynamics of an unregulated river floodplain in the southeastern U.S. Coastal Plain. *Ecology* 81(10):2730-2741.
- Bernal, S., J.L.J. Ledesma, X. Peñarroya, C. Jativa, N. Catalán, E.O. Casamayor, A. Lupon, R. Marcé, E. Martí, X. Triado-Margarit, and G. Rocher-Ros. 2025. Expanding towards contraction: the alternation of floods and droughts as a fundamental component in river ecology. *Biogeochemistry* 168(11):1-22.
- Bjorn, L.O., B.A. Middleton, M. Germ, and A. Gaberšcik. 2022. Ventilation systems in wetland plant species. *Diversity* 14(517):1-21.
- Blackmore, S., M. Takahashi, K. Uehara, and A.H. Wortley. 2012. Development of megaspores and microspores in *Isoetes japonica* A. Br. (Lycopodiophyta: Isoëtaceae). *Grana* 51(2):84-96.
- Blocksom, K., J. Flotemersch, H. Ferriby, B. Rabon, and D. Chestnut. 2025. Intra-annual and ecoregional differences in physico-chemical data between blackwater and non-blackwater streams in the coastal plains of South Carolina. *Frontiers in Water* 7(1540456):1-15.
- Bolin, J. 2026. Thin-wall quillwort questionnaire response. [Accessed March 18, 2026].
- Bolpagni, R., S. Magrini, A. Coppi, A. Troia, J. Alahuhta, M. Mjelde, and M.M. Azzella. 2021. *Isoetes sabatina* (Isoëtaceae, Lycopodiopsida): taxonomic distinctness and preliminary ecological insights. *Aquatic Conservation: Marine and Freshwater Ecosystems* 31(10):1-7.
- Bornette, G., and S. Puijalon. 2011. Response of aquatic plants to abiotic factors: a review. *Aquatic Science* 73:1-14.
- Boston, H.L., M.S. Adams, and T.P. Pienkowski. 1987. Utilization of sediment CO₂ by selected North American isoetids. *Annals of Botany* 60(5):485-494.
- Bray, R.D., P.W. Schafran and L.J. Musselman. 2018. Interesting, provocative, and enigmatic: morphological observations on southeastern quillworts (*Isoetes*, Isoëtaceae, Lycopodophyta). *Castanea* 83(2):263-269.
- Bren, L.J., T.M. Williams, and D.M. Amatya. 2025. Pp. 124-142. In L.J. Bren et al. (eds.), *Forest Hydrology: Processes, Management, and Assessment*, 2nd Edition. CAB International, Oxfordshire, UK and Boston, MA. 356 pp.

- Brinson, M.M. 1993. Changes in the functioning of wetlands along environmental gradients. *Wetlands* 13(2):65-74.
- Brizova, E. 2011. Quillwort (*Isoëtes*), a mysterious plant from the Czech Republic. *Acta Musei Nationalis Pragae* 67(1-2):25-34.
- Brock, J.M.R. 2025. Effective dispersal of fern spore and the ecological relevance of zoochory. *Biological Reviews* 100:2116-2130.
- Brown, C.N., M.A. Mallin, and A.N. Loh. 2020. Tracing nutrient pollution from industrialized animal production in a large coastal watershed. *Environmental Monitoring and Assessment* 192(515):1-17.
- Brunton, D.F. 2001. Quillwort dispersal; which way is the wind blowing? *Chinquapin* 9(3):20.
- Brunton, D.F. 2015. Key to the quillworts ("*Isoëtes*": Isoëtaceae) of the southeastern United States. *American Fern Journal* 105(2):86-100.
- Brunton, D.F. 2021. Email communication between D. Brunton (*Isoëtes* Researcher, Ontario, CAN) and J. Schaner (NCNHP Botanist, Raleigh, NC). [Accessed March 24, 2026].
- Brunton, D.F., D.M. Britton, and W.C. Taylor. 1994. *Isoëtes hyemalis*, sp. nov. (Isoëtaceae): a new quillwort from the southeastern United States. *Castanea* 59(1):12-21.
- Brunton, D.F., and D.M. Britton. 1996. The status, distribution, and identification of Georgia Quillwort (*Isoëtes georgiana*: Isoëtaceae). *American Fern Society* 86(4):105-113.
- Brunton, D.F., and D.M. Britton. 1998. *Isoëtes microvela* (Isoëtaceae), a new quillwort from the coastal plain of the southeastern United States. *Rhodora* 100:261-275.
- Brunton, D.F., P.C. Sokoloff, and P. Aymerich. 2020. The taxonomy, status, and origin of *Isoëtes xbrochonii* and *I. creussensis* (Isoëtaceae), two Pyrenean endemic taxa. *Botany Letters* 167(4):391-408.
- Brunton, D.F., J.T. Kartesz, M. Nishino, and P.C. Sokoloff. 2025. A Conspectus of the North American Isoëtaceae. BRIT Press, Fort Worth, TX. 112 pp.
- Brunton, D.F., and P.C. Sokoloff. 2020. A review of North American *Isoëtes engelmannii* (Isoëtaceae) complex hybrids, including the description of *I. × fernaldii*, hyb. nov. and *I. × karenae*, hyb. nov. *Webbia: Journal of Plant Taxonomy and Geography* 75(1):35-50.
- Brunton, D.F., and A. Troia. 2018. Global review of recent taxonomic research into *Isoëtes* (Isoëtaceae), with implications for biogeography and conservation. *The Fern Gazette* 20(8):309-333.
- Burandt, P., M. Grzybowski, K. Glińska-Lewczuk, W. Gotkiewicz, M. Szymańska-Walkiewicz, and K. Obolewski. 2023. Hydrology as a determinant of riparian habitat structure in lowland river floodplains. *Water* 16(164):1-17.
- Burke, M.K., S.L. King, D. Gartner, and M.H. Eisenbies. 2003. Vegetation, soil, and flooding relationships in a blackwater floodplain forest. *Wetlands* 23(4):988-1002.
- Burkholder, J.M., M.A. Mallin, H.B. Glasgow, Jr., L.M. Larsen, M.R. McIver, G.C. Shank, N. Deamer-Melia, D.S. Briley, J. Springer, B.W. Touchette, and E.K. Hannon. 1997. Impacts to a coastal river and estuary from rupture of a large swine waste holding lagoon. *Journal of Environmental Quality: Environmental Issues* 26:1451-1466.
- Caldeira, C.F., A.V.S. Lopes, K.C. Aquiar, A.L. Ferreira, J.V.S. Araujo, V.M.S. Gomes, D.B. Zandonadi, C.B. Abranches, S.J. Ramos, M. Gastauer, N.V. Campos, L.M.S. Gestinari, L.A. Prado,

- F.M.G. Santos, R.L. Martins, F.A. Esteves, G. Oliveira, and M.P. Santos. 2021. Distinct reproductive strategy of two endemic Amazonian quillworts. *Diversity* 13(348):1-14.
- Campbell, D.H. 1891. Contributions to the life-history of *Isoëtes*. *Annals of Botany* 5(19):231-258.
- Caplen, C.A., and C.R. Werth. 2000. Isozymes of the *Isoëtes riparia* complex, I. Genetic variation and relatedness of diploid species. *Systematic Botany* 25(2):235-259.
- Casanova, M.T., and M.A. Brock. 2000. How do depth, duration, and frequency of flooding influence the establishment of wetland communities? *Plant Ecology* 147(2):237-250.
- Catford, J.A., and R. Jansson. 2014. Drowned, buried, and carried away: effects of plant traits on the distribution of native and alien species in riparian ecosystems. *New Phytologist* 204:19-36.
- Cavalheiro-Filho, S.L., L.M.S. Gestinari, T.U.P. Konno, M.P. Santos, E.N. Calderon, M.C.H. Marques, F.M.G. dos Santos, A. Castilho, R.L. Martins, F.A. Esteves, and N.V. Campos. 2021. Morphological plasticity in the endemic *Isoëtes* species from Serra dos Carajás, Amazonia, Brazil. *American Fern Journal* 111(3):174-195.
- Cavalheiro-Filho, S.L., N.V. Campos, C.F. Caldeira, J.B.S. Pereira, T.U.P. Konno, S.D. Matos, A.B. Cavalcante, F.A. Esteves, R.L. Martins, and L.M.S. Gestinari. 2023. Anatomical studies of Brazilian Amazonian *Isoëtes* species: inferences on habitat adaptation. *Rodriguésia* 74(e00562023):1-18.
- Cavalheiro-Filho, S.L., N.V. Campos, A.R. Gripp, L.B. Jordão, A.B. Cavalcante, F.A. Esteves, R.L. Martins, and L.M.S. Gestinari. 2025. A simple and effective method for the cryopreservation of quillwort spores. *Aquatic Conservation: Marine and Freshwater Ecosystems* 35(e70070):1-12.
- Center for Biological Diversity (CBD). 2010. *Isoëtes microvela* pages (pp. 628-629) from Petition to List 404 Aquatic, Riparian and Wetland Species from the Southeastern United States as Threatened or Endangered under the Endangered Species Act. https://www.biologicaldiversity.org/programs/biodiversity/1000_species/the_southeast_freshwater_extinction_crisis/pdfs/SEPetition.pdf [Accessed January 16, 2026].
- Chaffin, A.G., L.B. Cahoon, M.A. Mallin, C. Lane, and K. Mapes. 2025. Aquatic nitrogen source elucidation using stable isotope analysis in the North Carolina Coastal Plain. *Journal of Environmental Quality* 54:499-512.
- Chappuis, E., A. Lumbreras, E. Ballesteros, and E. Gacia. 2015. Deleterious interaction of light impairment and organic matter enrichment on *Isoëtes lacustris* (Lycopodiophyta, Isoëtales). *Hydrobiologia* 760:145-158.
- Chappuis, E., E. Ballesteros, and E. Gacia. 2016. Temporary emersion enhances amphibious *Isoëtes* production. *Limnetica* 35(2):373-384.
- Chaya, T.M., T.B. Schafer, and T.Z. Osborne. 2023. Assessing water quality dynamics in tidally influenced blackwater rivers along a rural-urban gradient. *Water* 15(4154):1-15.
- Chen, J.M., X. Liu, J.Y. Wang, R.W. Gituru, and Q.F. Wang. 2005. Genetic variation within the endangered quillwort *Isoëtes hypsophila* in China as evidenced by ISSR analysis. *Aquatic Botany* 82:89-98.
- Chen, Y.Y., D.R. Kong, C.H. Huang, Y.X. Xu, and Z.Z. Li. 2012. Microsatellite analysis reveals the genetic structure and gene flow of the aquatic quillwort *Isoëtes sinensis*, a critically endangered species in China. *Aquatic Botany* 96:52-57.

- Christensen, J.R., M.S. Nash, and A. Neale. 2013. Identifying riparian buffer effects on stream nitrogen in southeastern Coastal Plain watersheds. *Environmental Management* 52:1161-1176.
- Ciu, J., Y. Zhu, H. Du, Z. Liu, S. Shen, T. Wang, W. Cui, R. Zhang, S. Jiang, Y. Wu, X. Gu, H. Yu, and Z. Liang. 2023. Chromosome-level reference genome of tetraploid *Isoetes sinensis* provides insights into evolution and adaption of lycophytes. *GigaScience* 12:1-15.
- Colby, B.R. 1964. Scour and Fill in Sand-Bed Streams. Geological Survey Professional Paper 462-D, Washington, DC. 32 pp.
- Cooray, A., J. Wu, and J. Dorfman. 2025. Surface water impairment from storms and swine in North Carolina. Selected Presentation: 2025 AAEE and WAEA Joint Annual Meeting, Denver, CO. 37 pp.
- Cox, P.A., and R.J. Hickey. 1984. Convergent megaspore evolution and *Isoetes*. *The American Naturalist* 124(3):437-441.
- Craine, J.M., and R. Dybzinski. 2013. Mechanisms of plant competition for nutrients, water, and light. *Functional Ecology* 27(4):833-840.
- Crandall, K.A., O.R.P. Bininda-Emonds, G.M. Mace, and R.K. Wayne. 2000. Considering evolutionary processes in conservation biology. *Trends in Ecology and Evolution* 15:290-295.
- Čtvrtlíková, M., P. Znachor, J. Nedoma, and J. Vrba. 2012. Effects of temperature on the phenology of germination of *Isoetes echinospora*. *Preslia* 84:141-153.
- Čtvrtlíková, M., P. Znachor, and J. Verba. 2014. The effect of temperature on the phenology of germination of *Isoetes lacustris*. *Preslia* 86:279-292.
- Čtvrtlíková, M., J. Kopáček, J. Nedoma, P. Znachor, and J. Vrba. 2020. Only the adults survive- a long-term resistance of *Isoetes lacustris* to acidity and aluminium toxicity stress in a Bohemian Forest lake. *Ecological Indicators* 111:1-11.
- Čtvrtlíková, M., J. Kopáček, J. Nedoma, P. Znachor, P. Hekera, and J. Vrba. 2023. Aquatic quillworts, *Isoetes echinospora* and *I. lacustris* under acidic stress- a review from a temperate refuge. *Ecology and Evolution* 13(e9878):1-14.
- Cuffney, T.F. 1988. Input, movement, and exchange of organic matter within a subtropical coastal blackwater river-floodplain system. *Freshwater Biology* 19:305-320.
- Culbertson, D.M., L.E. Young, and J.C. Brice. 1967. Scour and Fill in Alluvial Channels, With Particular Reference to Bridge Sites. U.S. Geological Survey Open-File Report, Washington, DC. 58 pp.
- Dalapicolla, J., R. Alves, R. Jaffe, S. Vasconcelos, E. Soares Pires, G. Lopes Nunes, J. Bernardino de Souza Pereira, J. Tasso F. Guimarães, Ma. C. Dias, T. Nogueira Fernandes, D. Scherer, F. Marino Gomes dos Santos, A. Castilho, M. Pupo Santos, E. Nicolas Calderón, R. Lemes Martins, R. Nunes da Fonseca, Francisco de Assis Esteves, C. Frois Caldeira, and G. Oliveira. 2021. Conservation implications of genetic structure in the narrowest endemic quillwort from the Eastern Amazon. *Ecology and Evolution* 11:10119-10132.
- Davies-Colley, R.J. and D.G. Smith. 2001. Turbidity, suspended sediment, and water clarity: a review. *Journal of the American Water Resources Association* 37(5):1085-1101.
- de Oliveira, P.C., J.M.D. Torezan, and C. Nunes da Cunha. 2015. Effects of flooding on the spatial distribution of soil seed and spore banks of native grasslands of the Pantanal wetland. *Acta Botanica Brasilica* 29(3):400-407.

- de Ruiter, M.C., A. Couasnon, M.J.C. van den Homberg, J.E. Daniell, J.C. Gill, and P.J. Ward. 2020. Why we can no longer ignore consecutive disasters. *Earth's Future* 8(e2019EF001425):1-19.
- Deb, M., N. Sun, Z. Yang, T. Wang, D. Judi, Z. Xiao, and M.S. Wigmosta. 2023. Interacting effects of watershed and coastal processes on the evolution of compound flooding during Hurricane Irene. *Earth's Future* 11(e2022EF002947):1-23.
- Deng, Y., H.M. Solo-Gabriele, M. Laas, L. Leonard, D.L. Childers, G. He, and V. Engel. 2010. Impacts of hurricanes on surface water flow within a wetland. *Journal of Hydrology* 392(3-4):1-33.
- Dewitz, J., and U.S. Geological Survey (USGS). 2021. National Land Cover Database (NLCD) 2019 Products (ver. 3.0, February 2024): U.S. Geological Survey data release, <https://doi.org/10.5066/P9KZCM54> [Accessed April 28, 2026].
- Engelmann, G. 1882. The genus *Isoëtes* in North America. *Transactions of the Academy of Science of St. Louis* 4(2):358-390.
- Environmental Working Group (EWG). 2022. Animal Facilities in North Carolina in 2022 data viewer. Environmental Working Group, Washington, DC. <https://ewg.maps.arcgis.com/apps/instant/interactivelegend/index.html?appid=2b4da8e5d2594be586650be071d4a759> [Accessed July 1, 2026].
- Fletcher, D.E., B.E. Lindell, A.H. Lindell, P.T. Stankus, N.D. Fletcher, J.V. McArthur, and J.C. Seaman. 2019. Basins, beaver ponds, and the storage and redistribution of trace elements in an industrially impacted coastal plain stream on the Savannah River Site, SC, USA. *Environment International* 133(105174):1-11.
- Flick, R.E., D.B. Chadwick, J. Briscoe, and K.C. Harper. 2012. "Flooding" versus "inundation." *Eos, Transactions American Geophysical Union* 93(38):365-366.
- Flotemersch, J.E. 2023. Conservation of blackwater rivers and streams of the coastal plains of United States: knowledge and research needs. *Ambio* 52:665-677.
- Flotemersch, J.E., K.A. Blocksom, A.T. Herlihy, P.R. Kaufmann, R.M. Mitchell, and D.V. Peck. 2024. Distribution and characteristics of blackwater rivers and streams of the contiguous United States. *Water Resources Research* 60(e2023WR035529):1-17.
- Freund, F.D. 2016. Characterizing quantitative variation in the glossopodia of three western North American *Isoëtes* species. *American Fern Journal* 106(2):87-115.
- Freund, F.D. 2022. The genus *Isoëtes* L., evolution, diversification, and population structure in a free-sporing heterosporous lycophyte. Doctoral dissertation, University of California-Berkeley, Berkeley, CA. 155 pp.
- Fricko, O., P. Havlik, J. Rogelj, Z. Klimont, M. Gusti, N. Johnson, P. Kolp, M. Strubegger, H. Valin, M. Amann, and T. Ermolieva. 2017. The marker quantification of the Shared Socioeconomic Pathway 2: a middle-of-the-road scenario for the 21st century. *Global Environmental Change* 42:251-267.
- Gabriel y Galán, J.M., and C. Prada. 2010. Chapter 14: Pteridophyte Spores Viability. In: A. Kumar, H. Fernández, and M.A. Revilla (eds.), *Working with Ferns: Issues and Applications*. Springer, NY. 386 pp.
- Gacia, E., and E. Ballesteros. 1994. Production of *Isoëtes lacustris* in a Pyrenean lake: seasonality and ecological factors involved in the growing period. *Aquatic Botany* 48(1):77-89.

- Gacia, E., and E. Ballesteros. 1998. Effects of building up a dam in a shallow mountain lake (Baciver, Central Pyrenees). *Oecologia Aquatica* 11:55-66.
- Gacia, E., T. Buchaca, N. Bernal-Mendoza, I. Sabas, E. Ballesteros, and M. Ventura. 2018. Non-native minnows threaten quillwort populations in high mountain shallow lakes. *Frontiers in Plant Science* 9(329):1-11.
- Gaertner, M., R. Biggs, M. Te Beest, C. Hui, J. Molofsky, and D.M. Richardson. 2014. Invasive plants as drivers of regime shifts: identifying high-priority invaders that alter feedback relationships. *Diversity and Distributions* 20:733-744.
- Gao, B., X. Li, Y. Liang, M. Chen, H. Liu, Y. Liu, J. Wang, J. Zhang, Y. Zhang, M.J. Oliver, and D. Zhang. 2023. Drying without dying: a genome database for desiccation-tolerant plants and evolution of desiccation tolerance. *Plant Physiology* 00:1-14.
- Garrett, M., and G. Kantvilas. 1992. Morphology, ecology, and distribution of *Isoëtes* L. in Tasmania. *Papers and Proceedings of the Royal Society of Tasmania* 126:115-122.
- Gentili, R., T. Abeli, G. Rossi, M. Li, C. Varotto, and S. Sgorbati. 2010. Population structure and genetic diversity of the threatened quillwort *Isoëtes malinverniana* and implication for conservation. *Aquatic Botany* 93:147-152.
- Gioria, M., P.E. Hume, D.M. Richardson, and P. Pyšek. 2023. Why are invasive plants successful? *Annual Review of Plant Biology* 74:635-670.
- Gold, A.C., S.P. Thompson, C.L. Magel, and M.F. Piehler. 2020. Urbanization alters coastal plain stream carbon export and dissolved oxygen dynamics. *Science of the Total Environment* 747(141132):1-48.
- Goodman, S. and P. Baudau. 2023. Climate Change as a “Threat Multiplier”: History, Uses and Future of the Concept. In: E. Sikorsky and F. Femia (eds.), *Center for Climate and Security Brief*. Washington, DC. 35 pp.
- Gori, A., N. Lin, and J. Smith. 2020. Assessing compound flooding from landfalling tropical cyclones on the North Carolina coast. *Water Resources Research* 56(e2019WR026788):1-21.
- Grimley, L.E., A. Sebastian, T. Leijnse, D. Eilander, J. Ratcliff, and R. Luetlich. 2025. Determining the relative contributions of runoff, coastal, and compound processes to flood exposure across the Carolinas during Hurricane Florence. *Water Resources Research* 61(e2023WR036727):1-19.
- Grimley, L.E., A. Sebastian, and A. Gori. 2026. Emerging importance of compound flooding in future tropical cyclone hazard profiles. *Earth’s Future* 14(e2025EF007367):1-19.
- Grzybowski, M., Z. Endler, and B. Jaworska. 2008. Disappearance of *Isoëtes lacustris* localities and changes in the phytolittoral of Lake Długie near Łukta. *Limnological Review* 8(1-2):69-77.
- Gu, L., J. Yin, L.J. Slater, J. Chen, H.X. Do, H.M. Wang, L. Chen, Z. Jiang, and T. Zhao. 2023. Intensification of global hydrological droughts under anthropogenic climate warming. *Water Resources Research* 59(e2022WR032997):1-18.
- Halls, J.N. and J.L. Magolan. 2019. A methodology to assess land use development, flooding, and wetland change as indicators of coastal vulnerability. *Remote Sensing* 11(19):1-23.
- Harden, S.L. 2015. Surface-Water Quality in Agricultural Watersheds of the North Carolina Coastal Plain with Concentrated Animal Feeding Operations: Scientific Investigations Report 2015-5080. Reston, VA. 55 pp.

- Harms, N.E., I.A. Knight, A.B. DeRossette, and D.A. Williams. 2023. Intraspecific trait plasticity to N and P of the wetland invader, *Alternanthera philoxeroides* under flooded conditions. *Ecology and Evolution* 13(e9966):1-13.
- Hegewisch, K.C. and J.T. Abatzoglou. 2026. Climate Toolbox. University of California, Merced and University of Washington. <https://climatetoolbox.org/tool/Climate-Mapper> [Accessed July 31, 2026].
- Heydon, P. 2011. Engelmann's Quillwort. *Newsletter of the North American Native Plant Society* 12(3):10.
- Hickey, R.J. 1986. *Isoëtes* megaspore surface morphology: nomenclature, variation, and systematic importance. *American Fern Journal* 76(1):1-16.
- Hickey, R.J., W.C. Taylor, and N.T. Luebke. 1989. The species concept in Pteridophyta with special reference to *Isoëtes*. *American Fern Journal* 79(2):78-89.
- Hino, M., T.K. BenDor, J. Branham, N. Kaza, A. Sebastian, and S. Sweeney. 2024. Growing safely or building risk? *Journal of the American Planning Association* 90(1):50-62.
- Hoot, S.B., A. Buerkle, W.C. Taylor, and N. T. Luebke. 2000. Genetic Diversity, Population Structure, and Phylogenetic Patterns in *Isoëtes louisianensis*, a Critically Endangered Plant. Report to U.S. Fish and Wildlife Service, Jackson, MS. 36 pp.
- Hoot, S.B., N.S. Napier, and W.C. Taylor. 2004. Revealing unknown or extinct lineages within *Isoëtes* (Isoëtaceae) using DNA sequences from hybrids. *American Journal of Botany* 91(6):899-904.
- Hoot, S.B., W.C. Taylor, and N.S. Napier. 2006. Phylogeny and biogeography of *Isoëtes* (Isoëtaceae) based on nuclear and chloroplast DNA sequence data. *Systematic Botany* 31:449-460.
- Hopper, G.W., E.M. Smith, J. Dormoy-Boulanger, and C.R. Benitez-Nelson. 2025. Urbanization impacts dissolved organic matter concentration and quality in a southeastern United States watershed. *ACS ES&T Water* 5:1281-1294.
- Huang, Y.M., Y.Q. Yap, and C.W. Li. 2018. A semiaquatic but desiccation-tolerant plant, *Isoëtes taiwanensis* DeVol (Isoëtaceae; Lycopphyta). *International Journal of Plant Reproductive Biology* 10(1):10-13.
- Huang, Y.M., W.L. Chiou, and Y.H. Chang. 2021. Distribution and viability of dormant corms of *Isoëtes taiwanensis* DeVol in Menghuan Pond, Taipei, Taiwan. *Journal of National Park* 31(1):1-8.
- Hupp, C.R. 2000. Hydrology, geomorphology, and vegetation of Coastal Plain rivers in the southeastern United States. *Hydrological Processes* 14(16-17):1-34.
- Husák, S., M. Vöge, and C. Weilner. 2000. *Isoëtes echinospora* and *I. lacustris* in the Bohemian Forest lakes in comparison with other European sites. *Silva Gabreta* 4:245-252.
- Husband, B.C., S.J. Baldwin, and J. Suda. 2013. Chapter 16: The Incidence of Polyploidy in Natural Plant Populations: Major Patterns and Evolutionary Processes. In: I.J. Leitch, J. Greilhuber, J. Dolezel, and J.F. Wendel (eds.), *Plant Genome Diversity (Volume 2): Physical Structure, Behaviour and Evolution of Plant Genomes*, Springer-Verlag Wien, Vienna, AUS. 356 pp.
- Inamdeen, F., M. Larson, G. Thiere, and C. Karlsson. 2021. Local scour in rivers due to bridges and natural features: a case study from Ronne River, Sweden. *VATTEN: Journal of Water Management and Research* 77(3):143-161.
- Integrated Taxonomic Information System (ITIS). 2026. ITIS Taxonomic Report: *Isoëtes microvela*. <https://www.itis.gov/> [Accessed January 28, 2026].

- Intergovernmental Panel on Climate Change (IPCC). 2021. Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge, UK and New York, NY, US. 110 pp.
- Janauer, G.A., U. Schmidt-Mumm, and W. Reckendorfer. 2013. Chapter 14: Ecohydraulics and Aquatic Macrophytes: Assessing the Relationship in River Floodplains. In: I. Maddock, A. Harby, P. Kemp, and P. Wood (eds.), Ecohydraulics: An Integrated Approach. John Wiley & Sons, Ltd., West Sussex, UK. 464 pp.
- Joint Venture GMI AECOM (JV-GMI-AECOM). 2015. 2015-2020 Integrated Natural Resources Management Plan: Marine Corps Base Camp Lejeune, North Carolina. Prepared for Naval Facilities Engineering Command, Mid-Atlantic Division, Norfolk, VA. 844 pp.
- Jones, P.E., S. Consuegra, L. Börger, J. Jones, and C. Garcia de Leaniz. 2020. Impacts of artificial barriers on the connectivity and dispersal of vascular macrophytes in rivers: a review. *Freshwater Biology* 65:1165-1180.
- Junk, W.J., P.B. Bayley, and R.E. Sparks. 1989. The Flood Pulse Concept in River-floodplain Systems. Pp. 110-127. In: D.P. Dodge (ed.), Proceedings of the International Large River Symposium, Ontario, CA. Canadian Special Publication of Fisheries and Aquatic Sciences 106.
- Kang, M., Q. Ye, and H. Huang. 2005. Genetic consequence of restricted habitat and population decline in endangered *Isoëtes sinensis* (Isoëtaceae). *Annals of Botany* 96(7):1265-1274.
- Karrfalt, E.E. 1977. Substrate penetration by the corm of *Isoëtes*. *American Fern Journal* 67(1):1-4.
- Keeley, J.E. 1982. Distribution of diurnal acid metabolism in the genus *Isoëtes*. *American Journal of Botany* 69(2):254-257.
- Keeley, J.E. 1998. CAM photosynthesis in submerged aquatic plants. *The Botanical Review* 64(2):121-175.
- Keeley, J.E., and D.R. Sandquist. 1991. Diurnal photosynthesis cycle in CAM and non-CAM seasonal-pool aquatic macrophytes. *Ecology* 72(2):716-727.
- Kelly, D.J., and I. Hawes. 2005. Effects of invasive macrophytes on littoral-zone productivity and foodweb dynamics in a New Zealand high-country lake. *Journal of the North American Benthological Society* 24(2):300-320.
- Kerbiriou, C., I. Leviol, F. Jiguet, and R. Jilliard. 2008. The impact of human frequentation on coastal vegetation in a Biosphere Reserve. *Journal of Environmental Management* 88(4):715-728.
- Keskin, M., and F. Harahan. 2025. *Isoëtes volkanii* (Isoëtaceae): a new aquatic quillwort from Türkiye. *Frontiers in Life Sciences and Related Technologies* 6(3):187-191.
- Kim, C., H. Shin, and H.K. Choi. 2009. Genetic diversity and population structure of diploid and polyploid species of *Isoëtes* in East Asia based on amplified fragment length polymorphism markers. *International Journal of Plant Science* 170(4):496-504.
- Klimaszyk, P., D. Borowiak, R. Piotrowicz, J. Rosińska, E. Szelag-Wasielewska, and M. Kraska. 2020. The effect of human impact on the water quality and biocoenoses of the soft water lake with isoetids: Lake Jeleń, NW Poland. *Water* 12(945):1-20.
- Koga, H., S. Ikematsu, and S. Kimura. 2024. Diving into the water: amphibious plants as a model for investigating plant adaptations to aquatic environments. *Annual Review of Plant Biology* 75:579-604.

- Kott, L.S., and D.M. Britton. 1982. A comparative study of spore germination of some species in northeastern North America. *Canadian Journal of Botany* 60:1679-1687.
- Kriegler, E., N. Bauer, A. Popp, F. Humpenöder, M. Leimbach, J. Strefler, L. Baumstark, B.L. Bodirsky, J. Hilaire, D. Klein, I. Mouratiadou, I. Weindl, C. Bertram, J. Dietrich, G. Luderer, M. Pehl, R. Pietzcker, F. Piontek, H. Lotze-Campen, A. Biewal, and O. Edenhofer. 2017. Fossil-fueled development (SSP5): an energy and resource intensive scenario for the 21st century. *Global Environmental Change* 42:297-315.
- Kroes, D.E., and C.W. Bason. 2015. Sediment-trapping by beaver ponds in streams of the mid-Atlantic Piedmont and Coastal Plain, USA. *Southeastern Naturalist* 14(3):577-595.
- Kunkel, K.E., D.R. Easterling, A. Ballinger, S. Bililign, S.M. Champion, D.R. Corbett, K.D. Dello, J.P. Dissen, G.M. Lackmann, R.A. Luettich, Jr., L.B. Perry, W.A. Robinson, L.E. Stevens, B.C. Stewart, and A.J. Terando. 2020. 2020: North Carolina Climate Science Report. North Carolina Institute for Climate Studies, Raleigh, NC. 233 pp.
- Kurki-Fox, J.J., M.R. Burchell, and B.J. Kamrath. 2019. The potential long-term impacts of climate change on the hydrologic regimes of North Carolina's Coastal Plain non-riverine wetlands. *American Society of Agricultural and Biological Engineers* 62(6):1591-1606.
- Larke, J. 1997. Status Survey Report for *Isoëtes louisianensis* Thieret in Louisiana. Louisiana Natural Heritage Program report prepared to U.S. Fish and Wildlife Service, Jackson, MS. 11 pp.
- LaMotte, C. 1937. Conditions affecting germination of the megaspores of *Isoëtes lithophila*. *American Journal of Botany* 24(7):454-456.
- Larsén, E., and C. Rydin. 2016. Disentangling the phylogeny of *Isoëtes* (Isoëtaceae), using nuclear and plastid data. *International Journal of Plant Sciences* 177(2):157-174.
- Larsén, E., N. Wikström, A. Khodabandeh, and C. Rydin. 2022. Phylogeny of Merlin's grass (Isoëtaceae): revealing an “*Amborella* syndrome” and the importance of geographic distribution for understanding current and historical diversity. *BMC Ecology and Evolution* 22(32):1-17.
- Larsén, E., A. Khodabandeh, and C. Rydin. 2025. Spore morphology and evolution in *Isoëtes* (Isoëtaceae). *Botanical Journal of the Linnean Society* 00:1-34.
- LeBlond, R.J., and D.F. Brunton. 2010. Noteworthy collections: North Carolina; *Isoëtes microvela* D.F. Brunt. & D.M. Britton. *Castanea* 75:501-502.
- LeBlond, R.J. 2001. Endemic plants of the Cape Fear Arch Region. *Castanea* 66(1/2):83-97.
- Lecce, S.A. 2000. Seasonality of flooding in North Carolina. *Southeastern Geographer* 40(2):168-175.
- Lecce, S.A., P.P. Pease, P.A. Gares, and C.A. Rigsby. 2004. Floodplain sedimentation during an extreme flood: the 1999 flood on the Tar River, eastern North Carolina. *Physical Geography* 25(4):334-346.
- Lecce, S.A., P.P. Pease, P.A. Gares, and J. Wang. 2005. Seasonal controls on sediment delivery in a small coastal plain watershed, North Carolina. *Geomorphology* 73(3-4):246-260.
- Leonard, S. 1998. Louisiana Quillwort (*Isoëtes louisianensis*) Survey of Camp Shelby Training Site, MS. Unpublished Report to the Mississippi Military Department, Jackson, MS. 9 pp.
- Leonard, S. 2011. Review of Quillworts (Isoëtaceae) of Southeast United States with Specific Reference to Louisiana Quillwort (*Isoëtes louisianensis*). Gulf Coast Biological Surveys, Inc., Report to U.S. Fish and Wildlife Service, Jackson, MS. 38 pp.

- Link-Pérez, M.A., and S.W. Laffan. 2018. Fern and lycopphyte diversity in the Pacific Northwest: patterns and predictors. *Journal of Systematics and Evolution* 56(5):498-522.
- Liu, X., W.R. Gituru, and Q.F. Wang. 2004. Distribution of basic diploid and polyploid species of *Isoëtes* in East Asia. *Journal of Biogeography* 31:1239-1250.
- Liu, X., J.Y. Wang, and Q.F. Wang. 2005. Current status and conservation strategies for *Isoëtes* in China: a case study for the conservation of threatened aquatic plants. *Oryx* 39(3):335-338.
- Liu, Y., and P. Wang. 2024. Drag in vegetation canopy: considering shelter and blockage effects. *Water Resources Research* 60(e2023WR036521):1-17.
- Looy, C.V., J.H.A. van Konijnenburg-van Cittert, and I.A.P. Duijnste. 2021. Proliferation of isoëtalean lycophytes during the Permo-Triassic biotic crises: a proxy for the state of the terrestrial biosphere. *Frontiers in Earth Science* 9(615370):1-19.
- Lopes, A.V. Sant'anna, L.R.G. Neto, L.P. Cunha, A.J.D. Cogo, C.F. Caldeira, G. Oliveira, R. Lemes Martins, F. de Assis Esteves, A.B. Cavalcante, H.M. Duarte, D.B. Zandonadi, Rodrigo Nunes da Fonseca, M.P. Santos. 2023. The resilience of the aquatic *Isoëtes cangae* to terrestrial environment: insights into molecular and ecophysiological adaptations. *Aquatic Botany* 189(103704):1-21.
- Lopez, H., S.K. Lee, R. West, D. Kim, G.R. Foltz, G.J. Alaka Jr., and H. Murakami. 2024. Projected increase in the frequency of extremely active Atlantic hurricane seasons. *Science Advances* 10(46):eadq7856.
- Lu, Y., G. Yang, Y. Zhang, B. Wei, Q. He, H. Yu, and Y. Wang. 2024. The influence of management practices on plant diversity: a comparative study of three urban wetlands in an expanding city in eastern China. *PeerJ* 12(e16701):1-30.
- Lucassen, E.C.H.E.T, A.J.P. Smolders, and J.G.M. Roelofs. 2012. Liming induces changes in the macrophyte vegetation of Norwegian softwater lakes by mitigating carbon limitation: results from a field experiment. *Applied Vegetation Science* 15:166-174.
- Lyman, M. 2017. Louisiana Quillwort (*Isoëtes louisianensis*): Permanent Plot Monitoring for Measuring Potential Effects of MPRC-H Construction, Final Report. The Nature Conservancy, Camp Shelby Field Office, Hattiesburg, MS. 9 pp.
- Ma, M., X. Dong, K. Liao, F. Hao, R. Qin, and H. Liu. 2019. Outcrossing rates and gene flow in natural population of the endangered endemic aquatic lycopphyte *Isoëtes yunguiensis* as revealed by ISSR markers. *Notulae Botanicae Horti Agrobotanici Cluj-Napoca* 47(2):339-346.
- Madsen, J.D., P.A. Chamber, W.F. James, E.W. Koch, and D.F. Westlake. 2001. The interaction between water movement, sediment dynamics and submersed macrophytes. *Hydrobiologia* 444:71-84.
- Mäemets, H. 2023. Chapter 5: Natural and Anthropogenic Impacts on the Macrophytes of Soft-Water Lakes of Estonia. In: L. Hufnagel (ed.), *New Insights into Protected Area Management and Conservation Biology*. IntechOpen, London, UK. 160 pp.
- Majszak, M., A. Zaki, and O. Wani. 2026. What defines a flood. *Communications Earth and Environment* 7(342):1-7.
- Mallin, M.A. 2000. Impacts of industrial-scale swine and poultry production on rivers and estuaries. *American Scientist* 88(1):26-37.
- Mallin, M.A., and C.A. Corbett. 2006. How hurricane attributes determine the extent of environmental effects: multiple hurricanes and different coastal systems. *Estuaries and Coasts* 29(6):1046-1061.

- Mallin, M.A., and L.B. Cahoon. 2020. The hidden impacts of phosphorus pollution to streams and rivers. *BioScience* 70(4):315-329.
- Mallin, M.A., L.B. Cahoon, D.C. Parson, and S.H. Ensign. 1998. Effect of Organic and Inorganic Nutrient Loading on Photosynthetic and Heterotrophic Plankton Communities in Blackwater Rivers. Water Resources Research Institute Report UNC-WRRI-98-315. Wilmington, NC. 54 pp.
- Mallin, M.A., M.H. Posey, G.C. Shank, M.R. McIver, S.H. Ensign, and T.D. Alphin. 1999. Hurricane effects on water quality and benthos in the Cape Fear watershed: natural and anthropogenic impacts. *Ecological Applications* 9(1):350-362.
- Mallin, M.A., M.H. Posey, M.R. McIver, D.C. Parson, S.H. Ensign, and T.D. Alphin. 2002. Impacts and recovery from multiple hurricanes in a Piedmont-Coastal Plain river system: human development of floodplains greatly compounds the impacts of hurricanes on water quality and aquatic life. *BioScience* 52(11):999-1010.
- Mallin, M.A., and M.R. McIver. 2018. Season matters when sampling streams for swine CAFO waste pollution impacts. *Journal of Water and Health* 16.1:78-86.
- Mallin, M.A., M.R. McIver, S.H. Ensign, and L.B. Cahoon. 2004. Photosynthetic and heterotrophic impacts of nutrient loading to blackwater streams. *Ecological Applications* 14(3):823-838.
- Mallin, M.A., V.L. Johnson, and S.H. Ensign. 2009. Comparative impacts of stormwater runoff on water quality of an urban, a suburban, and a rural stream. *Environmental Monitoring and Assessment* 159:475-491.
- Mallin, M.A., V.L. Johnson, S.H. Ensign, and T.A., MacPherson. 2006. Factors contributing to hypoxia in rivers, lakes, and streams. *Limnology and Oceanography* 51(1, part 2):690-701.
- Marble, S.C. 2018. Native weedy pests of the Deep South. *HortScience* 53(9):1244-1249.
- Martin, K.L., R.E. Emanuel, and J.M. Vose. 2018. Terra incognita: the unknown risks to environmental quality posed by the spatial distribution and abundance of concentrated animal feeding operations. *Science of the Total Environment* 642:887-893.
- Maslovat, C.Y., R. Batten, D.F. Brunton, and P.C. Sokoloff. 2021. Distribution, status, and habitat characteristics of Columbia Quillwort (*Isoetes minima*, Isoëtaceae) in Canada. *The Canadian Field-Naturalist* 135(3):293-304.
- May, R.W.P., J.C. Ackers, and A.M. Kirby. 2002. Manual on Scour at Bridges and Other Hydraulic Structures. Construction Industry Research and Information Association (CIRIA) C551, London, UK. 225 pp.
- Mayrose, I., S.H. Zhan, C.J. Rothfels, K. Magnuson-Ford, M.S. Barker, L.H. Rieseberg, and S.P. Otto. 2011. Recently formed polyploid plants diversify at lower rates. *Science* 333:1257 (plus Supporting Online Material).
- McCabe, K.M., E.M. Smith, S.Q. Lang, C.L. Osburn, and C.R. Benitez-Nelson. 2021. Particulate and dissolved organic matter in stormwater runoff influences oxygen demand in urbanized headwater catchments. *Environmental Science and Technology* 55:952-961.
- McInnis, N.C. 1991. Status Survey of *Isoetes louisianensis* Thieret. Louisiana Natural Heritage Program Report to The Nature Conservancy, Arlington, VA. 7 pp.

- Meentemeyer, R.K., W. Tang, M.A. Dorning, J.B. Vogler, N.J. Cunniffe, and D.A. Shoemaker. 2013. FUTURES: multilevel simulations of emerging urban–rural landscape structure using a stochastic patch-growing algorithm. *Annals of the Association of American Geographers* 103(4):785-807.
- Mengitsu, B.B., D. Unbushe, and E. Abebe. 2017. Invasion of water hyacinth (*Eichhornia crassipes*) is associated with decline in macrophyte biodiversity in an Ethiopian rift-valley lake—Abaya. *Open Journal of Ecology* 7:667-681.
- Meyer, J.L. 1990. A blackwater perspective on riverine ecosystems. *BioScience* 40(9):643-651.
- Meyer, J.L., A.C. Benke, R.T. Edwards, and J.B. Wallace. 1997. Organic matter dynamics in the Ogeechee River, a blackwater river in Georgia, USA. *Journal of the North American Benthological Society* 16(1):82-87.
- Mochalova, O.A., A.A. Bobrov, and D.F. Brunton. 2015. *Isoëtes* in Kamchatka (northern Russian Far East), with the description of a new hybrid *I. × paratunica* (*I. asiatica* × *I. maritima*). *American Fern Journal* 105(2):101-112.
- Mommer, L., and E.J.W. Visser. 2005. Underwater photosynthesis in flooded terrestrial plants: a matter of leaf plasticity. *Annals of Botany* 96:581-589.
- Montiel-Molina, J.A.M., J.P. Sexton, J.M. Beman, and A.C. Frank. 2023. The microbiome of an amphibious plant shifts dramatically across the host life cycle. *bioRxiv* 1-28.
- Mørk, O., T. Kragh, E. Kristensen and K. Sand-Jensen. 2018. Adaptation and growth performance of four endangered amphibious freshwater species. *Aquatic Botany* 150:16-22.
- Moulton, M., J.B. Zambon, Z.G. Xue, J.C. Warner, D. Bao, D. Yin, Z. Defne, R. He, and C. Hegermiller. 2024. Modeled coastal-ocean pathways of land-sourced contaminants in the aftermath of Hurricane Florence. *Journal of Geophysical Research* 129(e2023JC019685):1-19.
- Mulholland, P.J., G.R. Best, C.C. Coutant, G.M. Hornberger, J.L. Meyer, P.J. Robinson, J.R. Stenberg, R.E. Turner, F. Vera-Herrera, and R.G. Wetzel. 1997. Effects of climate change on freshwater ecosystems of the south-eastern United States and the Gulf Coast of Mexico. *Hydrological Processes* 11:949-970.
- Musselman, L.J. 2001. Georgia quillworts. *Tipularia*: *Journal of the Georgia Botanical Society* 16:2-19.
- Musselman, L.J. 2002. Ornamentation of *Isoëtes* (Isoëtaceae, Lycopphyta) microspores. *The Botanical Review* 68(4):474-487.
- Musselman, L.J., and D.A. Knepper. 1994. Quillworts of Virginia. *American Fern Journal* 84(2):48-68.
- National Aquatic Connectivity Collaborative (NACC). 2026. National Aquatic Barrier Inventory v4.1.0 (3/30/2026). Compiled by the Southeast Aquatic Resources Partnership (SARP) and the National Fish Habitat Partnership (NFHP). <https://tool.aquaticbarriers.org> [Accessed May 22, 2026].
- National Oceanic and Atmospheric Administration (NOAA) National Centers for Environmental Information (NCEI). 2026. Climate at a Glance City Time Series: Wilmington, North Carolina. <https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/city/time-series> [Accessed January 27, 2026].
- NOAA Office of Coastal Management (OCM). 2026. NOAA Office for Coastal Management Coastal Inundation Digital Elevation Model. <https://coast.noaa.gov/slrdata/DEMs/index.html> [Accessed July 1, 2026].

- Natural Resources Conservation Service (NRCS). 2026a. Soil Survey Geographic (SSURGO) State Database for North Carolina. U.S. Department of Agriculture, Natural Resources Conservation Service. <https://www.nrcs.usda.gov/resources/data-and-reports/gridded-soil-survey-geographic-gssurgo-database> [Accessed February 23, 2026].
- Natural Resources Conservation Service (NRCS). 2026b. PLANTS Database Wetland Rating (*Isoëtes microvela*). U.S. Department of Agriculture, Natural Resources Conservation Service. <https://plants.sc.egov.usda.gov/plant-profile/ISMI3/wetland> [Accessed July 8, 2026].
- NatureServe. 2020. Habitat-based Plant Element Occurrence Delimitation Guidance. NatureServe Biotics 5, Arlington, VA. 15 pp.
- NatureServe. 2026. NatureServe Explorer: *Isoëtes microvela* (Thin-wall Quillwort). Arlington, VA. https://explorer.natureserve.org/Taxon/ELEMENT_GLOBAL.2.137971/Isoetes_microvela [Accessed April 16, 2026].
- Neville, J.A., R.E. Emanuel, E.G. Nichols, and J. Vose. 2021. Extreme flooding and nitrogen dynamics of a blackwater river. *Water Resources Research* 57(e2020WR029106):1-16.
- Nicotra, A.B., E.A. Beever, A.L. Robertson, G.E. Hofmann, and J. O’Leary. 2015. Assessing the components of adaptive capacity to improve conservation and management efforts under global change. *Conservation Biology* 29:1268-1278.
- Nilsson, C. and M. Svedmark. 2002. Basic principles and ecological consequences of changing water regimes: riparian plant communities. *Environmental Management* 30(4):468-480.
- North Carolina Department of Agriculture and Consumer Services (NCDACS). 2021. Plant Conservation Laws and Regulations (Subchapter 48F: Plant Conservation). <https://www.ncagr.gov/divisions/plant-industry/plant-protection/plant-conservation-program/laws-and-regulations> [Accessed April 16, 2026].
- North Carolina Department of Environmental Quality (NCDEQ). 2026. NCDEQ GIS Open Data Portal: Animal Feed Operation Permits. NCDEQ Division of Water Resources, Animal Feeding Operations Program. <https://data-ncdenr.opendata.arcgis.com/datasets/ncdenr::animal-feed-operation-permits-view/about> [Accessed May 6, 2026].
- North Carolina General Assembly (NCLEG). 2025. 2025 North Carolina General Statutes Chapter 106, Agriculture Article 19B: Plant Protection and Conservation Act. (§ 106-202.19: Unlawful acts; penalties; enforcement). https://www.ncleg.gov/EnactedLegislation/Statutes/PDF/ByArticle/Chapter_106/Article_19B.pdf [Accessed April 16, 2026]
- North Carolina Natural Heritage Program (NCNHP). 2015. Chapter 143B, Article 2, Part 42: Nature Preserves Act. (§ 143B-135.250 to § 143B-135.262). <https://www.ncnhp.org/documents/files/nature-preserves-act-ss-143b-135250-ss-143b-135272/open> [Accessed April 16, 2026].
- North Carolina Natural Heritage Program (NCNHP). 2026a. *Isoëtes microvela* Element Occurrence Report. North Carolina Natural Heritage Program, N.C. Department of Natural and Cultural Resources, Raleigh, NC. 25 pp.
- North Carolina Natural Heritage Program (NCNHP). 2026b. *Isoëtes microvela* Element Occurrence Report (EO5 Update). North Carolina Natural Heritage Program, N.C. Department of Natural and Cultural Resources, Raleigh, NC. 3 pp.

- North Carolina Office of State Budget and Management (NCOSBM). 2022. 2020 Certified County Population Estimates Ranked by Percentage Change: 2010-2020. <https://www.osbm.nc.gov/county-pop-estimates-fastest-growth/download?attachment> [Accessed April 15, 2026].
- North Carolina Office of State Budget and Management (NCOSBM). 2025. Estimates of the Municipal and Non-Municipal Population of North Carolina and North Carolina Counties for July 1, 2024. <https://www.osbm.nc.gov/facts-figures/population-demographics/state-demographer/municipal-population-estimates/municipal-and-non-municipal-population> [Accessed April 15, 2026].
- North Carolina Office of State Budget and Management (NCOSBM). 2026. 2024 Certified County Population Estimates. <https://www.osbm.nc.gov/facts-figures/population-demographics/state-demographer/county-population-estimates/certified-county-population-estimates> [Accessed April 15, 2026].
- North Carolina Wildlife Resources Commission (NCWRC). 2025. North Carolina State Wildlife Action Plan. Raleigh, NC. 1633 pp. <https://www.ncwildlife.gov/state-wildlife-action-plan> [Accessed April 16, 2026].
- Nunez-Mir, G.C., and M.A. McCary. 2024. Invasive plants and their root traits are linked to the homogenization of soil microbial communities across the United States. *Proceedings of the National Academy of Sciences* 121(44):1-9.
- O'Driscoll, M., S. Clinton, A. Jefferson, A. Manda, and S. McMillan. 2010. Urbanization effects on watershed hydrology and in-stream processes in the southern United States. *Water* 2:605-648.
- Ogg, C.M., M.A. Wilson, J.M. Reed, and C.D. Gulley. 2013. Sediment provenance and geomorphic development of soils in a blackwater ecosystem, South Carolina Coastal Plain. *Geoderma* 192:394-406.
- Oh, M.J., C. Kim, H.R. Na, H. Shin, J.R. Liu, H.K. Choi, and S.W. Kim. 2013. High frequency sporophytes regeneration from the spore culture of the endangered aquatic fern *Isoetes coreana*. *American Journal of Plant Sciences* 4:14-20.
- O'Hare, M.T., A. Baattrup-Pedersen, I. Baumgarte, A. Freeman, I.D.M. Gunn, A.N. Lázár, R. Sinclair, A.J. Wade, and M.J. Bowes. 2018. Responses of aquatic plants to eutrophication in rivers: a revised conceptual model. *Frontiers in Plant Science* 9(451):1-13.
- Paerl, H.W., R.S. Fulton III, P.H. Moisaner, and J. Dyble. 2001a. Harmful freshwater algal blooms, with an emphasis on cyanobacteria. *The Scientific World* 1:76-113.
- Paerl, H.W., J.D. Bales, L.W. Ausley, C.P. Buzzelli, L.B. Crowder, L.A. Eby, J.M. Fear, M. Go, B.L. Peierls, T.L. Richardson, and J.S. Ramus. 2001b. Ecosystem impacts of three sequential hurricanes (Dennis, Floyd, and Irene) on the United States' largest lagoonal estuary, Pamlico Sound, NC. *Proceedings of the National Academy of Sciences* 98(10):5655-5660.
- Paerl, H.W., N.S. Hall, A.G. Hounshell, R.A. Luettich Jr., K.L. Rossignol, C.L. Osburn, and J. Bales. 2019. Recent increase in catastrophic tropical cyclone flooding in coastal North Carolina, USA: long-term observations suggest a regime shift. *Scientific Reports* 9(10620):1-9.
- Paller, M., J. Feminella, E. Kosnicki, S. Sefick, M. Jarrell, D. Fletcher, T. Tuberville, S. Sterrett, A. Grosse, and B. Prusha. 2010. Ecological reference models for blackwater streams: a prerequisite for successful ecosystem recovery and management. *Proceedings of the 2010 South Carolina Water Resources Conference*, Columbia, SC. 5 pp.

- Pannard, A., P. Souchu, C. Chauvin, M. Delabuis, C. Gascuel-Oudou, E. Jeppesen, M. Le Moal, A. Ménesguen, G. Pinay, N.N. Rabalais, Y. Souchon, and E.M. Gross. 2024. Why are there so many different definitions of eutrophication? *Ecological Monographs* 94(e1616):1-25.
- Pedersen, O., C. Pulido, S.M. Rich, and T.D. Colmer. 2011. In situ O₂ dynamics in submerged *Isoëtes lacustris*: varied leaf gas permeability influences underwater photosynthesis and internal O₂. *Journal of Experimental Botany* 62(13):4691-4700.
- Petrasova, A., G.M. Sanchez Salas, M.A. Lawrimore, J.B. Vogler, E.L. Collins, V. Petras, T. Harper, E. Butzler, and R.K. Meentemeyer. 2023. FUTURES v2: Status Quo Projections of Future Patterns of Urbanization Across the Conterminous United States from 2020 to 2100. U.S. Geological Survey (USGS) Data Release, 704. <https://doi.org/10.5066/P94N3ICH> [Accessed July 29, 2026].
- Pfeiffer, N.E. 1922. Monograph of the Isoëtaceae. *Annals of the Missouri Botanical Garden* 9(2):79-233.
- Pfingsten, I.A., K.M. Reaver, E.M. Dean, M.E. Neilson, B. Chivoiu, and W.M. Daniel. 2024. Combining storm flood water level and topography to prioritize inter-basin transfer of non-native aquatic species in the United States. *Biological Invasions* 26:4105-4120.
- Phillips, J.D. 2022a. Geomorphic impacts of Hurricane Florence on the lower Neuse River: portents and particulars. *Geomorphology* 397(108026):1-16.
- Phillips, J.D. 2022b. Geomorphology of the fluvial-estuarine transition zone, lower Neuse River, North Carolina. *Earth Surface Processes and Landforms* 47:2044-2061.
- Pietsch, W. 1991. On the phytosociology and ecology of *Isoëtes asiatica* (Makino) in oligotrophic water bodies of South Sakhalin. *Vegetatio* 97(2):99-115.
- Rapp, J., T. Shear, and D. Robison. 2001. Soil, groundwater, and floristics of a southeastern United States blackwater swamp 8 years after clearcutting with helicopter and skidder extraction of the timber. *Forest Ecology and Management* 149:241-252.
- Ren, G., Y. Du, B. Yang, J. Wang, M. Cui, Z. Dai, M.O. Adomako, S. Rutherford, and D. Du. 2023. Influence of precipitation dynamics on plant invasions: response of alligator weed (*Alternanthera philoxeroides*) and co-occurring native species to varying water availability across plant communities. *Biological Invasions* 25:519-532.
- Rhazi, M., P. Grillas, A. Charpentier, and F. Médail. 2004. Experimental management of Mediterranean temporary pools for conservation of the rare quillwort *Isoëtes setacea*. *Biological Conservation* 118:675-684.
- Rhazi, M., P. Grillas, L. Rhazi, A. Charpentier, and F. Médail. 2009. Competition in microcosm between a clonal plant species (*Bolboschoenus maritimus*) and a rare quillwort (*Isoëtes setacea*) from Mediterranean temporary pools of southern France. *Hydrobiologia* 634:115-124.
- Riahi, K., D.P. Van Vuuren, E. Kriegler, J. Edmonds, B.C. O'Neill, S. Fujimori, N. Bauer, K. Calvin, R. Dellink, O. Ricko, W. Lutz, A. Popp, J. C. Cuaresma, S. KC, M. Leimbach, L. Jiang, T. Kram, S. Rao, J. Emmerling, K. Ebi, T. Hasegawa, P. Havlik, F. Humpenöder, L. Aleluia Da Silva, S. Smith, E. Stehfest, V. Bosetti, J. Eom, D. Gernaat, T. Masui, J. Rogelj, J. Streffer, L. Drouet, V. Krey, G. Luderer, M. Harmsen, K. Takahashi, L. Baumstark, J.C. Doelman, M. Kainuma, Z. Klimont, G. Marangoni, H. Lotze-Campen, M. Obersteiner, A. Tabeau, and M. Tavoni. 2017. The Shared Socioeconomic Pathways and their energy, land use, and greenhouse gas emissions implications: an overview. *Global Environmental Change* 42:153-168.

- Richardson, K., H. Griffiths, M.L. Reed, J.A. Raven, and N.M. Griffiths. 1984. Inorganic carbon assimilation in the isoetids, *Isoetes lacustris* L. and *Lobelia dortmanna* L. *Oecologia* 61(1):115-121.
- Riggsbee, J.A., C.H. Orr, D.M. Leech, M.W. Doyle, and R.G. Wetzel. 2008. Suspended sediments in river ecosystems: photochemical sources of dissolved organic carbon, dissolved organic nitrogen, and adsorptive removal of dissolved iron. *Journal of Geophysical Research* 113(G03019):1-12.
- Riis, T., and B. Biggs. 2003. Stream vegetation and flow regimes. *Water and Atmosphere* 11(1):18-20.
- Riis, T., and I. Hawes. 2003. Effect of wave exposure on vegetation abundance, richness and depth distribution of shallow water plants in a New Zealand lake. *Freshwater Biology* 48:75-87.
- Ronowski, R., R. Chmara, and J. Szymeja. 2023. Structural and functional changes in macrophyte species composition in softwater lakes after 60 years of land use. *Aquatic Sciences* 85(104):1-12.
- Rorslett, B., and S.W. Johansen. 1995. Dynamic response of the submerged macrophyte, *Isoetes lacustris*, to alternating light levels under field conditions. *Aquatic Botany* 51:223-242.
- Salgado, J., C.D. Sayer, S.J. Brooks, T.A. Davidson, B. Goldsmith, I.R. Patmore, A.G. Baker, and B. Okamura. 2018. Eutrophication homogenizes shallow lake macrophyte assemblages over space and time. *Ecosphere* 9(9):1-15.
- Sand-Jensen, K. 2008. Drag forces on common plant species in temperate streams: consequences of morphology, velocity and biomass. *Hydrobiologia* 610:307-319.
- Sand-Jensen, K., and C.L. Møller. 2014. Reduced root anchorage of freshwater plants in sandy sediments enriched with fine organic matter. *Freshwater Biology* 59:427-437.
- Sand-Jensen, K., and O. Pedersen. 1999. Velocity gradients and turbulence around macrophyte stands in streams. *Freshwater Biology* 42:315-328.
- Sandquist, D.R., and J.E. Keeley. 1990. Carbon uptake characteristics in two high elevation populations of the aquatic CAM plant *Isoetes bolanderi* (Isoëtaceae). *American Journal of Botany* 77(5):682-688.
- Santos, M.P., J.V.S. Rabelo Araujo, A.V. Sant'anna Lopes, J.C. Fiorio Vettorazzi, M.S. Bastos Boechat, F.A. Santana Arêdes, N.V. Campos, E.N. Calderon, F.M. Gomes Santos, T. Nogueira Fernandes, R. Nunes da Fonseca, M. Gonzaga Pereira, G. Oliveira, D.B. Zandonadi, R. Lemes Martins, and F. de Assis Esteves. 2020. The genetic diversity and population structure of two endemic Amazonian quillwort (*Isoetes* L.) species. *PeerJ* 8(e10274):1-15.
- Schafale, M.P. 2024. Guide to the Natural Communities of North Carolina: Fourth Approximation. Department of Environmental and Natural Resources, North Carolina Natural Heritage Program, Division of Parks and Recreation, Raleigh, NC. 228 pp.
- Schafer, T., N. Ward, P. Julian, K.R. Reddy, and T.Z. Osborne. 2020. Impacts of hurricane disturbance on water quality across the aquatic continuum of a blackwater river to estuary complex. *Journal of Marine Science and Engineering* 8(412):1-20.
- Schafraan, P.W., S.W. Leonard, R.D. Bray, W.C. Taylor, and L.J. Musselman. 2016. *Isoetes mississippiensis*: a new quillwort from Mississippi, USA. *PhytoKeys* 74:97-106.
- Schafraan, P.W. 2019. Molecular systematics of *Isoetes* (Isoëtaceae) in eastern North America. Doctoral dissertation, Old Dominion University, Norfolk, VA. ProQuest Dissertations Publishing. 244 pp.
- Schaner, J. 2026. Peer Review of Thin-wall Quillwort (*Isoetes microvela*) SSA. Comments provided via U.S. Fish and Wildlife Service Peer Review Portal [Accessed August 31, 2026].

- Schaner, J., B. Wichmann, and L. Hamon. 2021. A Range-wide Status Survey of Thin-wall Quillwort *Isoetes microvela* (Isoëtaceae). N.C. Natural Heritage Program Report to U.S. Fish and Wildlife Service, Raleigh, NC. Cooperative Agreement F19AC0013. 32 pp.
- Schneider, R.L., and R.R. Sharitz. 1989. Hydrochory and regeneration in a bald cypress-water tupelo swamp forest. *Ecology* 69(4):1055-1063.
- Schneider, M.J., T. Newsom, and C. Schildknecht. 2025. "What's right for the environment is right in every other way"- environmental volunteering as a gateway to environmental frameworks. *Sociological Spectrum* 45(6):344-357.
- Schutten, J., J. Dainty, and A.J. Davy. 2005. Root anchorage and its significance for submerged plants in shallow lakes. *Journal of Ecology* 93:556-571.
- Sgro, C.M., A.J. Lowe, and A.A. Hoffmann. 2011. Building evolutionary resilience for conserving biodiversity under climate change. *Evolutionary Applications* 4:326-337.
- Sharma, B.D., and R. Singh. 1984. The ligule in *Isoetes*. *American Fern Journal* 74(1):22-28.
- Sheil, D. 2026. How forests may reduce the incidence of destructive tropical cyclones, hurricanes and typhoons. *Forests* 17(359):1-19.
- Shih, S.S., and K.Y. Lee. 2023. Time-series-based habitat model development for surface water management on endangered aquatic plant *Isoetes taiwanensis* conservation in mountain wetlands. *Ecological Indicators* 154(110489):1-11.
- Simmons, C.E. 1993. Sediment Characteristics of North Carolina Streams, 1970-79. U.S. Geological Survey Water-Supply Paper 2364, Washington, DC. 84 pp.
- Small, R., and R.J. Hickey. 1997. Levels and pattern of genetic variation in *Isoetes karstenii* with observations on *I. palmeri*. *American Fern Journal* 87(3):104-115.
- Smith, R.W. 1900. The structure and development of the sporophylls and sporangia of *Isoetes*. *Botanical Gazette* 29(4):225-258.
- Smith, D., N.L. Allan, C.P. McGowan, J. Szymanski, S. R. Oetker, and H. M. Bell. 2018. Development of a species status assessment process for decisions under the U.S. Endangered Species Act. *Journal of Fish and Wildlife Management* 9(1):1-19.
- Sorrell, B.K. 2004. Regulation of root anaerobiosis and carbon translocation by light and root aeration in *Isoetes alpinus*. *Plant, Cell and Environment* 27:1102-1111.
- Sorrie, B.A., and S.W. Leonard. 1999. Noteworthy records of Mississippi vascular plants. *SIDA, Contributions to Botany* 18(3):889-908.
- Southeast Regional Network of Expertise and Collections (SERNEC) Data Portal. 2026. *Isoetes microvela* (thin-wall quillwort) herbarium collection records (Drexel University, Harvard University Herbaria, Illinois Natural History Survey Herbarium, New York Botanical Garden, Old Dominion University Herbarium, Pringle Herbarium at University of Vermont, University of Florida Herbarium, University of Michigan Herbarium, University of North Carolina at Chapel Hill Herbarium, University of South Florida Herbarium, William Sherman Turrell Herbarium at Miami University). <http://sernecportal.org> [Accessed March 17, 2026].
- Spierenburg, P., J.G.M. Roelofs, T.J. Andersen, and A.F. Lotter. 2010. Historical changes in the macrophyte community of a Norwegian softwater lake. *Journal of Paleolimnology* 44:841-853.

- Stegen, J., A.J. Burgin, M.H. Busch, J.B. Fisher, J. Ladau, J. Abrahamson, L. Kinsman-Costello, L. Li, X. Chen, T. Datry, N. McDowell, C. Tatariw, A. Braswell, J.M. Deines, J.A. Guimond, P. Regier, K. Rod, E.K.P. Bam, E. Fluett-Chouinard, I. Forbrich, K.L. Jaeger, T. O'Meara, T. Scheibe, E. Seybold, J.N. Sweetman, J. Zheng, D.C. Allen, E. Herndon, B.A. Middleton, S. Painter, K. Roche, J. Scamardo, R. Vander Vorste, K. Boye, E. Wohl, M. Zimmer, K. Hondula, M. Laan, A. Marshall, and K.F. Patel. 2025. Reviews and syntheses: variable inundation across Earth's terrestrial systems. *Biogeosciences* 22:995-1034.
- Steuer, J.J., J.D. Bales, and E.M.P. Giddings. 2009. Relationship of stream ecological conditions to simulated hydraulic metrics across a gradient of basin urbanization. *Journal of the North American Benthological Society* 28(4):955-976.
- Strickland, B.A., C.J. Patrick, F.R. Carvallo, S.K. Kinard, A.T. Solis, B.K. Reese, and J.D. Hogan. 2024. Long-term climate and hydrologic regimes shape stream invertebrate community responses to a hurricane disturbance. *Journal of Animal Ecology* 93:823-835.
- Striker, G.G. 2012. Chapter 1: Flooding Stress on Plants: Anatomical, Morphological and Physiological Responses. In: J.K. Mworira (ed.), *Botany*. IntechOpen, London, UK. 238 pp.
- Suir, G.M., and J.F. Berkowitz. 2021. Inundation Depth and Duration Impacts on Wetland Soils and Vegetation. U.S. Army Corps of Engineers Report ERDC/EL SR-21-5, Washington, DC. 76 pp.
- Suissa, J.S., and W.A. Green. 2020. CO₂ starvation experiments provide support for the carbon-limited hypothesis on the evolution of CAM-like behaviour in *Isoetes*. *Annals of Botany* 127:135-141.
- Suissa, J.S., S.P. Kinoshian, P.W. Schafran, J.F. Bolin, W.C. Taylor, and E.A. Zimmer. 2022. Homoploid hybrids, allopolyploids, and high ploidy levels characterize the evolutionary history of a western North American quillwort (*Isoetes*) complex. *Molecular and Phylogenetics and Evolution* 166(107332):1-13.
- Sweet, W.V., B.D. Hamlington, R.E. Kopp, C.P. Weaver, P.L. Barnard, D. Bekaert, W. Brooks, M. Craghan, G. Dusek, T. Frederikse, G. Garner, A.S. Genz, J.P. Krasting, E. Larour, D. Marcy, J.J. Marra, J. Obeysekera, M. Osler, M. Pendleton, D. Roman, L. Schmied, W. Veatch, K.D. White, and C. Zuzak. 2022. Global and Regional Sea Level Rise Scenarios for the United States: Updated Mean Projections and Extreme Water Level Probabilities Along U.S. Coastlines. NOAA Technical Report NOS 01. National Oceanic and Atmospheric Administration, National Ocean Service, Silver Spring, MD. 111 pp. <https://oceanservice.noaa.gov/hazards/sealevelrise/noaa-nostechrpt01-global-regional-SLR-scenarios-US.pdf> [Accessed July 29, 2026].
- Szmeja, J. 1994. Effect of disturbances and interspecific competition in isoetid populations *Aquatic Botany* 48(3-4):225-238.
- Tamang, B.G., and T. Fukao. 2015. Plant adaptation to multiple stresses during submergence and following desubmergence. *International Journal of Molecular Sciences* 16:30164-30180.
- Tan, B., and S. Chen. 2023. Defining mechanisms of C₃ to CAM photosynthesis transition toward enhancing crop stress resilience. *International Journal of Molecular Sciences* 24:1-15.
- Taylor, W.C., and R.J. Hickey. 1992. Habitat, evolution, and speciation in *Isoetes*. *Annals of the Missouri Botanical Garden* 79(3):613-622.
- Taylor, W.C., and N.T. Luebke. 1986. Germinating spores and growing sporophytes of aquatic *Isoetes*. *American Fern Journal* 76(1):21-24.
- Taylor, W.C. and J.E. Schwegman. 1992. *Isoetes butleri* in Illinois. *American Fern Journal* 82:82-83.

- Ten Brink, C. 2026. Email communication between C. Ten Brink (Wildlife Biologist, MCB Camp Lejeune, Jacksonville, NC) and E. Polzer [Accessed February 3, 2026].
- Thorp, J.H., E.M. McEwan, M.F. Flynn, and F.R. Hauer. 1985. Invertebrate colonization of submerged wood in a cypress-tupelo swamp and blackwater stream. *The American Midland Naturalist* 113(1):56-68.
- Thurman, L.L., B.A. Stein, E.A. Beever, W. Foden, S.R. Geange, N. Green, J.E. Gross, D.J. Lawrence, O. LeDee, J.D. Olden, L.M. Thompson, and B.E. Young. 2020. Persist in place or shift in space? Evaluating the adaptive capacity of species to climate change. *Frontiers in Ecology and the Environment* 18(9):520-528.
- Todd, M.J., G. Vellidis, R.R. Lowrance, and C.M. Pringle. 2009. High sediment oxygen demand within an instream swamp in southern Georgia: implications for low dissolved oxygen levels in coastal blackwater streams. *Journal of the American Water Resources Association* 45(6):1493-1507.
- Todd, M.J., R.R. Lowrance, P. Goovaerts, G. Vellidis, and C.M. Pringle. 2010. Geostatistical modeling of the spatial distribution of sediment oxygen demand within a Coastal Plain blackwater watershed. *Geoderma* 159(1-2):53-62.
- Treher, A. 2024. *Isoëtes microvela*. The IUCN Red List of Threatened Species 2024: e.T195185678A195185681. <https://www.iucnredlist.org/species/195185678/195185681> [Accessed February 18, 2026].
- Troia, A. 2016. Dispersal and colonization in heterosporous lycophytes: palynological and biogeographical notes on the genus *Isoëtes* in the Mediterranean region. *Webbia: Journal of Plant Taxonomy and Geography* 71(2):277-281.
- Uehling, J. and C.J. Schreck III. 2024. Observed changes in extreme precipitation associated with U.S. tropical cyclones. *Journal of Climate* 37:3661-3674.
- U.S. Army Corps of Engineers (USACE). 2022. National Wetland Plant List, Version 3.6 (*Isoëtes microvela*). <https://nwpl.sec.usace.army.mil/plant/ISMI3/> [Accessed July 8, 2026].
- U.S. Drought Monitor (USDM): National Drought Mitigation Center, the U.S. Department of Agriculture, and the National Oceanic and Atmospheric Administration. 2026a. National Drought Monitor data. University of Nebraska-Lincoln, Lincoln, NE. <https://droughtmonitor.unl.edu/> [Accessed June 15, 2026].
- U.S. Drought Monitor (USDM). 2026b. Drought Classification. University of Nebraska-Lincoln, Lincoln, NE. <https://droughtmonitor.unl.edu/About/AbouttheData/DroughtClassification.aspx> [Accessed July 30, 2026].
- U.S. Environmental Protection Agency (USEPA). 2009. Land-Use Scenarios: National-Scale Housing-Density Scenarios Consistent with Climate Change Storylines. Global Change Research Program, National Center for Environmental Assessment, Office of Research and Development, Washington, DC. EPA/600/R-08/076F. 137 pp.
- U.S. Environmental Protection Agency (USEPA). 2017. Updates to the Demographic and Spatial Allocation Models to Produce Integrated Climate and Land Use Scenarios (ICLUS) Version 2. National Center for Environmental Assessment, Office of Research and Development, Washington, DC. EPA/600/R-16/366F. 134 pp.

- U.S. Environmental Protection Agency (USEPA). 2025. Water Quality Assessment and Total Maximum Daily Loads Information System (ATTAINS). U.S. EPA Office of Water, Office of Wetlands, Oceans and Watersheds. <https://www.epa.gov/waterdata/attains> [Accessed June 1, 2026].
- U.S. Environmental Protection Agency (USEPA). 2026. ATTAINS: Geospatial Data. U.S. EPA Office of Water, Office of Wetlands, Oceans and Watersheds. https://gispub.epa.gov/arcgis/rest/services/OW/ATTAINS_Assessment/MapServer [Accessed June 1, 2026].
- U.S. Fish and Wildlife Service (Service). 1996. Recovery Plan for the Louisiana quillwort (*Isoëtes louisianensis* Thieret). U.S. Fish and Wildlife Service, Southeast Region, Atlanta, GA. 26 pp.
- U.S. Fish and Wildlife Service (Service). 2016. USFWS Species Status Assessment Framework: An Integrated Analytical Framework for Conservation. Version 3.4 (dated August 2016). 21 pp.
- U.S. Fish and Wildlife Service (Service). 2019. Three Granite Outcrop Plants: Black-spored Quillwort (*Isoëtes melanospora*), Mat-forming Quillwort (*Isoëtes tegetiformans*), and Little Amphianthus (*Amphianthus pusillus*): Five-Year Review: Summary and Evaluation Report. Athens, GA. 57 pp.
- U.S. Fish and Wildlife Service (Service). 2026. Characterizing Representation in SSAs: Putting Concepts into Practice. U.S. Fish and Wildlife Service, Headquarters, Ecological Services. <https://doimsp.sharepoint.com/sites/fws-FF09E00000/SitePages/Putting-Concepts-into-Practice.aspx> [Accessed June 16, 2026].
- U.S. Forest Service (USFS). 2002a. Croatan National Forest Land and Resource Management Plan: Forest Plan Appendices. U.S. Department of Agriculture Forest Service, Southern Region, Washington, DC. 121 pp.
- U.S. Forest Service (USFS). 2002b. Croatan National Forest Revised Land and Resource Management Plan: Final Environmental Impact Statement. U.S. Department of Agriculture Forest Service, Southern Region, Washington, DC. 531 pp.
- U.S. Forest Service (USFS). 2016. Croatan National Forest Monitoring Program Administrative Change to Revised Land and Resource Management Plan (05/02/2016). U.S. Department of Agriculture Forest Service, Southern Region, Washington, DC. 25 pp.
- U.S. Geological Survey (USGS). 2025. Annual National Land Cover Database (NLCD) Collection 1.0 Reference Data Product: U.S. Geological Survey data release, <https://doi.org/10.5066/P13EDMAF>.
- Van den Broeck, M., L. Rhazi, A. Waterkeyn, M. El Madihi, P. Grillas, J.M. Kneitel, and L. Brendonck. 2019. Livestock disturbances in Mediterranean temporary ponds: a mesocosm experiment with sheep manure and simulated trampling. *Freshwater Biology* 64(5):1-14.
- Vander Stelt, E., J.B. Fant, S. Masi, and D.J. Larkin. 2017. Assessing habitat requirements and genetic status of a rare ephemeral wetland plant species, *Isoëtes butleri* Englem. *Aquatic Botany* 138:1-31.
- Vöge, M. 1997. Number of leaves per rosette and fertility characteristics of the quillwort (*Isoëtes lacustris* L.) in 50 lakes of Europe: a field study. *Archive of Hydrobiology* 139(3):415-431.
- Vöge, M. 2006. The reproductive phenology of *Isoëtes lacustris* L.: results of field studies in Scandinavian lakes. *Limnologia: Ecology and Management of Inland Waters* 36:228-233.
- Vöge, M. 2014. Monitoring the vitality of by using a non-destructive method. *Limnological Review* 14(3):153-158.

- Vöge, M. 2015. Monitoring the vitality of *Isoëtes lacustris* by using a non-destructive method. *Limnological Review* 14:153-158.
- Wahl, M.H., H.N. McKellar, and T.M. Williams. 1997. Patterns of nutrient loading in forested and urbanized coastal streams. *Journal of Experimental Marine Biology and Ecology* 213:111–131.
- Wahl, T., S. Jain, J. Bender, S.D. Meyers, and M.E. Luther. 2015. Increasing risk of compound flooding from storm surge and rainfall for major US cities. *Nature Climate Change* 5:1093-1098.
- Walker, A., J. Amoroso, and M. East. 2024. List of Rare Plant Species of North Carolina 2024. North Carolina Natural Heritage Program, North Carolina Department of Natural and Cultural Resources, Division of Land and Water Stewardship, Raleigh, NC. 55 pp.
- Wang, J.Y., R.W. Gituru, and Q.F. Wang. 2005. Ecology and conservation of the endangered quillwort *Isoëtes sinensis* in China. *Journal of Natural History* 39(48):4069-4079.
- Wang, Y., H. Fukuda, P. Zhang, T. Wang, G. Yang, W. Gao, and Y. Lu. 2022. Urban wetlands as a potential habitat for an endangered aquatic plant, *Isoëtes sinensis*. *Global Ecology and Conservation* 34(e02012):1-12.
- Weakley, A.S., M. Lee, S. Ward, J.C. Ludwig, D.B. Poindexter, B.A. Sorrie, E. Ungberg, R. LeBlond, J.C. Kees, C.A. McCormick, S. Oberreiter, R. Lance, H. Ballard, R. Carter, M. Pyne, W. Knapp, K. Bradley, S. Zona, W. Barger, L. Musselman, P. Schafran, D. Spaulding, J. Nelson, E. Keith, G. Fleming, D. Estes, L. Majure, H. Medford, E. Bridges, R. Folk, Z. Murrell, R.K. Peet, J. Townsend, G. Nesom, A. Floden, J. Horn, E.E. Schilling, J. Triplett, T. Murphy, Z. Irick, K. Schoonover McClelland, M. Alford, M. Brock, B. Keener, H. LeGrand, T. O'Brien, T. Witsell, and others. 2025. Flora of the Southeastern United States. University of North Carolina at Chapel Hill Herbarium, Chapel Hill, NC. 2329 pp.
- Wen, M.Z., X.A. Pang, Q.F. Wang, and W.C. Taylor. 2003. Relationship between water chemistry and the distribution of the endangered aquatic quillwort *Isoëtes sinensis* Palmer in China. *Journal of Freshwater Ecology* 18:361-367.
- Wharton, C.H., and M.M. Brinson. 1978. Characteristics of Southeastern River Systems. In: J.S. Larson, V. Carter, D.A. Jones, and M.C. Kolipinski (eds.), *Strategies for Protection and Management of Floodplain Wetlands and Other Riparian Ecosystems*. U.S. Forest Service General Technical Report WO-12. Washington, DC. 410 pp.
- Wharton, C.H., W.M. Kitchens, and T.W. Sipe. 1982. The Ecology of Bottomland Hardwood Swamps of the Southeast: A Community Profile. FWS/OBS-81/37 Report for U.S. Fish and Wildlife Service, Washington, DC. 133 pp.
- Wickell, D. 2026. Peer Review of Thin-wall Quillwort (*Isoëtes microvela*) SSA. Comments provided via U.S. Fish and Wildlife Service Peer Review Portal [Accessed August 28, 2026].
- Wickell, D., L.Y. Kuo, H.P. Yang, A.D. Ashok, I. Irisarri, A. Dadras, S. de Vries, J. de Vries, Y.M. Huang, Z. Li, M.S. Barker, N.T. Hartwick, T.P. Michael, and F.W. Li. 2021. Underwater CAM photosynthesis elucidated by *Isoëtes* genome. *Nature Communications* 12(6348):1-13.
- Wickell, D., J. Landis, E. Zimmer, and F.W. Lei. 2023. Population genomics of the *Isoëtes appalachiana* (Isoëtaceae) complex supports a ‘diploids-first’ approach to conservation. *Annals of Botany* 133:261-272.
- Windham-Meyers, L., T.L. Root, M.D. Petkewich, M.L. Musgrove, A.C. Gill, J.C. Weaver, C.H. Conaway, B.D. Lindsey, F. Parchaso, N. Knowles, and E.J. Tomaszewski. 2024. State of Science, Gap Analysis, and Prioritization for Southeastern United States Water-Quality Impacts from Coastal

Storms—Fiscal Year 2023 Program Report to the Water Resources Mission Area from the Water Availability Impacts of Extreme Events Program—Hurricanes. U.S. Geological Survey Open-File Report 2024-1048, Reston, VA. 64 pp.

Wing, S., S. Freedman, and L. Band. 2002. The potential impact of flooding on confined animal feeding operations in eastern North Carolina. *Environmental Health Perspectives* 110(4):387-391.

Wolf, S., B. Hartl, C. Carroll, M.C. Neel, and D.N. Greenwald. 2015. Beyond PVA: why recovery under the Endangered Species Act is more than population viability. *Bioscience* 65(2):200-207.

Wood, D.P., J.K. Olofsson, S.W. McKenzie, and L.T. Dunning. 2018. Contrasting phylogeographic structures between freshwater lycopods and angiosperms in the British Isles. *Botany Letters* 165(3-4):476-486.

Wood, D., G. Besnard, D.J. Beerling, C.P. Osborne, and P.A. Christin. 2020. Phylogenomics indicates the “living fossil” *Isoëtes* diversified in the Cenozoic. *PLoS ONE* 15(6):1-28.

World Flora Online (WFO). 2026. WFO Taxonomic Report: *Isoëtes microvela*. <https://www.worldfloraonline.org> [Accessed January 28, 2026].

Wu, H., Y. Liu, T. Zhang, M. Xu, and B. Rao. 2024. Impacts of soil properties on species diversity and structure in *Alternanthera philoxeroides*-invaded and native plant communities. *Plants* 13(1196):1-17.

Wubs, E.R.J., R.G.A. Fraaije, G.A. de Groot, R.H.J. Erkens, A.G. Garssen, E. Kleyheeg, B.M. Raven, and M.B. Soons. 2016. Going against the flow: a case for upstream dispersal and detection of uncommon dispersal events. *Freshwater Biology* 61:580-595.

Xu, Z., J. Hilton, J. Yu, P.B. Wignall, A. Farnsworth, I.P. Montañez, N. Peng, Q. Liang, X. Sun, B.J.W. Mills, and B.H. Lomax. 2026. CAM photosynthesis may have conferred an advantage during the Permian-Triassic mass extinction event. *Nature Ecology and Evolution* 10:997-1010.

Yang, H., Y.Y. Chen, Y.X. Xu, and Z.Z. Li. 2011. Gene flow dynamics of ex-situ conservation populations in two endangered *Isoëtes* species: genetic implications for reintroduction, conservation and management. *Plant Science Journal* 20(5):319-330.

Yang, T., and X. Liu. 2015. Comparing photosynthetic characteristics of *Isoëtes sinensis* Palmer under submerged and terrestrial conditions. *Nature Scientific Reports* 5(17783):1-13.

Yin, D., Z.G. Xue, J.C. Warner, D. Bao, Y. Huang, and W Yu. 2021. Hydrometeorology and hydrology of flooding in Cape Fear River Basin during Hurricane Florence in 2018. *Journal of Hydrology* 603(D):1-41.

Zhang, J., W. Li, G. Yang, Y. Wang, X. Ji, X. Zhang, and Y. Lu. 2024. Rural development drives temporal turnover in plant diversity in a China’s scenic area. *Environmental Research Communications* 6(115031):1-19.

Zhu, G., W. Li, M. Zhang, L. Ni, and S. Wang. 2012. Adaptation of submerged macrophytes to both water depth and flood intensity as revealed by their mechanical resistance. *Hydrobiologia* 696:1-17.

Appendix A. Glossary of Terms

A-1 Terms used in SSA report.

Term	Definition
Anemochory	dispersal by wind
Corm	short, bulb-like, underground structure (rootstock) from which leaves (sporophylls) and roots radiate; may be 2 or 3-lobed
Dehisce	natural splitting or opening of a sporangium to release spores into the environment
Gametophyte	highly reduced, microscopic stage in quillwort alternation of generation life cycle that develops entirely inside the spore
Heterosporous	the condition of producing two different types of spores: larger female spores (megaspores) and smaller male spores (microspores) = dimorphic
Hydrochory	dispersal by water
Iteroparous/iteroparity	having multiple reproductive cycles throughout one's their lifetime
Lanceolate	significantly longer than wide, with widest point positioned below the midpoint and tapering to a sharp, pointed apex
Ligule	small, delicate, scale-like or membranous appendage located just above the sporangium on the inner leaf surface
Meiosis	cell division process yielding sexual gametes (sperm and egg cells); occurs within quillwort sporangia
Ornamentation	surface sculpturing/texture of spores
Ploidy	the number of sets of chromosomes present in a cell
Diploid	having two complete sets of chromosomes
Hexaploid	having six complete sets of chromosomes
Polyploid	having more than two complete sets of chromosomes
Polyploidization	process by which organisms acquire multiple sets of chromosomes = whole-genome duplication
Proximal	the side or surface of the spore that was originally facing inward and attached to the center of the spore-producing structure during development
Proximal ridge	three raised, linear seams that meet at the top (the proximal face) of a female reproductive spore
Reticulate	net-like or honeycomb-like surface pattern formed by a network of raised ridges
Scour	the direct removal of bank material at or below water levels by the physical action of flowing water and the sediment it carries
Senescence	natural aging and degradation process that occurs as a result of environmental cues; quillwort senescence is heavily dictated by water availability
Sporangium/a	large, sac-like reproductive capsule situated in a hollow cavity at the expanded, swollen base of each fertile leaf
Spore	microscopic reproductive cell produced by quillworts

Megaspore	larger, female spore; mature megaspore size and ornamentation are the primary tools used for species identification
Microspore	much smaller, male spore; produced in large quantities
Sporeling	juvenile sporophyte (plant)
Sporophyll	leaf that bears a sporangium
Sporophyte	dominant, visible, mature plant stage in quillwort alternation of generation life cycle
Synaptospory	the adhesion of microspores to megaspore outer coats, thus increasing the likelihood of spores landing together and facilitating fertilization even after long-distance dispersal
Tubercle	minute, mound-like projections found on the surface of quillwort spores often used for identification
Velum	thin, transparent membrane that partially or completely covers the sporangium (for protection)
Zoochory	dispersal by animals

Appendix B. *Isoëtes* Taxa

B-1 *Isoëtes* taxa (and life form(s)) reviewed and referenced in SSA report. Based on Brunton et al. 2025 (pp. 3-4); () indicate secondary life form; * indicates taxa occupying the southeastern United States.

	Aquatic	Amphibious	Terrestrial
Within North America	appalachiana*	(appalachiana)*	butleri*
	bolanderi	(butleri)*	(melanopoda ssp. melanopoda)*
	engelmannii*	(engelmannii)*	(melanospora)*
	(howellii)	georgiana*	minima
	hyemalis*	howellii	orcuttii
	(melanopoda ssp. silvatica)*	(hyemalis)*	(tegetiformans)*
	(microvela)*	louisianensis*	
	(mississippiensis)*	melanopoda ssp. melanopoda*	
	riparia ssp. riparia	melanopoda ssp. silvatica*	
		melanospora*	
		microvela*	
		(minima)	
		mississippiensis*	
		(orcuttii)	
		tegetiformans*	

Outside North America	alpinus	(alpinus)	
	cangae	asiatica	
	creussensis	coreana	
	echinospora	durieui	
	japonica	histris	
	lacustris	(japonica)	
	malinverniana	(muelleri)	
	muelleri	setacea	
	sabatina	sinensis	
	yunguiensis	taiwanensis	
	x brochonii	velata	
	x paratunica	volkanii	
		(yunguiensis)	

Appendix C. Occupied HUC-12 and HUC-10 Watersheds

C-1 HUC-12 sub-watersheds and HUC-10 watersheds known to be occupied by thin-wall quillwort.

AU	RU	HUC-12 Name	HUC-12 Code	HUC-10 Name	HUC-10 Code	HUC-6 River Basin
1	1	Rowan Creek-Black River	030300060805	Black River (Lower)	0303000608	Cape Fear
2	1	Rowan Creek-Black River	030300060805	Black River (Lower)	0303000608	Cape Fear
3	2	Middle Swamp	030402080101	Lockwoods Folly River	0304020801	Lumber
4	2	Middle Swamp	030402080101	Lockwoods Folly River	0304020801	Lumber
5	3	Headwaters Sandy Run Swamp	030300070602	Angola Swamp-NE Cape Fear River (Lower)	0303000706	Cape Fear
6	4	Town of Maysville-White Oak River	030203010102	Headwaters White Oak River	0302030101	White Oak
7	4	Town of Maysville-White Oak River	030203010102	Headwaters White Oak River	0302030101	White Oak

Appendix D. Cause and Effects Review

D-1 Cause and effects evaluation for Hydrological Alteration.

ESA Factor(s)	3.2.1 Flooding + Inundation
SOURCE(S)	A) Compound Flooding; B) Inundation + Submergence
Activity(ies)	A/B) Altered hydrodynamics: Multiple co-occurring flood drivers > increased precipitation + surface water flow ("flash flooding") > increased water velocity/water depth/duration greater than the species' natural range of variation + often accompanied by increased turbidity/sedimentation ("flash flood" = heavy rainfall-induced rapid water-level rise for a variable period of time, esp. as runoff moves to low-lying areas) (Deng et al. 2010, entire)
STRESSOR(S)	A) Hydraulic disturbance: Elevated water flow velocity + increased total suspended solids in water column. B) Hydraulic disturbance: Inability of light to penetrate water column
Affected Resource(s)	A) Increased variable flow velocity > increased drag forces on plant tissues. B) Reduced light availability > Reduced photosynthesis; Reduced C availability > reduced photosynthesis; Reduced O (hypoxia/temporary anoxia) (Davies-Colley and Smith 2001, entire; Mommer and Visser 2005, entire). EXTRA: Damming results in a prompt reduction of light because of water level increase and re-suspension of sediments and particles coming especially from the newly flooded area (Gacia & Ballesteros, 1998). In parallel, water column nutrients and suspended organic matter increases because of the decomposition of flooded terrestrial vegetation and the eventual die-off of aquatic vegetation (Gacia and Ballesteros 1998; Camargo et al. 2005). (Chappuis et al. 2015, p. 5).
Exposure of Stressor(s)	All TWQ life stages.
Immediacy of Stressor(s)	A) Present, future. B) Past, present, future.
Changes in Resource(s)	A) Augmented drag forces > plant sheer stress, breakage, and uprooting (Sand-Jensen et al. 1999; Sand-Jensen 2008; Bornette and Puijalon 2011; Sand-Jensen and Moller 2014). B) Diminished gas diffusion > root zone toxicity.
Response to Stressors: GENERAL INDIVIDUALS	A) Unclear. Adults, Sporelings: In saturated/submergent sediments with substantial OM, plants demonstrate reduced root anchorage due to reduced cohesive binding between root surfaces and surrounding sediment (Sand-Jensen and Moller 2014). B) Unclear. Species-specific response anticipated based on unique hydrological requirements. Different species likely respond along a "tolerance/mortality continuum" depending on respective needs and concomitant environmental factors. Many <i>Isoetes</i> are capable of crassulacean acid metabolism, which enhances Ps during inundation ('underwater CAM') (Keeley and Sandquist 1991, entire; Pedersen et al. 2011, entire; Yang and Liu 2015, p. 2; Wickell et al. 2021, entire). Adults, Sporelings, Embryos (not buoyant): if Ps limited (absence of CAM) > biomass loss > (eventual) mortality BUT if Ps possible (CAM-enabled) > biomass loss > reduced growth. Spores (buoyant): short/long-distance dispersal (Mommer and Visser 2005, entire; Catford and Jansson 2014, pp. 21-24).
Response to Stressors: AQUATIC INDIVIDUALS	ISLA (Spain): deeper inundation of submersed species originally at 0.2-2.3m flooded to 5.7-7.8m (dam construction) > combination of reduced leaf number with increased leaf length, die off (Gacia and Ballesteros 1996, p. 57-58). ISAU (Australia): (Pedersen et al. 2011, p.). Percent surface irradiance of submerged <i>Isoetes</i> for colonization

	estimated as 12.9%, suggesting a relatively high requirement relative to other submerged macrophytes (Middelboe and Markager 1997, entire, esp. p. 559).
Response to Stressors: AMPHIBIOUS INDIVIDUALS	ISOR (CA, USA): increasing frequency of occurrence with increased period of inundation (Bauder 2000, p. 52). ISDE (Spain): increased leaf production with prolonged emersion (Chappuis et al. 2016, entire). ISSE (France): Adults: increased leaf length and microsporangia number in flooded conditions (Rhazi et al. 2009, entire, esp. pp. 119-120).
Effects of Stressors: POPULATIONS (Resiliency)	
GEOGRAPHIC SCOPE	<i>What is the geographic extent of the stressor relative to the range of the species/populations? In other words, this stressor effects what proportion of the rangewide populations?</i> Unclear. We surmise that all populations would be affected equally. While there is an abundance of literature citing the influence of 1) storms across the region and the watersheds of interest, we have no direct evidence indicating current populations have sustained negative impacts as a result of this factor.
MAGNITUDE	<i>How large of an effect do you expect it to have on the populations?</i> Unclear. <i>Isoëtes</i> are known as general tolerators across a continuum of variable meteorologic/hydrologic conditions, including some magnitude of inundation/drought; however, their adaptation to shed sporophylls under inclement conditions imposes a challenge to verify if plants/populations are simply dormant or extirpated in response to extreme events.
SUMMARY	Has the potential to have a significant negative effect on the species. Propose to carry forward.
ESA Factor(s)	3.2.2 Sediment + Debris Flux
SOURCE(S)	A) Flood Erosion + Scouring; B) Flood Debris + Sedimentation
Activity(ies)	A) Under catastrophic/compound storm conditions, standard dynamic flood patterns may transition to >> Altered hydrodynamics: Increased surface water flow > substrate and/or plant erosion/destruction/removal via high-velocity floodwaters. B) Altered hydrodynamics: Increased wind + surface water flow > efflux/influx and redistribution of in-stream plant material (coarse woody debris (CWD), branches, leaves, etc.) and substrate movement/de-compaction resulting from fast moving floodwaters
STRESSOR(S)	A) Hydraulic disturbance: Increased variable water velocity > elevated suspended sediment in water column. B) Hydraulic disturbance: Augmented wind and water velocity > uplift and transport of vegetative (and other) material + sediment re-suspension in water column (TSS, total suspended sediments) (Mallin et al. 2009, entire)
Affected Resource(s)	A) Sediment flux > sediment transport away from roots and/or across entirety of habitat. B) Plant debris flux > storm debris carried overland/aquatically (debris flow) to new locales AND Sediment flux > sediment transport to new locales; increased water turbidity
Exposure of Stressor(s)	All TWQ life stages.
Immediacy of Stressor(s)	Past, present, future.

Changes in Resource(s)	A) Sediment transport > substrate failure/decreased intact substrate for root anchorage > plant uprooting (Sand-Jensen and Moller 2014); plant sheer stress, breakage. B) Vegetative debris and sediment transport > sediment and plant material deposition to other locales, including atop existing vegetation/plant tissue abrasion > "benthic smothering" by sediments/overlying vegetation > decreased photosynthesis (Davies-Colley and Smith 2001, entire, esp. p. 1085; Phillips 2022, entire)
Response to Stressors: GENERAL INDIVIDUALS	A) Adults, Sporelings, Spores: exogenous dispersal to other locales (dispersal >> fragmentation). Adults, Sporelings: forced uprooting/displacement of entrenched individuals, yielding complete elimination and/or patchiness/reduced biomass/density (Brunton and Britton 1996, p. 110; Catford and Jansson 2014, entire). B) Unclear. 1) Depends on structural characteristics of landscape and local setting (where plant exists- e.g., low riparian shore, high riparian shelf, etc.) > If high structural diversity exists, greater extant in-stream CWD may generate a sheltering effect, which reduces water velocity and sediment flow, thereby mitigating drag effects on plants/plant tissue (Liu and Wang 2024, entire). 2) Depends on substrate integrity. Abundant loose, coarser-grained sediments may bury inhabited vegetation (Phillips 2022, entire). Adults, Sporelings: Tolerance > mortality via suffocation (sediment smothering) ; phenotypic (stress) response: increasing aboveground biomass in response to augmented sedimentation in tandem with impaired light levels due to algal overgrowth (Gacia et al. 2018, pp. 6-8) > increased positive buoyancy > augmented uprooting (Sand-Jensen and Moller 2014, p. 435; Gacia et al. 2018, p. 8). Spores (buoyant): short/long-distance dispersal. (Catford and Jansson 2014, 21-24)
Response to Stressors: AQUATIC INDIVIDUALS	
Response to Stressors: AMPHIBIOUS INDIVIDUALS	ISGA (GA, USA): plants flattened and shaped by impact of rapidly flowing water (Brunton and Britton 1996, p. 110). Isx HYBRID-5 (VA, USA): site severely impacted by storm-associated (Hurricane Ernesto) flash flooding, resulting in absence of plants in subsequent survey years (Brunton and Sokoloff 2020, pp. 43-44).
Effects of Stressors: POPULATIONS (Resiliency)	Excessive litter and sediment accumulation may negatively influence <i>Isoëtes</i> regeneration dynamics (Leonard 1998, pp. 7-8; Hoot et al. 2000, pp. 9-10; Rhazi et al. 2004, pp. 681-682; Chappuis et al. 2015, entire; Vander Stelt et al. 2017, pp. 4-5)
GEOGRAPHIC SCOPE	<i>What is the geographic extent of the stressor relative to the range of the species/populations? In other words, this stressor effects what proportion of the rangewide populations?</i> A) Extensive; however, channel scouring is a natural ecological process of blackwater riverine systems and is likely experienced by all/majority TWQ populations. One TWQ site was noted to have been heavily scoured by hurricane stormwaters and debris, with Brunton and Britton (1998, p. 268) positing that the resultant erosion destroyed many plants (Schaner et al. 2021, pp. 9-10, 13). B) Unclear. One site has been noted as being subject to erosion as the result of recreational use.
MAGNITUDE	<i>How large of an effect do you expect it to have on the populations?</i> A) Existing in dynamic riverine-swamp systems, all TWQ populations are likely to be, at minimum, seasonally influenced by flood scouring. B) Aside from the TWQ site receiving heavy use, it is not anticipated that erosion will have a large-scale negative influence on TWQ populations in the near-term.
SUMMARY	We have no data to support that scouring and erosion are having a significant deleterious effect on the species. Propose not to carry forward.

ESA Factor(s)	3.2.3 Drying + Desiccation
SOURCE(S)	Prolonged Drought
Activity(ies)	A) Altered hydrodynamics: Long-term drying and desiccation of water bodies
STRESSOR(S)	A) Hydraulic disturbance: Substrate drying
Affected Resource(s)	A) Unavailability of water for metabolic activity
Exposure of Stressor(s)	All TWQ life stages.
Immediacy of Stressor(s)	Present, future.
Changes in Resource(s)	A) Water scarcity > plant senescence > plant mortality
Response to Stressors: GENERAL INDIVIDUALS	A) Species-specific response anticipated based on unique hydrological requirements. Different species likely respond along a "tolerance/mortality continuum" depending on respective needs and concomitant environmental factors. Many <i>Isoetes</i> capable of crassulacean acid metabolism, which proffers an ability to prevent desiccation when drought conditions arise ('dry-adapted CAM') (Keeley 1982, entire; Suissa and Green, 2020, p. 135; Wickell et al. 2021, pp. 1, 4). Amphibious Adults, Sporelings: drought deciduous (drop leaves) > dormancy (underground corm desiccation tolerance) > mortality (Keeley 1998, p. 144; Alejo-Jacuinde and Herrera-Estrella 2022, pp. 3-4; Gao et al. 2023, p. 2). Amphibious Spores: viability yielding eventual germination following desiccation to 180 d (air-dried), 275 d (low-temp storage), and 12 mo (Lopes et al. 2023, p. 10; Cavaleiro-Filho et al. 2025, p. 9).
Response to Stressors: AQUATIC INDIVIDUALS	Kott and Britton 1981?
Response to Stressors: AMPHIBIOUS INDIVIDUALS	ISME/ISTE (GA, USA): Observed resuscitation of individual plants following lengthy dry periods after extraction from habitat (Engelmann 1882, p. 26; Musselman 2001, p. 12). ISTA (Taiwan): although, drought tolerant, a prolonged drought period may diminish the competitive advantage over other aquatic plants (Huang et al. 2018, entire; Shih and Lee 2023, p. 1). Yu et al. (2013) pointed out that a consecutive drought of 1+ month adversely impacts ISTA survival (Shih and Lee 2023, p. 9). ISCA (Brazil): prolonged (3-6 mo) terrestrial conditions (drought) > increased proline/total antioxidant capacity/MDA content/carotenoid content and decreased chlorophyll a/b ratio leading to reductions in leaf size and root length BUT continued to produce reproductive structures and, after 12 mo desiccation, Spores (Lopes et al. 2023, pp. 9-10); genetic (transcription factors LEC1, LFY and hormones AUX1, GIB1) component to desiccation tolerance (Lopes et al. 2023, entire). ISTA (Taiwan): after more drought-tolerant, emergent leaves desiccate and shed, energy supplies are directed to dormant corms to survive drought stress, which may survive upwards of 6 mo (Huang et al. 2021, pp. 5-7) and can produce new "tougher" roots and leaves within 2-3 weeks of exposure to improved conditions (Huang et al. 2018, entire; Huang et al. 2021, p. 5, 7).
Effects of Stressors: POPULATIONS (Resiliency)	Under conditions of prolonged surface/soil water absence, or reduced ambient humidity, <i>Isoetes</i> regeneration dynamics may be affected > amphibious <i>Isoetes</i> plants develop mature spores before the reduction of soil humidity

	in order to maintain viability (Taylor & Hickey 1992, p. 621) AND with increasing temps associated with reduced water availability, possible de-synchronization of macro/microspores (Čtvrtlíková et al. 2011, pp. 11).
GEOGRAPHIC SCOPE	<i>What is the geographic extent of the stressor relative to the range of the species/populations? In other words, this stressor affects what proportion of the rangewide populations?</i> Unclear. Insufficient evidence to indicate presence of effects of major drought/desiccation in TWQ habitat or confirmation of purported effects on TWQ populations.
MAGNITUDE	<i>How large of an effect do you expect it to have on the populations?</i> Unclear. We surmise that all TWQ populations would be affected equally, although there is no direct evidence indicating current populations have sustained negative impacts as a result of this factor. In addition, <i>Isoëtes</i> readily shed aboveground biomass in variable climatic conditions, including drought, which masks the full magnitude of the species' response to extreme events.
SUMMARY	May have a significant negative effect on the species in the future. Propose to carry forward.
ESA Factor(s)	3.2.4 Climate Change
SOURCE(S)	Climate Change
Activity(ies)	Altered dynamics: Increased hydrological variability/storm intensity/ frequency > increased catastrophic/flash flooding + longer periods of extended drought (Gu et al. 2023, entire; Uehling and Schreck III 2024, entire; Bernal et al. 2025, p. 11). Synergistic/compound effects of multiple coexisting threats = regional-scale + localized threats acting together (Mulholland et al. 1997; Kunkel et al. 2020; Goodman and Baudau 2023)
STRESSOR(S)	Altered hydrodynamics: augmented flow velocity/inundation OR drying > <i>see Hydrological Alteration: 3.2.1-3.2.3</i> . Altered chemical composition: increased sediment/nutrient suspension > <i>see Water Quality: 3.3</i> . Altered floristic dynamics: enhanced competition/invasion > <i>see Vegetation Dynamics: 3.4</i> .
Affected Resource(s)	<i>See Hydrological Alteration: 3.2.1-3.2.3. See Water Quality: 3.3. See Vegetation Dynamics: 3.4.</i>
Exposure of Stressor(s)	All TWQ life stages.
Immediacy of Stressor(s)	Present? Future
Changes in Resource(s)	<i>Isoëtes</i> (polyploid): significant reduction in available suitable habitat under projected climate change (models SSP3-7.0 and SSP5-8.5) (Babin and Bell 2023, pp. 8, 10 ; Figure S1, p. 2). <i>See Hydrological Alteration: 3.2.1-3.2.3. See Water Quality: 3.3. See Vegetation Dynamics: 3.4.</i>
Response to Stressors: GENERAL INDIVIDUALS	Species-specific response anticipated based on unique hydrological requirements. Different species likely respond along a "tolerance/mortality continuum" depending on respective needs and concomitant environmental factors. Adults, Sporelings, Embryos: Tolerance > phenotypic (stress) response: increasing/decreasing aboveground biomass. Spores (buoyant): short/long-distance dispersal OR desiccation. <i>See Hydrological Alteration: 3.2.1-3.2.3. See Water Quality: 3.3. See Vegetation Dynamics: 3.4.</i>
Effects of Stressors: POPULATIONS (Resiliency)	ISSE (France): Adults: while exhibiting a high tolerance to desiccation, attributes (few macro/microsporangia, minimal leaf growth) under dry conditions reveal that it is unlikely to maintain itself throughout an extended period of

	dry climate (Rhazi et al. 2009, p. 122-123). <i>See Hydrological Alteration: 3.2.1-3.2.3. See Water Quality: 3.3. See Vegetation Dynamics: 3.4.</i>
GEOGRAPHIC SCOPE	<i>What is the geographic extent of the stressor relative to the range of the species/populations? In other words, this stressor effects what proportion of the rangewide populations?</i> Unclear. We have no direct evidence that extant populations are currently being affected, although they may be in the future as hydrological cycles continue to become more variable.
MAGNITUDE	<i>How large of an effect do you expect it to have on the populations?</i> Dependent on future time step and severity of incorporated factors. We do not anticipate the current magnitude of effect to be significant on current populations.
SUMMARY	May have a significant negative effect on the species in the future. Propose to carry forward.?

D-2 Cause and effects evaluation for Water Quality.

ESA Factor(s)	3.3.1 CAFOs
SOURCE(S)	Concentrated Animal Feeding Operations
Activity(ies)	Nutrient loading of neighboring water bodies via CAFO lagoon leakage/breaching/inundation (following facility structural failure, storms) (Burkholder et al. 1997, entire; Mallin et al. 1998, entire; Mallin et al. 2002, entire; Wing et al. 2002, entire; Mallin et al. 2004, entire; Christensen et al. 2013, entire; Harden 2015, entire; Martin et al. 2018, p. 892; Brown et al. 2020, entire; Chaffin et al. 2025, entire; Cooray et al. 2025, entire)
STRESSOR(S)	Rapid influx of N/P/OM > increase phytoactive algal/organismal growth (algal bloom/eutrophication) > algal death > augmented BOD/decreased DO
Affected Resource(s)	Hypoxia/anoxia of water bodies
Exposure of Stressor(s)	All TWQ life stages.
Immediacy of Stressor(s)	Present, future.
Changes in Resource(s)	Unavailability of O for metabolic activity
Response to Stressors: GENERAL INDIVIDUALS	Extent to which deleterious influence of CAFOs may impact downstream TWQ populations will likely vary depending on watershed setting factors (e.g., basin size, CAFO density and proximity to TWQ populations, stream morphology, and streamflow conditions) (Harden 2015, entire; Martin et al. 2018, entire). Blackwater systems are characteristically oligotrophic (low-nutrient) and elevated DOC systems (Mallin et al. 2004, entire; Hopper et al. 2025, p. 1288). ISLA/ISEC (Europe): Adults, Sporelings, Embryos, Spores: in presence of more competitive plant taxa resulting from nutrient influx: possible relegation to sub-optimal habitat > "density stress" > reduced leaf and megaspore production > discontinuation of spore production > decrease in occurrence (Grzybowski et al. 2008, entire; Voge 2014, pp. 153-154, 157). ISMA (NW Italy): While eutrophication by contamination with domestic or agricultural effluents is generally regarded as the main predictor of <i>Isoetes</i> decline (Smolders et al. 2002; Arts 2002; Wen et al. 2003; Liu et al. 2005; Pedersen et al. 2006), sites with high N/P/K/OC had greatest ISMA density and productivity (Barni et al. 2013, p. 11). <i>No amphibious taxa represented in available literature.</i>

Effects of Stressors: POPULATIONS (Resiliency)	No documented reports of CAFO impacts to native amphibious/terrestrial plant populations. However, existing documentation suggests that some aquatic plant communities (eelgrass, <i>Zostera</i> spp.) have disappeared along coastal NC due to reduced light availability and nitrate loading resulting from large upstream agricultural enhancements (Mallin et al. 2000).
GEOGRAPHIC SCOPE	<i>What is the geographic extent of the stressor relative to the range of the species/populations? In other words, this stressor affects what proportion of the rangewide populations?</i> CAFOs exist in many HUC-10s inhabited by TWQ, although not necessarily in close proximity. Further, CAFOs occur in HUC-12s inhabited by 2 TWQ populations (28% of total known populations), making the potential threat more pronounced, but with only 1 of these populations having upstream CAFOs in relatively close proximity (14%).
MAGNITUDE	<i>How large of an effect do you expect it to have on the populations?</i> Moderate impact anticipated depending on proximity in space (distance to TWQ sub-populations) and time (recency of breach/storm-inundation), as well as concurrent effects of other stressors.
SUMMARY	Has the potential to have a significant negative effect on the species. Propose to carry forward.
ESA Factor(s)	3.3.2 Urbanization
SOURCE(S)	Urbanization
Activity(ies)	Aquatic chemical flux: C export from stream system; N/P/chemical residue loading from urban stormwater runoff, esp. during rainfall events (Mallin et al. 2009, entire; O'Driscoll et al. 2010, entire; Gold et al. 2020, entire; McCabe et al. 2021, entire; Chaya et al. 2023, entire).
STRESSOR(S)	Chemical flux > decreased DOM/particulate C (Gold et al. 2020, entire); increased turbidity, fecal coliform bacteria, orthophosphate, P. ...??
Affected Resource(s)	Aquatic: O depletion/hypoxia/toxicity of water bodies. Terrestrial/amphibious: O depletion > plant mortality
Exposure of Stressor(s)	All TWQ life stages.
Immediacy of Stressor(s)	Present, future.
Changes in Resource(s)	Aquatic: When respiration drives daily average DO below 5 mg L ⁻¹ (≤4 mg L ⁻¹ for blackwater systems), the threshold often used to define DO impairment, aquatic life becomes stressed. Once DO concentrations reach hypoxic levels (<2 mg L ⁻¹), major mortality events occur (McCabe et al. 2021, p. 952). Urbanized streams behave in a flashier fashion than rural counterparts and have a greater occurrence of extreme flow events, thereby increasing sediment transport, erosion, channel depth/width, water temperature (O'Driscoll et al. 2010, entire, esp. pp. 612, 624).
Response to Stressors: GENERAL INDIVIDUALS	Adults, Sporelings, Embryos, Spores (ISSE, China): While untested, it has been posited that the species' habitat requirements make it amenable to planting in urban settings, as long as resident water bodies are neither too high nor too low in ammonium-nitrogen concentration (Wang et al. 2022c; Lu et al. 2024; Zhang et al. 2024, pp. 13-14).

Effects of Stressors: POPULATIONS (Resiliency)	Lack of data is apparent re: effects of urbanization-water quality on primary production and community succession, especially when trying to move from an understanding of how an individual plant might respond to an overall population or plant community response. Liming as a temporary measure to de-acidify eutrophic/urbanizing /Isoëtes lake habitat shown to result in increased root/shoot elongation, thereby easing their ability to be uprooted during storm events, thus having a population-level impact (Lucassen et al. 2012; Klimaszyk et al. 2020).
GEOGRAPHIC SCOPE	<i>What is the geographic extent of the stressor relative to the range of the species/populations? In other words, this stressor affects what proportion of the rangewide populations?</i> Current NLCD data demonstrate 8% imperviousness for a single population (14%), demonstrating low direct impact from urbanization.
MAGNITUDE	<i>How large of an effect do you expect it to have on the populations?</i> Negligible direct impact from urbanization anticipated.
SUMMARY	No direct negative effects anticipated in the near-term. Do not carry forward.
ESA Factor(s)	Other
SOURCE(S)	Catastrophic Storm Flooding
Activity(ies)	Rapid N loading via hurricane (precipitation)-induced flooding (sources: fossil fuel combustion, soil erosion, agricultural/residential fertilizer) (Mallin et al. 1999, entire; Mallin et al. 2002, entire; Schafer et al. 2020, entire; Neville et al. 2021, entire; Windham-Meyers et al. 2024, entire)
STRESSOR(S)	Rapid influx of N > increase phytoactive algal/organismal growth (algal bloom/eutrophication) > algal death > augmented BOD/decreased DO
Affected Resource(s)	Hypoxia/anoxia of water bodies
Exposure of Stressor(s)	All TWQ life stages.
Immediacy of Stressor(s)	Present, future.
Changes in Resource(s)	Unavailability of O for metabolic activity
Response to Stressors: GENERAL INDIVIDUALS	Adults, Sporelings, Embryos: (ISLA, Spain): fertilizer increases leaf production until (if nutrient increase persists) reaches eutrophic threshold > "stress effect" > reduced leaf lengthening+new leaf production > decreased leaf turgidity/increased necrosis > leaf mortality (exacerbated by reduced light) (Chappuis et al. 2015, pp. 4, 11-13). Spores: (ISLA, Spain): As low leaf numbers indicative of few(er) sporangia, also indicative of low spore quantity > low proportion of fertile individuals in population > reduced reproductive performance (Voge 1997, entire; Chappuis et al. 2015, p. 12). ISLA (Norway): eco-engineers with typically high root radial O losses (ROL), which maintain aerobic and nutrient-poor sediment conditions (thereby avoiding competition with other plants) > high root ROL is related to presence of many gas-filled lacunae in leaves (causing high buoyancy); however, in more fertile sediments (nutrient enriched) with increased CO2 availability, decreased root:shoot ratio reduces plant grip on soil due to lower cohesive strength on more loose (organic or degraded) sediments, which promotes uprooting (Schutten et al. 2005; Pedersen et al. 2011; Lucassen et al. 2012, p. 173). No amphibious taxa represented in available literature.

Effects of Stressors: POPULATIONS (Resiliency)	Increasing mechanical stress/damage to plants and increasing plant toxicity are anticipated during and after catastrophic flood events, resulting in deep, turbid, nutrient-enriched waters; in some cases, likely leading to widespread vegetation mortality (Mallin et al. 1999; Neville et al. 2021, p. 11).
GEOGRAPHIC SCOPE	<i>What is the geographic extent of the stressor relative to the range of the species/populations? In other words, this stressor affects what proportion of the rangewide populations?</i> Unclear. We surmise that all populations would be affected equally. While there is an abundance of literature citing the influence of past and present storms across the region and the watersheds of interest, we have no direct evidence indicating current populations have sustained negative impacts as a result of this factor.
MAGNITUDE	<i>How large of an effect do you expect it to have on the populations?</i> Unclear. As it pertains to dynamic hydrological conditions, <i>Isoetes</i> are viewed as stress tolerators; however, the combined effect of nutrient enrichment and flooding may push them past a threshold into decline/mortality.
SUMMARY	Unclear.

D-3 Cause and effects evaluation for Vegetation Dynamics.

ESA Factor(s)	3.4 Interspecific Vegetation Dynamics
SOURCE(S)	3.4.1) Woody/Herbaceous Encroachment; 3.4.2) Exotic Invasive Plants
Activity(ies)	3.4.1/3.4.2) Increased density/cover/biomass ("dominance") of woody, ruderal, and/or exotic invasive vegetation
STRESSOR(S)	3.4.1/3.4.2) Light attenuation by live/standing dead woody/herbaceous and/or exotic invasive canopy; 3.4.1/3.4.2) Space and water/nutrient resource acquisition by live/standing dead woody/herbaceous and/or exotic invasive vegetation
Affected Resource(s)	3.4.1) Reduced light, water, and nutrient availability resulting from canopy dominance by woody/ruderal species (Craine and Dybzinski 2013, entire); 3.4.2) Decreased light/nutrient/water availability by exotic invasive plants (e.g., <i>Alternanthera philoxeroides</i> (ALPH, Amaranthaceae), which reproduces primarily by clonal fragmentation) (Harms et al. 2023, entire; Hao et al. 2024, p. 2)
Exposure of Stressor(s)	3.4.1/3.4.2) All TWQ life stages.
Immediacy of Stressor(s)	Present, future.
Changes in Resource(s)	3.4.1/3.4.2) Increased resource competition; 2) Rapid/lagged range expansion of exotic invasive taxon; 3.4.2) Possible facilitation of establishment and spread of other nonnative species (Gioria et al. 2013, entire); 3.4.2) Growth inhibition by ALPH secondary metabolites (alkaloids, organic acids, and flavonoids) and soil microbe allelopathy (Hao et al. 2023, pp. 2-3; Jia et al. 2026, pp. 8-10)
Response to Stressors: GENERAL INDIVIDUALS	3.4.1: ISSE (France): Adults, Sporelings, Embryos, Spores: light attenuation > "tolerance threshold" > reduced germination/growth/spore dispersal > decreased leaf /spore production and bulb mass (Rhazi et al. 2004, entire, esp. pp. 681-682); Spores: lower spore number and quality (Rhazi et al. 2004, p. 682). 3.4.2: While response is dependent on specific competitive strategy employed by invader as well as hydrological environment, <i>Isoetes</i> are recognized as notoriously poor competitors overall. ISSE (France): Adults, Sporelings: increased

	productivity/biomass by invader > space and resource dominance by invader > reduction of ISSE growth through restricting resource acquisition > diminished abundance > competitive exclusion by invader (enhanced in wet and flooded conditions) (Rhazi et al. 2009, entire). ISSE (China): competitive exclusion by invasive plants (Liu et al. 2005, p. 336).
Effects of Stressors: POPULATIONS (Resiliency)	3.4.2: ISSE (France): diminished abundance through competitive exclusion via plant invasion (Rhazi et al. 2009, entire).
GEOGRAPHIC SCOPE	<i>What is the geographic extent of the stressor relative to the range of the species/populations? In other words, this stressor effects what proportion of the rangewide populations?</i> Somewhat unclear. 3.4.1) Most habitats in which TWQ occurs are classified as (blackwater) riverine swamp, thus having a persistent expansive overstory (canopy) during the growing season. And we have no direct evidence demonstrating an effect of woody encroachment on TWQ populations. 3.4.2) Although there have been recent observations of invasive alligator-weed (<i>Alternanthera philoxeroides</i>) co-occurrence at 3 of 9 known TWQ sites (i.e., 33% of total, with 30-40% cover at one locale) (Schaner et al. 2021, p. 14), there has been no confirmation of its deleterious effects on TWQ populations.
MAGNITUDE	<i>How large of an effect do you expect it to have on the populations?</i> Depends on abundance and pressure of invading species, i.e., ALPH. Schaner et al. (2021, pp. 14, 17-18) noted presence of ALPH at 3 populations--one at low density, one negligible, and one having multiple patches, with one patch of 30-40% cover (affecting ~145 individuals) (NCNHP 2026a, unpaginated?).
SUMMARY	Negative functional impact of ALPH on TWQ unclear. Requires additional input/data. With increasing climate change, increasing advancement/invasion may be anticipated in the future. Propose not to carry forward.

D-4 Cause and effects evaluation for Habitat Modification and Loss.

ESA Factor(s)	3.5 Habitat Modification + Loss
SOURCE(S)	3.5.1) Recreation-Associated Trampling + Erosion; 3.5.2) Construction + Development
Activity(ies)	3.5.1) ("Small-scale") traversing TWQ habitat for access to shoreline recreational site(s) via human foot traffic; 3.5.1) Wave action disturbance on shoreline during low-water periods; 3.5.2) ("Large-scale") traversing, excavation, backfilling, removal of TWQ habitat via heavy equipment
STRESSOR(S)	3.5.1) Plant mechanical injury/uprooting and/or direct mortality; 3.5.1) Soil structural change(s) > substrate instability; 3.5.2) Elimination of habitat; 1/2) Alteration of surface-water flow. 3.5.2) Site/habitat destruction > plant destruction via paving/soil turnover ("smothering") (construction activities). ISAS (Kamchatka, Russia): lakeshore populations affected by active recreation when shallows are used as a swimming beach (Mochalova et al. 2015, p. 110); ISVO (Turkey): utilization pressure of lakeshore habitat for resting, natural wandering, picnicking and spending time (Keskin and Karahan 2025, p. 190)
Affected Resource(s)	3.5.1) Tissue abrasion (via trampling/wave action) > tissue degradation; 1) Plant death; 1) Plant dispersal?; 3.5.1) Erosion/Bioturbation (via trampling/wave exposure) > increased turbidity/sedimentation > diminished water quality; 3.5.2) Decreased habitat quality > reduced habitat suitability > decreased plant establishment; 3.5.2) Traversing: mechanical injury; 3.5.2) Excavation: soil/plant elimination; 2) Backfilling: plant smothering > mortality. ISAS (Kamchatka, Russia): direct mechanical disturbance of habitats and via strongly increasing pH/ mineralization

	of lake water > Physical disturbance often favorable for <i>Isoëtes</i> hybridization, because it creates habitats where the spores of two or more species can intermix (Mochalova et al. 2015, p. 111). ISLA/ISEC (Estonia): plants trampled and in pieces due to lakeshore use by swimmers > elimination of <i>Isoëtes</i> beds in areas of high use > erosion due to scarce vegetation (Maemets 2025, p. 15)
Exposure of Stressor(s)	All TWQ life stages.
Immediacy of Stressor(s)	Present, future.
Changes in Resource(s)	
Response to Stressors: GENERAL INDIVIDUALS	3.5.1/3.5.2) Unclear. Indication of opposing response mechanisms to physical disturbance, possibly dependent on species-specific traits > Adults, Sporelings, Embryos: mechanical damage to tissue > direct mortality OR dispersal. Spores (buoyant): if dispersed aquatically, short/long-distance dispersal; if dispersed terrestrially, possible desiccation/mortality. 3.5.1) Increased turbidity/sedimentation > possible reduction in plant photosynthetic capability > biomass loss > reduced growth. ISBR (France): Adults, Sporelings: plants destroyed by dam construction (unclear if this refers to inundation or the impacts of construction itself) (Brunton et al. 2020, p. 391). ISVE (Morocco): cover/abundance positively associated with physical disturbance (Van den Broeck et al. 2019, p. 8). ISHI (France): readily occurs in both trampled and natural conditions, with physical disturbance creating large numbers of new locales (n=40; i.e., +25%) (Kerbiriou et al. 2008, pp. 10, 12). ISAL (New Zealand): while reduced via exposure, one of most common littoral taxa in lake habitat exposed to wave action and associated sediment motion (Riis and Hawes 2003, p. 82). ISxBR (VA, USA): areas where road and/or bridge construction has taken place are suitable sites (Musselman and Knepper 1994, p. 56)
Effects of Stressors: POPULATIONS (Resiliency)	3.5.1/3.5.2) Possible fragmentation of impacted population; 3.5.1/3.5.2) If plant propagules (spores) successfully dispersed, possible establishment and creation of new population(s). ISHI (France): physical disturbance created large numbers of new locales (4n=40; i.e., +25%) (Kerbiriou et al. 2008, p. 12). ISSI (China): destruction from urban infrastructure construction (Wang et al. 2006, p. 4074)
GEOGRAPHIC SCOPE	<i>What is the geographic extent of the stressor relative to the range of the species/populations? In other words, this stressor affects what proportion of the rangewide populations?</i> Limited. We have evidence that 1 TWQ population has been/is being impacted by recreational activities (as well as mowing and wave exposure), which appears to have reduced the size (abundance) of the population or eliminated it entirely (Schaner et al. 2021, pp. 12-13, 18-19). A second population is within some distance of a recreation access point, but no further information is known.
MAGNITUDE	<i>How large of an effect do you expect it to have on the populations?</i> The magnitude of effect for the 1 impacted site (14% of total known populations) is large and remains dependent on ongoing shoreline use for recreation (which is likely); however, other TWQ sites remain untouched. Overall, this factor is restricted in scope, and we do not anticipate impacts to be sustained by 6 of 7 (86%) of extant populations.
SUMMARY	Does not appear to be having a significant negative effect on the species as a whole. Propose not to carry forward.

Appendix E. Current Land Cover

E-1 Current land cover for each analysis unit (AU/population) and HUC-12 watershed (USGS 2025, entire).

AU	Measure	Open Water	Developed	Barren	Forest	Shrub Grassland Herb	Pasture Cultivated	Woody Wetlands	Emergent Herb Wetlands
1	AU (ac)	42.9	109.0	63.4	529.5	102.5	26.7	611.6	5.8
	% of AU	2.9	7.3	4.3	35.5	6.9	1.8	41.0	0.4
	HUC-12 (ac)	176.4	1244.1	282.4	8746.1	2753.5	1875.5	6971.4	448.3
	% of HUC-12	0.8	5.5	1.3	38.9	12.2	8.3	31.0	2.0
2	AU (ac)	21.1	33.4	0	265.1	58.9	72.1	515.1	4.2
	% of AU	2.2	3.5	1.0	27.3	6.1	7.4	53.1	0.4
	HUC-12 (ac)	467.3	1039.3	142.8	7530.5	3391.7	2580.2	10182.3	861.3
	% of HUC-12	1.8	4.0	0.6	28.8	13.0	9.9	38.9	3.3
3	AU (ac)	0	284.0	0	707.4	97.2	149.7	380.3	0.7
	% of AU	1.0	17.5	1.0	43.7	6.01	9.2	23.5	0.04
	HUC-12 (ac)	35.1	1555.0	57.2	0	1228.3	1103.7	6622.5	239.3
	% of HUC-12	0.2	8.7	0.3	7110.4	6.8	6.2	36.9	1.3
4	AU (ac)	2.0	114.5	1.3	413	93.9	108.8	290.2	-1.1
	% of AU	0.2	11.2	0.1	40.3	9.2	10.6	28.3	-0.1
	HUC-12 (ac)	35.1	1555.0	57.2	0	1228.3	1103.7	6622.5	239.3
	% of HUC-12	0.2	8.7	0.3	7110.4	6.8	6.2	36.9	1.3
5	AU (ac)	1.3	25.1	0	332.9	39.4	0	381.2	0.2
	% of AU	0.2	3.2	1.0	42.7	5.1	0	48.9	0.03
	HUC-12 (ac)	15.4	647.8	5.3	5163.1	2574.9	923.8	7522.5	191.3
	% of HUC-12	0.1	3.8	0.03	30.3	15.1	5.4	44.1	1.1
6	AU (ac)	0	25.1	0	146.3	63.8	157.2	383.6	0.0
	% of AU	0	3.2	1.0	18.9	8.2	20.3	49.4	1.0
	HUC-12 (ac)	15.1	1139.8	24.5	6048.5	2486.8	3497.4	6490.4	588
	% of HUC-12	0.1	5.6	0.1	29.8	12.3	17.2	32.0	2.9

7	AU (ac)	82.3	171.9	71.8	81.2	58.7	115.9	192.4	2.7
	% of AU	10.6	22.1	9.3	10.5	7.6	14.9	24.8	0.3
	HUC-12 (ac)	244.6	661.2	446.1	3108.2	924.1	1955.7	4160.1	8.0
	% of HUC-12	2.1	5.8	3.9	27.0	8.0	17	36.2	0.1

Appendix F. Adaptive Capacity Review

F-1 Adaptive capacity (AC) tables for thin-wall quillwort (following Thurman et al. 2020, entire). Blue font indicates those AC attributes that best characterize species representation.

	AC Attribute	AC Assessment	Justification	Evidence
DISTRIBUTION	Extent of Occurrence {extent of occurrence, EOO, range, distribution, occurrence, occupancy}	MODERATELY LOW: 100-5000 km² = MODERATE	EOO is deemed equivalent to the species known range, which was estimated at ~2850 km ² (~1100 mi ²) via computing a minimum convex polygon around all known occurrences in GIS (Richman et al. 2015, p. 3). However, with moderate to broad habitat tolerances/requirements, TWQ's EOO is likely substantially larger than presented here.	LOW: expert best estimate/personal knowledge/opinion, unpublished data, or discrepancies in available information
	Area of Occupancy {area of occupancy, AOO range, distribution, occurrence, occupancy}	HIGH: >2000 km²	Due to the occurrence of TWQ within highly contiguous riverine swamp habitats of NC's Coastal Plain, much that has not been developed, it is presumed that suitable habitat for the species is fairly expansive and is likely greater than the current known 2850 km ² EOO.	LOW: expert best estimate/personal knowledge/opinion, unpublished data, or discrepancies in available information
	Habitat Specialization {habitat specialization, habitat utilization, habitat association, habitat}	HIGH: Having a clear preference for a particular habitat (contains >85% of occurrences), but the habitat is among the dominant types within the species' range; OR somewhat flexible in habitat utilization; OR described as a habitat generalist and/or occurrence has been documented on widely varied habitat types	TWQ is currently known to persist in fluvial blackwater stream swamps, with varying habitat forms classified as Coastal Plain Small Stream Swamp, Blackwater Bottomland Hardwood Forest, Blackwater Levee/Bar Forest, or Sand and Mud Bar (Schafale 2024). More or less believed to be a habitat generalist within these systems. Many quillwort taxa on other continents are classified as being specialists of oligotrophic aquatic systems.	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)
	Commensalism with Humans {human tolerance, human avoidance, disturbance}	MODERATE: Moderately tolerant of human influences, utilization of semi-natural landscapes (e.g., agricultural fields, suburban parks, etc.)	While not generally reported in heavily anthropogenically impacted or created habitats, has been observed in three roadside pools (w/i 50 ft of road) (Schaner et al. 2021). Generally, human interaction (i.e, recreation, trampling) with the species has resulted in poor outcomes (Schaner et al. 2021). Some quillwort taxa, however, have been suggested to be amenable to persistence in urbanized settings as long as water bodies maintain	MODERATE: one peer-reviewed publication (or information in support of assessment is location- or context-specific)

			acceptable quality (Capotorti et al. 2013; Wang et al. 2022c). Similarly, some <i>Isoetes</i> have been observed to thrive in agricultural settings, although supporting literature is limited (Thomas 1985).	
	Geographic Rarity {geographic rarity, rarity, occurrence, distribution, range, population size, connectivity}	MODERATE: Geographically restricted with highly connected populations (e.g., endemic species) AND/OR broadly distributed with sparse or isolated populations	Quillworts often exhibit endemism in response to limited dispersal, and thus have restricted geographic ranges, making gene flow between populations unlikely (Vander Stelt et al. 2017). TWQ is an endemic that appears somewhat rare within its range, with populations broadly spatially distributed throughout, and evidence of clustered and/or potentially connected populations as well as more isolated populations.	LOW: expert best estimate/personal knowledge/opinion, unpublished data, or discrepancies in available information

	AC Attribute	AC Assessment	Justification	Evidence
MOVEMENT	Dispersal Syndrome {dispersal capacity, dispersal behavior, dispersal, movement}	Passive, sessile HIGH: Wind or water currents	TWQ dispersal behavior has not been formally evaluated, but as a polyploid (hexaploid) is assumed to have low genetic diversity, indicating reduced spore dispersion among populations ("dispersal-limited") (Kim et al. 2009; Santos et al. 2020; Suissa et al. 2022; Wickell et al. 2023). The degree of (genetic) connectivity between spatially segregated TWQ populations remains unknown. Spore dispersion of amphibious quillwort taxa is more complicated than for aquatics, being limited by the rainy season and other factors, such as wind, thereby affecting the genetic structure of populations and direction of gene flow (Troia 2016, entire; Ma et al. 2019, entire).	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)
	Dispersal Distance {dispersal distance, dispersal capacity, dispersal behavior, dispersal, movement}	MODERATE: Species is characterized by limited to restricted dispersal or movement capability; species rarely disperses through unsuitable habitat more than about 10-100 m per dispersal event; OR dispersal beyond a very limited distance (or outside a small, isolated patch of	This aspect of TWQ life history has not been assessed (true for most quillwort taxa). However, based on expert input, we assume a population buffer distance of 1km upstream, downstream, and laterally from EOs, with the understanding that propagules (spores, sporangia, entire plants) likely readily disperse up to 1km. Being a semi-aquatic organism, disperses via water currents and is purportedly transported by wildlife (Brunton 2001; Hoot et al. 2006; Larsén and Rydin	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)

		suitable habitat) periodically or irregularly occurs, but is dependent on highly fortuitous or rare events; OR species has substantial movement capabilities but exhibits a very high degree of site fidelity	2016; Troia 2016). May conceivably “hitch-hike” on the gales of severe spring storms, but this is deemed uncertain and rare (Brunton 2001). In this manner, its dispersal/ movement capability may not be truly restricted.	
	Dispersal Phase {dispersal capacity, dispersal behavior, dispersal, movement}	HIGH: Long period or throughout its life	TWQ dispersal behavior has not been formally evaluated. However, maturation of individual sporelings to a reproductive stage likely varies by species/habitat and has been documented as within 12 months and upwards to ~2-3 yr, although sporangia size may be smaller in early reproductive cycles (Voge 2006; COSEWIC 2019; Caldeira et al. 2021; Cavaleiro-Filho et al. 2021). Because all sporophylls (leaves) are capable of becoming fertile (developing functional sporangia), once mature, individual plants remain capable of reproduction annually throughout the remainder of their lifetime (Jermy 1990; Čtvrtlíková et al. 2023, p. 4).	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)

	AC Attribute	AC Assessment	Justification	Evidence
EVOLUTIONARY POTENTIAL	Genetic Diversity {genetic diversity, genetic variability, genetic variation, genetics, heterozygosity, genetic adaptation, adaptation, phenotypic plasticity}	UNKNOWN	Recent literature suggests that TWQ is most closely related to <i>I. melanopoda</i> and <i>I. melanopoda</i> ssp. <i>silvatica</i> (progenitors) (Schafran 2019), although no further genetic information is known. Low gene flow between populations is assumed for all <i>Isoetes</i> , and often observed with increasing distance, which may increase genetic differentiation over time (thus, genetic diversity) (Chen et al. 2005; Kang et al. 2005; Kim et al. 2008; Chen et al. 2012; Ma et al. 2019). Research suggests that polyploid species have reduced genetic diversity relative to diploid relations (Kim et al. 2009; Suissa et al. 2022).	MODERATE: one peer-reviewed publication (or information in support of assessment is location- or context-specific)
	Population Size {population size, population, abundance}	LOW to MODERATE: Less than 250 to less than 2500 mature individuals	Known current TWQ populations encompass between 0-5 and 1828 mature individuals (Schaner et al. 2021; NCNHP 2026a; NCNHP 2026b)	MODERATE: one peer-reviewed publication (or information in support of

				assessment is location- or context-specific)
	Hybridization Potential {hybridization, hybrid vigor, fitness, interbreeding}	HIGH: Hybridization occurs; offspring are viable DON'T KNOW IF OFFSPRING ARE VIABLE	Hybridization is the rule for <i>Isoetes</i> plants, not the exception (with polyploidization a frequent associate) (Hickey et al. 1989; Liu et al. 2004; Wickell et al. 2023)!	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)
ECOLOGICAL ROLE	Competitive Ability {competition, interspecific competition, biotic interactions, species interactions, community dynamics}	LOW: Competitively inferior; strongly affected by a native or non-native species that is likely to be favored by climate change	While formal behavior has not been evaluated, most <i>Isoetes</i> species are regarded as stress-tolerators rather than successful competitors (Rhazi et al. 2009; Vander Stelt et al. 2017; Looy et al. 2021) although within the genus, polyploid taxa may be more competitive than diploids within their geographical ranges (Liu et al 2004). TWQ has been collected alongside various native herbaceous plants, but only one significant exotic invasive plant (alligatorweed) has been noted to exist at some populations (Schaner et al. 2021; NCNHP 2026a).	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)
	Diversity of Obligate Species {obligate species, mutualisms, symbionts, interspecific dependencies}	UNKNOWN	While unknown for TWQ or closely related taxa in the genus <i>Isoetes</i> , symbioses may occur (Fuentes-Jacques et al. 2022).	MODERATE: one peer-reviewed publication (or information in support of assessment is location- or context-specific)

	AC Attribute	AC Assessment	Justification	Evidence
ABIOTIC NICHE	Seasonal Phenology {phenology, seasonal behavior, life cycle, timing}	UNKNOWN	Like other quillworts, water fluctuations/depth, ambient temperature, and precipitation regimes likely playing a key role in proffering important environmental cues for TWQ reproductive activity, like spore germination and dispersal (Čtvrtlíková et al. 2012; Baskin and Baskin 1979; Troia 2016; Vander Stelt et al. 2017; Santos et al. 2020). Nonetheless, we have no documentation of such behavior.	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)

	Climatic Niche Breadth {climatic niche, niche, climatic tolerances, climate change}	MODERATE: Species is somewhat to moderately (10-90%) restricted to particular climatic/hydrological condition that may be lost or reduced in the assessment area as a result of climate change	Formal evaluation of TWQ climatic tolerance has not been performed; however, quillworts are known to respond to desiccation and drought conditions and/or other hydrological fluxes via undergoing leaf senescence and dormancy to survive (Rhazi et al. 2009; Huang et al. 2018; Alejo-Jacuinde and Herrera-Estrella 2022; Gao et al. 2023; Montiel-Molina et al. 2023).	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)
	Physiological Tolerances {physiological tolerances, physiology, thermal tolerance, thermal optima, stress response, phenotypic plasticity}	UNKNOWN, but assumed HIGH	Formal evaluation of TWQ physiological tolerance has not been conducted, and response to novel conditions is unknown. However, its polyploidy could engender it with augmented genetic variation, which could then potentially enhance its tolerance to a broad spectrum of ecological conditions (Liu et al. 2004; Husband et al. 2013; Lopes et al. 2023; Babin and Bell 2023; Almeida and Santos Leal 2024). Further, per other Isoetes, CAM photosynthesis may facilitate increased tolerance of underwater submergence (Pedersen et al. 2011; Yang and Liu 2015, entire). Outcomes are context-dependent, with polyploidy also linked with increased rates of extinction in some plants (Arrigo and Barker 2012). Similarly, much research demonstrates that <i>Isoetes</i> are readily desiccation tolerant (Rhazi et al. 2009; Alejo-Jacuinde and Herrera-Estrella 2022; Gao et al. 2023; Cavalheiro-Filho et al. 2025).	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)
	Behavioral Regulation of Physiology {behavior, flexibility, activity, exposure}	MODERATE: Moderate behavioral flexibility and reduction in exposure; behavior is infrequent, or is occasionally limited due to restricted access to resources; OR behavior temporarily restricts foraging or reproductive activities, but is not detrimental to survivability or fitness	Quillworts, including TWQ, regulate their physiology via leaf senescence and dormancy (Rhazi et al. 2009; Huang et al. 2018; Alejo-Jacuinde and Herrera-Estrella 2022; Gao et al. 2023; Montiel-Molina et al. 2023).	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)
	AC Attribute	AC Assessment	Justification	Evidence

LIFE HISTORY	Reproductive Phenology {phenology, seasonal behavior, life cycle, timing, reproduction, breeding, mating, life history}	LOW: Dependence on environmental cue; species is incapable of adjusting the timing or duration of reproductive events (i.e., detectable change in cue, but no detectable change in the phenological variable measured); OR timing indirectly dependent and linked to non-environmental cue (e.g., resource availability), which may result in fitness consequences due to misalignment between reproductive events and climate	We have no knowledge of specific TWQ reproductive behaviors, and it is not known if/how well the species adjusts to changing factors. And, while quillwort behavior as a whole generally tracks environmental conditions (e.g., earlier spore germination and sporeling development at higher temps, but not prolonged such that micro-/megaspore germination becomes desynchronized; cold stratification to break megaspore dormancy; spore/leaf maturation and release in rainy and dry seasons, respectively), there is a great deal of displayed variability and may be species and/or region-specific (Kott and Britton 1982; Voge 2006; Čtvrtlíková et al. 2012; Čtvrtlíková et al. 2014; Campos et al. 2021).	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined) Much variation across species
	Mating System {mating system, reproduction, breeding, mating, life history}		Quillwort species perform sexual reproduction via outcrossing and as well as self-fertilization and, in some cases, is capable of asexual reproduction (Chen et al. 2010; Dong et al. 2018; Ma et al. 2019; Caldeira et al. 2021; Thomas et al. 2026). Selfing is noted to be more common in polyploid taxa than diploids (Husband et al. 2013; Dar and Rehman 2017). As well, both underwater and terrestrial sexual reproduction has been observed in <i>Isoetes</i> (Wood et al. 2018; Santos et al. 2020).	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)
		HIGH: Cross-fertilization; sexual reproduction (but also self-fertilization, and asexual) DON'T KNOW WITH 100% CERTAINTY HOW TWQ REPRODUCES, BUT QUILLWORTS AS A WHOLE ARE CAPABLE OF ALL 3		
	Fecundity {fecundity, offspring, propagules, life history}	HIGH: Many offspring or propagules (>10)	TWQ reproduction has not been directly tracked. Adult quillworts are highly fecund, however, with the number of spores formed within a sporangium deemed high, with as many as 150,000 to 1,000,000 (avg. 300,000 to 400,000) microspores and up to hundreds (50-300) of megaspores produced during one sporulation event/season (Smith 1990; Husk et	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g.,

			al. 2000; Wang et al. 2005; Voge 2006; Blackmore et al. 2012; Nunes et al. 2018; Caldeira et al. 2019; Larsén et al. 2025).	taxonomically determined)
	Parity {parity, reproduction, life cycle, life history}	HIGH: Polycarpic; multiple reproductive events per lifetime	Quillworts have multiple reproductive events in their lifetime. While perennial, only a small portion of aboveground plant material persists from year to year; as such, fertile leaves/sporangia/spores are regenerated and produced annually (Smith 1900).	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)
	Sex Ratio {sex ratio, males, females, males to females, sex-biased survival, male-biased, female-biased}	HIGH	While there is no documentation for TWQ, we assume its monoecy because most <i>Isotetes</i> species are documented to be monoecious (e.g., single plant produces both male and female spores) (Turner et al. 2005, p. 86).	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g., taxonomically determined)

	AC Attribute	AC Assessment	Justification	Evidence
DEMOGRAPHY	Life Span {life span, longevity, life expectancy, mortality, survival, survivorship, demography, demographics}	MODERATELY HIGH: 1-10 years	TWQ lifespan has not been tracked; however, <i>Isotetes</i> plants may become rather old, with some species documented at 8-10 years, and upwards to 50+ years (Voge 2003; Voge 2006; Čtvrtlíková et al. 2020; Čtvrtlíková et al. 2023).	MODERATE: one peer-reviewed publication (or information in support of assessment is location- or context-specific)
	Generation Time {generation, maturity, reproduction, population growth, demography, demographics}	MODERATELY HIGH: 1-10 years	While we do not know the duration of maturation in TWQ, other quillworts mature into reproductive adults in various timeframes (sexual: 5 years; asexual: 1 year) (COSEWIC 2019; Caldeira et al. 2021).	LOW: expert best estimate/personal knowledge/opinion, unpublished data, or discrepancies in available information
	Age of Sexual Maturity {age of sexual maturity, age at recruitment, maturation, maturity, reproduction, population growth,}	HIGH: Rapid (early relative to lifespan)	While we do not know the age of sexual maturation in TWQ, other quillworts may become reproductively mature (produce spores) in 1 year, although may be influenced by environmental conditions (Voge 2003). Nonetheless, spore maturation may extend across seasons and into a second year (Voge	HIGH: accepted consensus from peer-reviewed literature (at least two publications in agreement), or general knowledge (e.g.,

	demography, demographics, life history}		2006). Some state that only after the development of sterile leaves for a few seasons will juvenile plants produce sporangia (Pfeiffer 1922). Quillworts are deemed slow growers (Gacia and Ballesteros 1994).	taxonomically determined)
	Age Structure {age structure, age class, population growth, demography, demographics}	UNKNOWN	We have no information about age structure in TWQ populations. However, some species demonstrate top-heavy (adult dominance) age structure (Čtvrtlíková et al. 2009; Čtvrtlíková et al. 2023).	LOW: expert best estimate/personal knowledge/opinion, unpublished data, or discrepancies in available information
	Recruitment {recruitment, survival, survivorship, demography, demographics, juvenile, population growth}	UNKNOWN	TWQ recruitment has not been formally investigated, but some sister taxa demonstrate low levels (due to their environmental setting) (Čtvrtlíková et al. 2016; Čtvrtlíková et al. 2020; Čtvrtlíková et al. 2023).	LOW: expert best estimate/personal knowledge/opinion, unpublished data, or discrepancies in available information

Appendix G. Future Land Cover Change

G-1 Future land cover change for each analysis unit (AU/population) at 2050, 2070, and 2100 under two future scenarios: scenario A (lower impacts) and scenario B (higher impacts). The value presented is the percentage of unsuitable land cover within each HUC-12.

AU	HUC-12 Code	A-2050	A-2070	A-2100	B-2050	B-2070	B-2100
1	030300060803	11.3	11.3	11.3	10.7	10.5	10.4
2	030300060805	12.0	12.0	12.0	12.0	11.8	10.9
3	030402080101	16.7	17.7	18.6	19.8	23.0	28.3
4	030402080101	16.7	17.7	18.6	19.8	23.0	28.3
5	030300070602	20.6	21.3	22.8	20.1	21.3	22.2
6	030203010102	13.1	13.6	14.3	15.2	16.2	17.2
7	030203010106	22.4	22.1	22.2	22.5	22.6	23.9